A LEAST SINGULAR VALUE BOUND FOR RANDOM BALANCED BANDED TOEPLITZ MATRICES

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Date: June 2026.

2020 *Mathematics Subject Classification.* Primary 15B05, 60B20; Secondary 15A18, 15A12, 42A05, 30C15.

Key words and phrases. banded Toeplitz matrices, least singular value, condition number, random polynomials, small-ball probability, negative second moment, Konyagin–Schlag.

ABSTRACT. Let $P_{r,r}(z) = \sum_{j=-r}^{r} \xi_j z^j$ be a random symbol with balanced band $r = s$, bandwidth $m = 2r$, and coefficients ξ_j that are independent, mean zero, variance one and have a bounded density; the Gaussian case is included. Write $G_m(z) = z^r P_{r,r}(z) = \sum_{j=0}^{m} \xi_j z^j$ for the associated Kac polynomial and $T_n = T_n(P_{r,r})$ for the $n \times n$ finite sections. The companion paper [1] proves that the stability probability of (T_n) is at most $\frac{1}{2} + o(1)$ for *every* band, through a parity obstruction, and proves exponential instability only for strongly unbalanced bands $|r - s| > 2m^{9/10}$; in the balanced regime it gives no quantitative bound on the smallest singular value on the stable subevent. This note isolates the missing conditioning mechanism. Importing the least singular value machinery of the phase retrieval paper [2] — the negative second moment small-ball identity and a good-family second moment philosophy — we formulate the boundary small-ball estimates for G_m on the unit circle that close the Gaussian balanced case up to the zero-winding event. The expected single-arc rate on arcs of length c/m is $\mathbb{P}\{\min_I |G_m| \le t\} \le C(1+c)\, t/\sqrt{m}$, governed by one-dimensional curve crossing rather than two-dimensional point anti-concentration; we prove this sharp rate away from $z = \pm 1$ by a Kac–Rice crossing count for $|G_m(e^{i\theta})|^2$, and prove the endpoint contribution by a separate endpoint Kac–Rice estimate. We also compute the exact two-endpoint Gaussian formula and the sharp second-order endpoint expansion through the t^2/N term. Together these give the sharp uniform Gaussian boundary small-ball upper law $\mathbb{P}\{\min_{\mathbb{T}} |G_m| \le t\} \le C\sqrt{m}\, t$. In the fixed bulk we compute the leading Kac–Rice crossing constant and the resulting first-moment probability upper bound at $t = \tau/\sqrt{N}$. At the continuum scale $t = \tau/\sqrt{N}$ the two parity endpoints contribute only $2\sqrt{2/\pi}\,\tau/N + O_\tau(N^{-2})$, and the full natural $O(1/m)$ endpoint neighborhoods have probability $O_{C_0}(\tau/N)$, so the $O(1)$ candidate law is a non-real-arc phenomenon. We also isolate the exact remaining lower-tail input: the fixed-bulk upper bound and endpoint-neighborhood separation give the sharp whole-circle continuum upper constant $\sqrt{\pi/3}\,\tau$, while a second-factorial-moment bound for the bulk down-crossing count implies, by Paley–Zygmund, a fixed-bulk lower tail of order $(2\pi - 4\delta)\tau/\sqrt{12\pi}$ up to the expected quadratic correction. Together with the sharp upper bounds, the same second-moment input gives the full first-order fixed-bulk law as $\tau \downarrow 0$, and a version uniform as $\delta \downarrow 0$ gives the corresponding first-order whole-circle lower law with leading constant $\sqrt{\pi/3}$. The sharp global upper tail also gives a useful endpoint-integrability consequence: the inverse scaled minimum $(\sqrt{N} \min_{\mathbb{T}} |G_m|)^{-1}$ is uniformly weak L^1, has sharp finite-window weak constant at most $\sqrt{\pi/3}$, and has $O(\alpha^{-1})$ inverse upper-tail quantiles with sharp limsup constant at most $\sqrt{\pi/3}\,\alpha^{-1}$. The same weak endpoint and quantile bounds hold simultaneously for finite batches, with the expected extra factor equal to the batch size; the corresponding finite-batch package also controls bad-clearance counts, clearance order-statistic quantiles, total bad-mass, powered and jump/Stieltjes-weighted deficit summability, subcritical inverse-mass, Stieltjes inverse-gauge budgets, confidence, menu, and aggregate-menu bounds, normalized deficit-budget schedules, and sparse bad-count Borel–Cantelli schedules, subcritical inverse-power moments, critical truncated expectation, and logarithmic moments with the same factor. It has only logarithmic critical truncated expectation with the same leading coefficient, and has uniform logarithmic inverse moments of every positive order, with limsup constant at most $\sqrt{\pi/3}\Gamma(p+1)$ for the p-th logarithmic moment. Therefore the scaled minimum has uniformly bounded negative moments of every order below one, with the sharp small-window upper coefficient inherited from $\sqrt{\pi/3}$. For bounded-density laws we prove the matching fixed-point and deterministic-grid anti-concentration, its independent-random grid conditioning consequence, plus local and uniform boundary bounds conditional

1. Introduction

A banded Toeplitz matrix is constant along its diagonals and supported on finitely many of them. Given nonnegative integers r, s and numbers $x_{-r}, \dots, x_s$, the $n \times n$ matrix T_n has entries $(T_n)_{jk} = x_{k-j}$ for $-r \le k-j \le s$ and 0 otherwise, and the asymptotic theory of $(T_n(P))_{n\ge 1}$ is governed by the symbol $P(z) = \sum_{j=-r}^{s} x_j z^j$; the reference is Böttcher–Grudsky [3]. The sequence $(T_n(P))$ is *stable* — eventually invertible with uniformly bounded inverses — if and only if P has no zeros on the unit circle $\mathbb{T}$ and zero winding number [3, Ch. 3]. Writing

$$G_m(z) := z^r P(z) = \sum_{j=0}^{m} \xi_{j-r} z^j, \qquad m := r + s, \tag{1.1}$$

and $\nu := \#\{z \in \mathbb{D} : G_m(z) = 0\}$ for the number of zeros in the open unit disk, stability is the requirement that $P \neq 0$ on $\mathbb{T}$ and that exactly r of the m zeros of G_m lie inside $\mathbb{T}$.

Randomizing the coefficients turns G_m into a Kac polynomial, and the stability question becomes a delicate question about the exact disk-count ν. The companion paper [1] settles the qualitative side of this question and locates a gap in the literature. Its parity obstruction shows that for any independent, mean-zero, unit-variance coefficients with a finite third moment, $\nu \equiv \mathbf{1}\{G_m(1)G_m(-1) < 0\} \pmod 2$ on the event $G_m(1)G_m(-1) \neq 0$, and the right-hand side is an asymptotically fair coin; hence

$$\mathbb{P}\big((T_n(P_{r,s}))_n \text{ stable}\big) \le \tfrac{1}{2} + o(1) \tag{1.2}$$

for *every* band. On the unstable side, [1] proves that strongly unbalanced bands $|r - s| > 2m^{9/10}$ are exponentially ill-conditioned: $\|T_n^{-1}\| \ge Ce^{\alpha_m n}$ and $\kappa(T_n)$ grows exponentially in n, with probability $1 - O((\log m)^{-1/2})$. The mechanism is operator-theoretic: a nonzero winding number forces an exponentially blowing-up inverse [3, Thm. 4.1], and (1.2) together with the zero-count window confines ν near $m/2$, so that a band far from balanced cannot have winding zero.

What [1] does not provide, and what it explicitly leaves open, is any quantitative control of the smallest singular value $\sigma_{\min}(T_n)$ in the *balanced* regime $r = s$ on the stable subevent. There the parity obstruction caps the stability probability at $\frac{1}{2}$ but says nothing about the size of $\sigma_{\min}(T_n)$ once the zero-winding condition $\nu = r$ actually holds. The operator-theoretic exponential lower bound of the unbalanced case is simply unavailable on that zero-winding event. This paper isolates the zero-winding replacement: for every bounded-density variance-one coefficient law satisfying the same circle-clearance estimate, balanced finite sections are polynomially conditioned on the stable subevent. In the Gaussian case the needed clearance is supplied by Konyagin–Schlag, so the balanced-conditioning conclusion is unconditional on the zero-winding event.

1.1. **The mechanism: from the circle to the finite section.** The link between $\sigma_{\min}(T_n)$ and the symbol is classical and local to the unit circle. If P does not vanish on $\mathbb{T}$, then the finite sections inherit a lower bound governed by how close P comes to zero on $\mathbb{T}$ together with the rate at which the nearest zeros of G_m clear the circle. Concretely, for banded symbols there is a constant $c_0 = c_0(m)$, polynomial in m, such that

$$\sigma_{\min}\big(T_n(P)\big) \;\geq\; c_0(m)^{-1} \min_{z\in\mathbb{T}} |P(z)| \;\big(\,\mathrm{dist}(\text{zeros of } G_m, \mathbb{T})\,\big)^{O(1)} \tag{1.3}$$

once n exceeds a threshold depending on m; we make a clean form of this precise in Section 4 (Proposition 4.1). Two independent random quantities therefore control $\sigma_{\min}(T_n)$:

(i) the *boundary minimum* $\min_{z\in\mathbb{T}} |G_m(z)|$, which one needs to keep away from 0; and
(ii) the *circle clearance* $\min\{\,|\,|\zeta| - 1\,| : G_m(\zeta) = 0\}$, the distance of the zero cloud to $\mathbb{T}$.

The clearance (ii) is exactly the Konyagin–Schlag estimate already used in [1, 4]: with high probability no zero of G_m lies within $m^{-2}(\log m)^{-3}$ of $\mathbb{T}$. The boundary minimum (i) is the genuinely new ingredient, and it is here that the phase-retrieval machinery enters.

1.2. **Importing the least singular value machinery.** The phase retrieval paper [2] develops, for a tall Gaussian matrix $A \in \mathbb{F}^{N\times m}$, two tools that we transplant from the linear-algebra setting to the boundary-evaluation setting on $\mathbb{T}$.

The first is the *negative second moment identity*. For an invertible $m \times m$ matrix G with rows $R_1, \dots, R_m$ and $d_i = \mathrm{dist}(R_i, \mathrm{span}\{R_j : j \neq i\})$,

$$\sum_{k=1}^{m} \sigma_k(G)^{-2} = \sum_{i=1}^{m} d_i^{-2}, \tag{1.4}$$

where σ_k denotes the k-th singular value [2, Lem. 2.1]. If $\sigma_{\min}(G) \leq t$ the left side is $\geq t^{-2}$, so some distance $d_i \leq \sqrt{m}\, t$; since each d_i is, conditionally on the other rows, a one-dimensional Gaussian projection, a union bound gives the one-row small-ball bound

$$\mathbb{P}\{\sigma_{\min}(G) \leq t\} \leq C_{\mathbb{F}}\, m^{1+d_{\mathbb{F}}/2}\, t^{d_{\mathbb{F}}}, \qquad d_{\mathbb{F}} = \dim_{\mathbb{R}} \mathbb{F}. \tag{1.5}$$

The point is that (1.4) converts a small *least singular value* (a global, geometric event) into a small *distance of a single row to a hyperplane* (a one-dimensional small-ball event), which is elementary to bound; the small-ball estimate for such a one-dimensional projection is the linear-image small-ball of Rudelson–Vershynin [7]. The same one-dimensional reduction reappears below as the curve-crossing mechanism on an arc, where the role of the projected coordinate is played by the planar curve $\{G_m(z) : z \in I\}$.

The second tool addresses the obstruction that makes the problem hard. To bound a minimum over an exponentially large family one cannot afford a union bound: in [2] there are $\binom{N-1}{m-1} = \exp(\Theta(m))$ hyperplanes, and in our

setting the analogue is a family of $\Theta(m)$ arcs together with the continuum of points within each. The resolution in [2] is a *good-family second moment argument*: one fixes an apex (one Gaussian row) and counts, via $Z_t = \sum_S \mathbf{1}\{|\langle x, u_S\rangle| \le t\}$, how many hyperplanes the apex nearly meets; the first moment $\mathbb{E}Z_t$ is large, and the pairwise overlap combinatorics (Lemmas on the two-slab estimate and the overlap distribution there) make $\operatorname{Var} Z_t/(\mathbb{E}Z_t)^2 \to 0$, so that $Z_t > 0$ with high probability by Chebyshev. The crucial structural fact that powers the second moment is *decorrelation*: two row sets sharing few rows give nearly independent hyperplane normals. In the Toeplitz problem the same role is played by the Dirichlet kernel

$$\operatorname{Cov}(G_m(z), G_m(w)) = \sum_{j=0}^{m} z^j \overline{w}^{\,j} = \frac{1-(z\overline{w})^{m+1}}{1-z\overline{w}}, \tag{1.6}$$

which for $z, w \in \mathbb{T}$ at angular separation $\gg 1/m$ is small: the evaluations of G_m at points of $\mathbb{T}$ that are more than a Dirichlet width apart are weakly correlated, so $\mathbb{T}$ splits into $\Theta(m)$ arcs that decorrelate, exactly the analogue of the weakly-overlapping row sets. For the upper bound proved below, the continuum of points inside one arc is handled by a curve-crossing argument; the good-family second moment is what the matching *lower* bound, the exponential law of [5, 6], and the sharp whole-circle picture require.

1.3. **Results.** The boundary small-ball estimates are the analytic input to the balanced-regime conditioning bound. We first give the proved Gaussian single-arc estimate, then the proved sharp uniform whole-circle estimate. The single-arc estimate is governed not by the two-dimensional point anti-concentration of $G_m(z)$ but by the geometry of the planar curve $\{G_m(z) : z \in I\}$: as z traverses an arc of length c/m this curve has length of order $\sqrt{m}$, so the event that it comes within t of the origin has the *one-dimensional* curve-crossing probability $t/\sqrt{m}$, rather than the pointwise rate t^2/m.

Theorem 1.1 (Boundary small-ball, single arc)**.** *Let $G_m(z) = \sum_{j=0}^{m} \xi_j z^j$ with ξ_j independent standard real Gaussians. For every fixed $\gamma > 0$ there is a constant $C_\gamma > 0$ such that for every $c \ge 1$, every arc $I \subset \mathbb{T}$ with arclength $|I| = c/m$ and angular distance at least γ/m from the two real points $z = \pm 1$, and every $t \in (0, 1]$,*

$$\mathbb{P}\Big\{\min_{z\in I} |G_m(z)| \le t\Big\} \;\le\; C_\gamma\,(1+c)\,\frac{t}{\sqrt{m}}. \tag{1.7}$$

The exponent of t is 1 and that of m is $-\frac{1}{2}$; the bound is the curve-crossing rate, not the pointwise t^2/m rate.

The proof, given below, is a one-dimensional Kac–Rice crossing count for $|G_m(e^{i\theta})|^2$. Proposition 3.21 records the older occupation-measure proof with a logarithmic speed loss because that argument is robust and is reused for the speed-conditional bounded-density estimates.

Theorem 1.2 (Uniform boundary small-ball)**.** *With G_m as above (independent standard real Gaussian coefficients), there is an absolute constant $C' > 0$ such that*

$$\mathbb{P}\Big\{\min_{z\in\mathbb{T}} |G_m(z)| \le t\Big\} \;\le\; C'\sqrt{m}\,t \qquad (0 < t \le 1), \tag{1.8}$$

equivalently $\mathbb{P}\{\min_{\mathbb{T}} |G_m| \le \tau/\sqrt{m}\} \le C'\tau$. The expected constant is governed by the Kac–Rice first moment for down-crossings, of order $t\sqrt{m}$, in agreement with the minimum-modulus picture of [4, 5, 6]; away from the real points this first moment is two-sided by Proposition 3.14, in the fixed bulk its leading density is computed in Proposition 3.15, and Corollary 3.16 gives the resulting continuum-scale fixed-bulk first-moment limit at $t = \tau/\sqrt{N}$. Corollary 3.17 converts that first moment into an explicit fixed-bulk probability upper bound with the down-crossing constant, while Proposition 3.18 proves the complementary conditional lower-tail transfer from a second-factorial-moment estimate for the down-crossing count, and Corollary 3.19 sandwiches the fixed-bulk probability to first order in τ. Corollary 3.20 records the endpoint-cutoff removal: if the same factorial second-moment estimate is uniform over fixed bulks as $\delta \downarrow 0$, then the whole-circle lower tail has leading constant $\sqrt{\pi/3}$. At the two real endpoints the local order is already sharp, by Proposition 3.25; Corollaries 3.26, 3.27, and 3.28 give the exact bivariate Gaussian endpoint formula, the sharp second-order expansion, and the fact that at $t = \tau/\sqrt{N}$ both the two endpoints and their natural $O(1/m)$ neighborhoods have only $O_\tau(N^{-1})$ probability. Corollary 3.29 combines this endpoint separation with the fixed-bulk upper bound to give the sharp whole-circle continuum-scale upper constant $\sqrt{\pi/3}$, and Corollary 3.30 gives the conditional whole-circle first-order law under the uniform factorial second-moment input. The missing part of the matching global lower tail is therefore precisely the anti-clustering and decorrelation estimate that bounds the second factorial moment of the $O(m)$ non-real down-crossing opportunities.

For the finite-section theorem one does not need the sharp log-free estimate at all scales. It is enough to know that a polynomially small boundary minimum is unlikely. The following corollary is the polynomial-scale form used in Theorem 1.10.

Corollary 1.3 (Uniform Gaussian boundary small ball at polynomial scales)**.** *Let $G_m(z) = \sum_{j=0}^m \xi_j z^j$ with ξ_j independent standard real Gaussians. For every $A, B > 0$ there are constants $C_{A,B} < \infty$ and $m_0(A, B)$ such that, for every $m \ge m_0(A, B)$ and every $m^{-B} \le t \le 1$,*

$$\mathbb{P}\Big\{\min_{z\in\mathbb{T}} |G_m(z)| \le t\Big\} \le C_{A,B}\sqrt{m}\,t. \tag{1.9}$$

In particular, Hypothesis 1.4 holds for standard real Gaussian coefficients.

Hypothesis 1.4 (Polynomial-scale uniform boundary small ball for the coefficient law)**.** *For the coefficient law in Theorem 1.6, and the associated $G_m(z) = \sum_{j=0}^m \xi_j z^j$, for every $A, B > 0$ there are constants $C_{A,B,K} < \infty$*

and $m_0(A, B, K)$, where K denotes the density bound of the law, such that, for every $m \geq m_0(A, B, K)$ and every $m^{-B} \leq t \leq 1$,

$$\mathbb{P}\Big\{\min_{z\in\mathbb{T}} |G_m(z)| \leq t\Big\} \leq C_{A,B,K}\sqrt{m\log m}\, t + C_{A,B,K} m^{-A}.$$

Theorem 1.2 implies Hypothesis 1.4 for Gaussian coefficients, and Corollary 1.3 records the polynomial-scale form used below. For non-Gaussian bounded-density laws, Hypothesis 1.4 is the natural polynomial-scale boundary small-ball input. The finite-dimensional part of that input is unconditional: Proposition 3.2 below proves the correct fixed-point and deterministic-grid small-ball estimates for every bounded-density law, and Corollary 3.57 gives the corresponding continuum bound after intersecting with a deterministic speed cutoff. Corollary 3.67 records the exact conditional upgrade: a polynomially small speed-tail estimate at the natural $\sqrt{\log m}$ scale implies Hypothesis 1.4; Corollaries 3.70 and 3.71 check that speed-tail input for all-polynomial-moment and subexponential bounded-density laws, respectively (hence also for subgaussian ones). For the finite-section conditioning theorem itself, Theorem 1.6 needs only a much cruder polynomial speed cutoff, available from variance one. What remains open in full bounded-density generality, without a tail assumption, is the sharp continuum boundary law itself: proving a natural-scale speed/crossing input, or replacing it by a genuine local-limit argument.

The full boundary law is stronger than the finite-section conditioning theorem needs. The conditioning proof uses only three high-probability inputs: circle clearance, one polynomial boundary-minimum event, and a crude coefficient-size bound. Corollary 3.60 supplies that one boundary-minimum event for every bounded-density variance-one law, at the cost of a deliberately polynomial speed cutoff in its proof; the exceptional probability is $O_{A,K}(m^{-2(A-1)/3})$ for the threshold m^{-A}. Thus the only coefficient-law input left in the conditioning theorem, beyond bounded density and variance one, is circle clearance. The point is elementary but important: the final probability is measured as

$$\mathbb{P}\{\nu = r\} - o(1),$$

not as a conditional probability given $\nu = r$. Hence a high-probability clearance event may be intersected with the zero-winding event at additive cost $o(1)$, without any independence or transport of the clearance law through the event $\{\nu = r\}$. The finite-section threshold is deterministic once a clearance floor is fixed; see Proposition 4.2.

We therefore isolate the remaining genuine probabilistic input.

Hypothesis 1.5 (Circle-clearance estimate for the coefficient law)**.** *Let the coefficients be as in Theorem 1.6 below. With $\rho := \min\{\,||\zeta| - 1| : G_m(\zeta) = 0\}$ the distance of the zero cloud of G_m to $\mathbb{T}$, there are a constant $C_0 > 0$*

and events $\mathcal{C}_m$ with $\mathbb{P}(\mathcal{C}_m) = 1 - o(1)$ such that, on $\mathcal{C}_m$,

$$\rho \geq \frac{1}{C_0\, m^2 (\log m)^3}$$

and in particular G_m has no zeros on $\mathbb{T}$.

For standard real Gaussian coefficients this hypothesis is exactly the circle-clearance theorem of Konyagin–Schlag [4, Theorem 1.1], as used in [1]. For other coefficient laws, Theorem 1.6 assumes this same clearance input explicitly.

Theorem 1.6 (Balanced conditioning on the stable event under clearance)**.** *Let $r = s$, $m = 2r$, and let the coefficients $\xi_{-r}, \dots, \xi_r$ be independent, mean zero, variance one, with a density bounded by K (the Gaussian case is included). Set*

$$\nu := \#\{z \in \mathbb{D} : G_m(z) = 0\}.$$

Assume Hypothesis 1.5. Then, for every $b > 0$, there exist a deterministic function $n_{0,b}(m) < \infty$ and events Ω_m with $\mathbb{P}(\Omega_m) = 1 - o(1)$ such that, on $\Omega_m \cap \{\nu = r\}$, the symbol $P_{r,r}$ has no zeros on $\mathbb{T}$, has winding number zero, and for all $n \geq n_{0,b}(m)$

$$\sigma_{\min}(T_n(P_{r,r})) \;\geq\; n^{-b} \qquad \text{and} \qquad \kappa(T_n(P_{r,r})) \;\leq\; n^{b}. \tag{1.10}$$

Equivalently, for any prescribed polynomial exponent, the conditioning conclusion holds with probability at least $\mathbb{P}\{\nu = r\} - o(1)$, after increasing the deterministic finite-section threshold. The theorem does not assert that a balanced random symbol is stable with high probability; that would require a separate local theorem for ν. This is the zero-winding counterpart to the exponential ill-conditioning of the strongly unbalanced regime [1, Thm. 1.2].

Corollary 1.7 (Quantitative stable-event conditioning)**.** *In the setting of Theorem 1.6, assume more quantitatively that for some $C_0 > 0$, events $\mathcal{C}_m$, and numbers $q_m \in [0, 1]$,*

$$\mathbb{P}(\mathcal{C}_m^c) \leq q_m, \qquad \rho \geq \frac{1}{C_0 m^2 (\log m)^3} \quad \text{on } \mathcal{C}_m.$$

Fix $A_0 > 2$, $b > 0$, and a deterministic envelope $R_m > 0$. Then there are deterministic thresholds $n_{0,b,A_0,R}(m) < \infty$ and events $\Omega_{m,A_0,R}$ such that

$$\mathbb{P}(\Omega_{m,A_0,R}^c) \leq q_m + C_{A_0,K} m^{-2(A_0-1)/3} + \frac{m+1}{R_m^2},$$

and, on $\Omega_{m,A_0,R} \cap \{\nu = r\}$, the symbol $P_{r,r}$ has no zeros on $\mathbb{T}$, has winding number zero, and for all $n \geq n_{0,b,A_0,R}(m)$,

$$\sigma_{\min}(T_n(P_{r,r})) \geq n^{-b}, \qquad \kappa(T_n(P_{r,r})) \leq n^{b}.$$

Consequently the same conclusion holds with probability at least

$$\mathbb{P}\{\nu = r\} - q_m - C_{A_0,K} m^{-2(A_0-1)/3} - \frac{m+1}{R_m^2}.$$

In particular, for every $D > 0$, choosing $R_m = m^{(D+1)/2}$ replaces the last loss by $2m^{-D}$ for $m \geq 1$, after changing only the deterministic threshold.

Corollary 1.8 (Prescribed polynomial stable-event loss)**.** *In the setting of Corollary 1.7, fix $b > 0$ and $D > 0$. Then there are deterministic thresholds $n_{0,b,D}(m) < \infty$ and events $\Omega_{m,D}$ such that*

$$\mathbb{P}(\Omega_{m,D}^c) \leq q_m + C_{D,K} m^{-D},$$

and, on $\Omega_{m,D} \cap \{\nu = r\}$, the symbol $P_{r,r}$ has no zeros on $\mathbb{T}$, has winding number zero, and for all $n \geq n_{0,b,D}(m)$,

$$\sigma_{\min}(T_n(P_{r,r})) \geq n^{-b}, \qquad \kappa(T_n(P_{r,r})) \leq n^b.$$

Consequently the same conclusion holds with probability at least

$$\mathbb{P}\{\nu = r\} - q_m - C_{D,K} m^{-D}.$$

Proof. Apply Corollary 1.7 with

$$A_0 = 1 + \frac{3}{2}(D+1), \qquad R_m = m^{(D+2)/2}.$$

Then $A_0 > 2$ and $2(A_0 - 1)/3 = D + 1$, so the boundary-separation loss is $C_{D,K} m^{-(D+1)}$. Also

$$\frac{m+1}{R_m^2} \leq 2m^{-(D+1)} \leq 2m^{-D} \qquad (m \geq 1).$$

Absorbing constants and renaming the deterministic threshold gives the stated event and probability bound. □

Corollary 1.9 (Conditional stable-event probability)**.** *In the setting of Corollary 1.8, suppose $p_m := \mathbb{P}\{\nu = r\} > 0$. Let $\mathcal{G}_{m,b,D}$ be the event that $P_{r,r}$ has no zeros on $\mathbb{T}$, has winding number zero, and for all $n \geq n_{0,b,D}(m)$,*

$$\sigma_{\min}(T_n(P_{r,r})) \geq n^{-b}, \qquad \kappa(T_n(P_{r,r})) \leq n^b.$$

Then

$$\mathbb{P}\left(\mathcal{G}_{m,b,D} \,|\, \nu = r\right) \geq 1 - \frac{q_m + C_{D,K} m^{-D}}{p_m}.$$

In particular, if $p_m \geq p_* > 0$ along a subsequence, then the conditional failure probability on that subsequence is at most $p_*^{-1}(q_m + C_{D,K} m^{-D})$.

Proof. Corollary 1.8 gives an event $\Omega_{m,D}$ with complement probability at most $q_m + C_{D,K} m^{-D}$ and $\Omega_{m,D} \cap \{\nu = r\} \subset \mathcal{G}_{m,b,D} \cap \{\nu = r\}$. Hence

$$\mathbb{P}\left(\mathcal{G}_{m,b,D}^c \cap \{\nu = r\}\right) \leq q_m + C_{D,K} m^{-D}.$$

Dividing by $p_m = \mathbb{P}\{\nu = r\}$ proves the conditional estimate. □

Theorem 1.10 (Gaussian balanced conditioning)**.** *In the setting of Theorem 1.6, assume the coefficients are independent standard real Gaussians. Then the conclusion of Theorem 1.6 holds unconditionally: Hypothesis 1.5 is supplied by the Konyagin–Schlag clearance theorem [4].*

Corollary 1.11 (Subexponential boundary law and conditioning)**.** *In the setting of Theorem 1.6, assume in addition that* $\|\xi_j\|_{\psi_1} \le \kappa$ *uniformly in* j *and* m*. Then Hypothesis 1.4 holds for this coefficient law. If the coefficient law also satisfies the circle-clearance estimate Hypothesis 1.5, then the conclusion of Theorem 1.6 holds. Thus, for subexponential bounded-density laws – in particular for subgaussian bounded-density laws – the full boundary-law input is available; this is stronger than what the conditioning proof under clearance ultimately needs.*

Corollary 1.12 (All-moment bounded-density boundary law and conditioning)**.** *In the setting of Theorem 1.6, assume instead that for every* $p < \infty$ *the coefficients satisfy*

$$\sup_{m\ge 1} \max_{0\le j\le m} \mathbb{E}\,|\xi_j|^p < \infty.$$

Then Hypothesis 1.4 holds for this coefficient law. If the coefficient law also satisfies the circle-clearance estimate Hypothesis 1.5, then the conclusion of Theorem 1.6 holds. Thus the exponential-tail assumption is not the boundary input required for the full polynomial-scale boundary law; uniformly bounded moments of all orders already supply the needed natural-scale continuum speed tail.

Indeed, Corollaries 3.71 and 3.70 verify Hypothesis 1.4 from, respectively, the subexponential derivative tail and the Fuk–Nagaev polynomial-moment derivative tail. Theorem 1.6 shows that the weaker conditioning conclusion only needs the one-threshold substitute packaged in Corollary 3.60, while Hypothesis 1.5 is assumed here.

The boundary estimate over the whole circle is what upgrades "one arc is good" to "all arcs are good simultaneously": a single favorable arc cannot give the finite-section bound (1.10), because Proposition 4.1 needs the full boundary minimum $\min_{z\in\mathbb{T}} |G_m(z)|$. For the full polynomial-scale boundary law this remains the role of Hypothesis 1.4; for the conditioning conclusion under clearance, the tail-free polynomial separation in Corollary 3.60 is already sufficient.

1.4. **The bridge to the parity obstruction.** The boundary evaluations that drive Theorem 1.1 are the same two evaluations that drive the parity obstruction of [1]. The parity of ν is the parity of the number of real zeros of G_m in $(-1, 1)$, which is controlled by the signs of the two boundary values $G_m(1) = \sum_j \xi_j$ and $G_m(-1) = \sum_j (-1)^j \xi_j$, the evaluations of G_m at the two real points of $\mathbb{T}$. The boundary small-ball estimate is the quantitative refinement of the same observable: instead of asking only for the *sign* of $G_m(\pm 1)$, it asks how close $|G_m|$ comes to zero along $\mathbb{T}$.

This shared dependence on boundary evaluations is precisely where the two papers meet, and the meeting point is the Bernoulli case. In the balanced setup of this paper $m = 2r$, so $m + 1$ is odd and the two sums $G_m(1)$ and $G_m(-1)$ cannot vanish for Rademacher coefficients. Nevertheless their lattice

spacing is a boundary obstruction to density-type small-ball estimates at very fine scales. In adjacent parities, where an even number of signs is present, exact boundary zeros occur with polynomial probability at $z = \pm 1$. This is the Toeplitz analogue of the obstruction recorded in [2, Rem. 6.2]: for ± 1 measurement matrices the least singular value over square submatrices can vanish by a coordinate hyperplane collision, so continuity of the entry law is necessary for the clean lower-bound theorem. We develop this correspondence in Section 5.

1.5. **What is proved and what is conjectured.** Because the bounded-density theorem still separates the coefficient-law inputs, we state plainly where the line falls.

- **Proved.** The finite-section symbol reduction (1.3) on the zero-winding event (Propositions 4.1 and 4.2), the sharp Gaussian single-arc bound away from $z = \pm 1$ (Theorem 1.1), the away-from-endpoint two-sided first moment for crossings, the explicit fixed-bulk Kac–Rice down-crossing constant $1/\sqrt{12\pi}$, the fixed-bulk continuum-scale crossing limit at $t = \tau/\sqrt{N}$, the corresponding fixed-bulk first-moment probability upper bound, the Paley–Zygmund transfer from a bulk down-crossing factorial second moment to a fixed-bulk lower tail, the resulting fixed-bulk first-order probability sandwich, the sharp global continuum-scale probability upper bound, the weak-L^1 inverse-scaled-minimum endpoint, single and finite-family inverse-minimum quantile bounds, finite-family clearance order-statistic quantile and total/sparse bad-count schedule bounds, finite-family Stieltjes-weighted/powered clearance-deficit summability, normalized deficit-budget schedules, Stieltjes inverse-gauge mass schedules, inverse-gauge confidence budgets, gauge-menu budgets, and aggregate gauge-menu summability, finite-family subcritical inverse-power and truncated/logarithmic endpoint moments, the logarithmic critical truncated inverse-moment bound, logarithmic inverse-moment bounds of all positive orders, and subcritical negative-moment bounds for the scaled global boundary minimum, the conditional global first-order probability sandwich under a uniform factorial moment bound, and the uniform-in-δ transfer from fixed bulks to the whole-circle lower-tail constant $\sqrt{\pi/3}$ (Propositions 3.14, 3.15, and 3.18, Corollaries 3.16, 3.17, 3.19, 3.29, 3.31, 3.32, 3.33, 3.35, 3.36, 3.37, 3.45, 3.46, 3.47, 3.48, 3.49, 3.50, 3.40, 3.41, 3.42, 3.43, 3.44, 3.51, 3.38, 3.39, 3.52, 3.53, 3.30, and 3.20), the endpoint lower bound showing the real-point arc scale is sharp, the exact two-endpoint Gaussian formula and sharp second-order expansion, the $O_\tau(N^{-1})$ endpoint and natural-endpoint-neighborhood contributions at the continuum scale, and the resulting global endpoint floor (Proposition 3.25 and Corollaries 3.26, 3.27, 3.28, and 3.54), the sharp uniform Gaussian boundary small-ball bound (Theorem 1.2), the exact Gaussian non-real and full

root-grid law, including endpoint survival factors, and its pointwise Rayleigh-limit/moment-expansion and exact continuum-scale grid-coefficient corollaries (Proposition 3.74 and Corollaries 3.75 and 3.76), the fixed-point/deterministic-grid bounded-density small-ball estimate, deterministic and independent-random subquadratic grid-miss consequences, matching quadratic-resolution detection under a speed cutoff, independent uniform random-grid detection with one extra covering logarithm, and Gaussian quadratic-log high-probability grid detection (Proposition 3.2, Corollaries 3.3, 3.4, 3.7, and 3.11, and Lemma 3.5), the bounded-density local and uniform estimates under a deterministic speed cutoff (Proposition 3.56 and Corollary 3.57), the variance-only polynomial derivative cutoff (Lemma 3.58), the resulting bounded-density conditioning theorem under circle clearance and its quantitative stable-event loss (Theorem 1.6 and Corollary 1.7), the reduction from a polynomial speed tail to the full polynomial-scale boundary law and to high-probability grid detection (Corollaries 3.67 and 3.68), the verification of that speed-tail hypothesis for all-polynomial-moment and subexponential bounded-density laws (Corollaries 3.70 and 3.71), the endpoint, order-q root-of-unity, fixed-radius lower, rational-grid, bounded-denominator, fixed-root, primitive-quadratic, primitive-quadratic second-order/overlap, degree-four, finite-cyclotomic-layer, and bounded-denominator exact-zero Rademacher lattice anti-concentration/atom scales, including the exact parity-residue criterion for zero atoms, the peeled layer formula, the first two exact-zero coefficients, the full unpeeled expansion through N^{-2}, and the endpoint-inclusive bounded-denominator expansion through order N^{-1} (Propositions 5.1, 5.2, 5.17, and 5.5, and Corollaries 5.3, 5.4, 5.6, 5.7, 5.8, 5.9, 5.10, 5.11, 5.12, 5.13, 5.14, 5.15, and 5.16), and the unconditional Gaussian deduction in Theorem 1.10. The finite-family endpoint package also records the weak tail in Corollary 3.34, bad-clearance counts in Corollary 3.36, clearance order statistics in Corollary 3.37, and the summable growing-batch first-bad, total bad-mass, deficit-mass, normalized deficit-budget, powered and jump/Stieltjes-weighted deficit, subcritical inverse-mass, Stieltjes inverse-gauge, inverse-gauge confidence, gauge-menu, aggregate gauge-menu, and sparse bad-count schedules of Corollaries 3.45, 3.46, 3.47, 3.48, 3.49, 3.50, 3.40, 3.41, 3.42, 3.43, 3.44, and 3.51. The same variance-only cutoff also gives polynomial-grid continuum-excursion detection with failure probability $O_\delta(m^{1-2\delta})$ (Corollaries 3.59 and 3.62); its finite-confidence form gives η-level grid detection at mesh $O(\tau\sqrt{\eta}/(m^2\sqrt{N}))$ and bounded-density boundary loss $\eta + C_{B,K} mt/\sqrt{\eta}$ (Corollaries 3.65 and 3.61); in fixed-power form, every uniform N^β-grid with $\beta > 5/2$ has miss probability $O_\beta(N^{5-2\beta})$ (Corollary 3.63), hence $O_A(m^{-A})$ failure after increasing the grid size

to $O_{A,\tau}(N^{(A+5)/2})$ (Corollary 3.64), and the combined subquadratic-obstruction/fine-grid-sufficiency sampling bracket (Corollary 3.66). Under the natural polynomial derivative-speed tail, the fine-grid side improves to $O_{A,\tau}(N^2\sqrt{\log N})$ points with the same $O_A(m^{-A})$ miss probability, while the subquadratic obstruction is unchanged (Corollary 3.69). The same natural-speed scale holds for stratified one-point-per-cell random grids and random shifts of a uniform lattice, without the iid coupon-collector overhead (Corollary 3.8). Under bounded density it also gives the tail-free polynomial separation $\min_{\mathbb{T}} |G_m| > m^{-A}$ with exceptional probability $O_{A,K}(m^{-2(A-1)/3})$ for every $A > 1$ (Corollary 3.60).

- **Conjectured/conditional.** The full continuum boundary-law version of universality beyond Gaussian coefficients remains open at bounded-density generality, without tail or all-moment input. The matching Gaussian lower-tail law is now reduced, on every fixed bulk B_δ, to the factorial second-moment/decorrelation estimate in (3.2); Proposition 3.18 proves the transfer from that estimate to the lower-tail probability, Corollary 3.19 combines it with the sharp upper bound to give the fixed-bulk first-order law as $\tau \downarrow 0$, Corollary 3.29 gives the sharp whole-circle upper constant, and Corollary 3.30 shows that a version uniform as $\delta \downarrow 0$ gives the whole-circle first-order law as $\tau \downarrow 0$. The conditioning theorem no longer needs that full law once circle clearance is assumed; for coefficient laws outside the available Konyagin–Schlag hypotheses the remaining conditioning input is the circle-clearance estimate in Hypothesis 1.5. There is no additional clearance-coupling hypothesis: the intersection with $\{\nu = r\}$ costs only the unconditional exceptional probability.

We do not use an asymptotic formula for $\mathrm{Var}(\nu)$ here; the disk-count fluctuation problem belongs to the companion Toeplitz-parity analysis.

Relation to the companions and the literature. This paper is a companion to [1] and [2]. From [1] we take the operator-theoretic framework (1.2), the unbalanced exponential instability, and the Konyagin–Schlag clearance [4]; we complement it by isolating the zero-winding conditioning mechanism in the balanced regime. From [2] we take the negative-second-moment small-ball engine and the good-family second moment philosophy, transplanted from row submatrices to boundary arcs. The conditioning of *dense* random Toeplitz and circulant matrices is treated in the references collected in [1]; the operator theory of banded symbols is [3].

2. Setup and the negative second moment engine

Throughout, $m = 2r$ is even, $\xi_0, \dots, \xi_m$ are independent with $\mathbb{E}\xi_j = 0$, $\mathbb{E}\xi_j^2 = 1$, and (unless stated otherwise) with a density bounded by K; $G_m(z) = \sum_{j=0}^m \xi_j z^j$. We write $\mathbb{T} = \{z : |z| = 1\}$ and $\mathbb{D} = \{z : |z| < 1\}$,

and parametrize $\mathbb{T}$ by $z = e^{i\theta}$. The symbol is $P_{r,r}(z) = z^{-r}G_m(z)$, so $|P_{r,r}(z)| = |G_m(z)|$ on $\mathbb{T}$. Constants $c, C, \dots$ are absolute or depend only on K, and may change line to line.

2.1. **Boundary values as a Gaussian field.** Fix points $z = e^{i\theta}, w = e^{i\phi} \in \mathbb{T}$. Then $G_m(z) = \sum_j \xi_j e^{ij\theta}$, and in the real-coefficient case the real and imaginary parts

$$\Re G_m(e^{i\theta}) = \sum_{j=0}^{m} \xi_j \cos(j\theta), \qquad \Im G_m(e^{i\theta}) = \sum_{j=0}^{m} \xi_j \sin(j\theta)$$

are linear forms in the ξ_j with covariance structure governed by (1.6). The diagonal is $\mathbb{E}\,|G_m(z)|^2 = m+1$, and the covariance between two points is the Dirichlet kernel

(2.1)

$$\mathbb{E}\big[G_m(z)\overline{G_m(w)}\big] = \sum_{j=0}^{m} e^{ij(\theta-\phi)} = D_m(\theta-\phi), \qquad |D_m(\psi)| \le \min\Big\{m+1, \frac{1}{|\sin(\psi/2)|}\Big\}.$$

Thus $|D_m(\theta-\phi)| \le 1/|\sin((\theta-\phi)/2)|$ is $o(m+1)$ once $|\theta-\phi| \gg 1/m$: points more than a Dirichlet width apart are weakly correlated. This is the decorrelation that organizes $\mathbb{T}$ into $\Theta(m)$ effectively independent arcs and is the analogue, in the present problem, of the weak overlap of well-separated row sets in [2].

2.2. **The negative second moment identity on an arc.** We will reduce the minimum of $|G_m|$ over an arc to a one-dimensional small-ball event by the same device as (1.4), applied not to a single matrix but to a well-chosen finite sample of the field on the arc. Fix an arc I of length $|I| \ge c/m$ and sample G_m at $L \asymp m\,|I| \ge c$ equally spaced points $z_1, \dots, z_L \in I$ at spacing $\asymp 1/m$. Consider the random vector $\mathbf{g} = (G_m(z_1), \dots, G_m(z_L))^\top = V\xi$, where $V \in \mathbb{C}^{L\times(m+1)}$ has entries $V_{\ell j} = z_\ell^j$ (a Vandermonde block) and $\xi = (\xi_0, \dots, \xi_m)^\top$.

Lemma 2.1 (Sample covariance is well-conditioned on a short arc)**.** *Fix $c' < \infty$ and spacing constants $0 < a \le A < \infty$. There are constants $0 < \lambda_- \le \lambda_+ < \infty$ and $m_0 < \infty$, depending only on a, A, c', such that the following holds for $m \ge m_0$. If $z_\ell = e^{i\theta_\ell}$ are sample points on an arc of length at most c'/m, with $L \le Ac'$ and pairwise angular separation at least a/m, then the $L \times L$ Gram matrix $\Sigma := \mathbb{E}[\mathbf{g}\,\mathbf{g}^*] = VV^*$ satisfies*

$$\lambda_-(m+1)\,I_L \preceq \Sigma \preceq \lambda_+(m+1)\,I_L. \tag{2.2}$$

Proof. The entries of Σ are $\Sigma_{\ell\ell'} = D_m(\theta_\ell - \theta_{\ell'})$ by (2.1). Write $\theta_\ell = \theta_0 + x_\ell/m$. The parameters x_ℓ lie in a compact interval of length c', are separated by at least a, and there are only boundedly many of them. Uniformly over this compact parameter set,

$$\frac{1}{m+1} D_m\Big(\frac{x_\ell - x_{\ell'}}{m}\Big) \longrightarrow \int_0^1 e^{iu(x_\ell - x_{\ell'})}\,du.$$

The limiting matrix is the Gram matrix of the distinct exponentials $u \mapsto e^{iux_\ell}$ in $L^2[0,1]$. It is positive definite, and compactness of the separated configuration space gives a positive lower bound for its smallest eigenvalue. The upper bound is immediate from bounded L and bounded entries. The same bounds therefore hold for $\Sigma/(m+1)$ for all $m \geq m_0(a, A, c')$. □

For circular complex Gaussian coefficients, Lemma 2.1 says that on a short arc the sampled field $\mathbf{g}$ is a non-degenerate Gaussian vector in $\mathbb{C}^L$ at scale $\sqrt{m+1}$, with condition number $O(1)$. In the real-coefficient problem of the present paper, the Hermitian covariance alone does not give full real $2L$-dimensional nondegeneracy; the pointwise real nondegeneracy away from $z = \pm 1$ is supplied by Lemma 3.1, and the endpoint degeneration is handled separately by Lemma 3.23. The Vandermonde lemma is included to record the decorrelation geometry behind the negative-second-moment analogy.

3. Boundary small-ball estimates and targets

We record the route from pointwise anti-concentration to arc estimates and then to the remaining uniform problem. The mechanism is not the discrete strategy (1.4)–(1.5) applied pointwise, which would give the pointwise t^2/m rate and lose the arc geometry; it is a one-dimensional curve-crossing argument. The sharp Gaussian single-arc theorem is proved below by Kac–Rice. We also keep the older occupation-measure argument, based on a derivative cutoff and Lemma 3.55, because it is the robust template for bounded-density laws.

3.1. The three ingredients.

Lemma 3.1 (One-point anti-concentration)**.** *Let ξ_j be independent standard real Gaussians and fix $z = e^{i\theta} \in \mathbb{T}$. Then $G_m(z)$ is a centered real Gaussian vector in $\mathbb{R}^2$. For every fixed $\gamma > 0$ there are constants $c_\gamma, C_\gamma > 0$ and $m_0(\gamma)$ such that, for $m \geq m_0(\gamma)$, if z is separated from ± 1 by an angular distance at least γ/m, its covariance has both eigenvalues at least $c_\gamma(m+1)$; at the two real points $z = \pm 1$, $G_m(z)$ is real Gaussian of variance $m+1$. Consequently, for $t \in (0, 2]$,*

$$\mathbb{P}\{|G_m(z)| \leq t\} \;\leq\; \frac{C_\gamma\, t^2}{m+1} \quad (\operatorname{dist}_{\mathbb{T}}(z, \{\pm 1\}) \geq \gamma/m), \qquad \mathbb{P}\{|G_m(z)| \leq t\} \;\leq\; \frac{C\, t}{\sqrt{m+1}} \quad (z = \pm 1).$$

Proof. $\Re G_m(z) = \sum_j \xi_j \cos(j\theta)$ and $\Im G_m(z) = \sum_j \xi_j \sin(j\theta)$ are jointly Gaussian. Its covariance matrix is

$$\frac{1}{2}\begin{pmatrix} m+1+\Re S_m(\theta) & \Im S_m(\theta) \\ \Im S_m(\theta) & m+1-\Re S_m(\theta) \end{pmatrix}, \qquad S_m(\theta) := \sum_{j=0}^{m} e^{2ij\theta}.$$

Thus the eigenvalues are $\frac{1}{2}(m+1 \pm |S_m(\theta)|)$. Write $u = \operatorname{dist}(2\theta, 2\pi\mathbb{Z})$. The separation from 0 and π gives $u \geq 2\gamma/m$, and

$$|S_m(\theta)| = \left|\frac{\sin((m+1)u/2)}{\sin(u/2)}\right|.$$

For fixed γ, the ratio $|S_m(\theta)|/(m+1)$ is bounded by some $q_\gamma < 1$ uniformly for $m \geq m_0(\gamma)$: on $u \leq u_0$ this follows from the uniform bound $\sup_{x\geq\gamma} |\sin x|/x < 1$, after putting $x = (m+1)u/2$ and choosing u_0 so that $(u/2)/\sin(u/2)$ is close to 1, while on $u \geq u_0$ the ratio is $O_{u_0}(1/m)$. Hence the smallest covariance eigenvalue is at least $c_\gamma(m+1)$, so the planar Gaussian density is bounded by $C_\gamma/(m+1)$. Integrating over a disk of radius $t \leq 2$ gives the first estimate. At $z = \pm 1$ the imaginary part vanishes identically and $G_m(\pm 1)$ is a single real Gaussian of variance $m+1$, whose density is $\leq C/\sqrt{m+1}$, giving the second estimate. □

The two real points are exactly the parity points $z = \pm 1$ of [1]: at those points the field is one-real-dimensional, the small-ball exponent drops from t^2 to t^1, and the anti-concentration is weaker by a factor $\sqrt{m+1}$. This is the first appearance, inside the small-ball calculation, of the boundary mechanism developed in Section 5, and it is exactly the two endpoints that Theorem 1.1 separates off through the γ/m condition.

Proposition 3.2 (Bounded-density fixed points and deterministic grids)**.** *Let $\xi_0, \dots, \xi_m$ be independent real random variables whose densities are bounded by K. Fix $\gamma > 0$. There are constants $C_{\gamma,K}, C_K < \infty$ and $m_0(\gamma)$ such that, for $m \geq m_0(\gamma)$, all $a \in \mathbb{C}$, and all $t > 0$,*

$$\mathbb{P}\{|G_m(z) - a| \leq t\} \leq \frac{C_{\gamma,K} t^2}{m+1} \qquad (\mathrm{dist}_{\mathbb{T}}(z, \{\pm 1\}) \geq \gamma/m),$$

while at the two real points

$$\mathbb{P}\{|G_m(\pm 1) - a| \leq t\} \leq \frac{C_K t}{\sqrt{m+1}}.$$

Consequently, for every deterministic finite set $Q \subset \mathbb{T}$,

$$\mathbb{P}\Big\{\min_{z\in Q_*} |G_m(z)| \leq t\Big\} \leq \frac{C_{\gamma,K}\,|Q_*|\,t^2}{m+1}, \qquad Q_* := \{z \in Q : \mathrm{dist}_{\mathbb{T}}(z, \{\pm 1\}) \geq \gamma/m\},$$

and the two endpoint evaluations contribute at most $2C_K t/\sqrt{m+1}$.

Proof. For $z = e^{i\theta}$, let

$$A_\theta : \mathbb{R}^{m+1} \to \mathbb{R}^2, \qquad A_\theta x = \left(\sum_{j=0}^{m} x_j \cos(j\theta), \sum_{j=0}^{m} x_j \sin(j\theta)\right).$$

The deterministic row Gram $A_\theta A_\theta^\top$ is the same matrix computed as the Gaussian covariance in Lemma 3.1. Hence, if $\mathrm{dist}_{\mathbb{T}}(z, \{\pm 1\}) \geq \gamma/m$, both singular values of A_θ are at least $c_\gamma\sqrt{m+1}$ for all large m. The linear-image small-ball estimate of Rudelson–Vershynin [7], applied to the two-dimensional image $A_\theta\xi$, gives uniformly in $y \in \mathbb{R}^2$

$$\mathbb{P}\{A_\theta\xi \in B_{\mathbb{R}^2}(y,t)\} \leq \frac{C_K\ \mathrm{vol}(B_{\mathbb{R}^2}(0,t))}{s_1(A_\theta)s_2(A_\theta)} \leq \frac{C_{\gamma,K} t^2}{m+1}.$$

Taking $y = (\Re a, \Im a)$ gives the first displayed estimate, since $A_\theta\xi = (\Re G_m(z), \Im G_m(z))$.

At $z = 1$ and $z = -1$, $G_m(z) = \sum_{j=0}^{m} a_j \xi_j$ with $|a_j| = 1$, so $\|a\|_2 = \sqrt{m+1}$. The one-dimensional linear-image small-ball estimate gives

$$\sup_{u \in \mathbb{R}} \mathbb{P}\{\left|\sum_j a_j \xi_j - u\right| \le t\} \le C_K \frac{t}{\sqrt{m+1}},$$

and the event $\{|G_m(\pm 1) - a| \le t\}$ is contained in $\{|G_m(\pm 1) - \Re a| \le t\}$. This proves the endpoint bound. The grid statement is the union bound over the deterministic points. □

Corollary 3.3 (Subquadratic deterministic grids miss the continuum scale)**.** *Let the coefficients have densities bounded by K, and fix $\gamma > 0$. Let $N = m+1$, and let $Q_m \subset \mathbb{T}$ be deterministic nonempty finite sets such that every point of $Q_m \setminus \{\pm 1\}$ has angular distance at least γ/m from $\{\pm 1\}$. For every fixed $T < \infty$ there are constants $C_{\gamma,K,T} < \infty$ and $m_0(\gamma)$ such that, for $m \ge m_0(\gamma)$,*

$$\sup_{0 \le \tau \le T} \mathbb{P}\Big\{ \min_{z \in Q_m} |G_m(z)| \le \frac{\tau}{\sqrt{N}} \Big\} \le C_{\gamma,K,T} \left(\frac{|Q_m|}{N^2} + \frac{1}{N} \right).$$

In particular, if $|Q_m| = o(N^2)$, then this probability tends to zero uniformly for $0 \le \tau \le T$.

Proof. Apply Proposition 3.2 with $t = \tau/\sqrt{N}$. The non-real points contribute at most

$$\frac{C_{\gamma,K}|Q_m|t^2}{N} = \frac{C_{\gamma,K}|Q_m|\tau^2}{N^2},$$

and the two endpoint evaluations contribute at most

$$\frac{2C_K t}{\sqrt{N}} = \frac{2C_K \tau}{N}.$$

Taking the supremum over $0 \le \tau \le T$ gives the stated bound. □

Corollary 3.4 (Independent randomized subquadratic grids also miss)**.** *Let the coefficients have densities bounded by K, and fix $\gamma > 0$. Let $N = m+1$, and let $\mathcal{Q}_m \subset \mathbb{T}$ be a nonempty finite random set, independent of the coefficients, such that every point of $\mathcal{Q}_m \setminus \{\pm 1\}$ has angular distance at least γ/m from $\{\pm 1\}$ almost surely. For every fixed $T < \infty$ there are constants $C_{\gamma,K,T} < \infty$ and $m_0(\gamma)$ such that, for $m \ge m_0(\gamma)$,*

$$\sup_{0 \le \tau \le T} \mathbb{P}\Big\{ \min_{z \in \mathcal{Q}_m} |G_m(z)| \le \frac{\tau}{\sqrt{N}} \Big\} \le C_{\gamma,K,T} \left(\frac{\mathbb{E}|\mathcal{Q}_m|}{N^2} + \frac{1}{N} \right).$$

In particular, if $\mathbb{E}|\mathcal{Q}_m| = o(N^2)$, then this probability tends to zero uniformly for $0 \le \tau \le T$.

Proof. Condition on the realized grid $\mathcal{Q}_m = Q$. The deterministic Corollary 3.3 gives

$$\sup_{0 \le \tau \le T} \mathbb{P}\left\{ \min_{z \in Q} |G_m(z)| \le \frac{\tau}{\sqrt{N}} \;\middle|\; \mathcal{Q}_m = Q \right\} \le C_{\gamma,K,T} \left(\frac{|Q|}{N^2} + \frac{1}{N} \right).$$

Integrating over Q gives the displayed estimate. □

Lemma 3.5 (Quadratic grids detect under a speed cutoff)**.** *Let $Q \subset \mathbb{T}$ be an h-net in arclength: for every $z \in \mathbb{T}$ there is $q \in Q$ with $\operatorname{dist}_{\mathbb{T}}(z,q) \le h$. For every coefficient vector, every $s > 0$, and every $R > 0$,*

$$\Big\{\min_{z\in\mathbb{T}} |G_m(z)| \le s\Big\} \cap \Big\{\sup_{z\in\mathbb{T}} |G_m'(z)| \le R\Big\} \subseteq \Big\{\min_{q\in Q} |G_m(q)| \le s + Rh\Big\}.$$

Consequently, if $N = m+1$, $L \ge 1$, $\tau > 0$, and

$$h \le \frac{\tau}{2Lm^{3/2}\sqrt{N}},$$

then

$$\Big\{\min_{z\in\mathbb{T}} |G_m(z)| \le \frac{\tau}{2\sqrt{N}}\Big\} \cap \Big\{\sup_{z\in\mathbb{T}} |G_m'(z)| \le Lm^{3/2}\Big\} \subseteq \Big\{\min_{q\in Q} |G_m(q)| \le \frac{\tau}{\sqrt{N}}\Big\}.$$

A uniform grid of $M \ge CLm^{3/2}\sqrt{N}/\tau$ points has this property. For fixed L and fixed positive τ, this is $O_L(N^2)$ points.

Proof. Choose $z_0 \in \mathbb{T}$ with $|G_m(z_0)| \le s$, and choose $q \in Q$ with $\operatorname{dist}_{\mathbb{T}}(z_0, q) \le h$. The fundamental theorem of calculus along the shorter arc from z_0 to q gives

$$|G_m(q)| \le |G_m(z_0)| + |G_m(q) - G_m(z_0)| \le s + h \sup_{\mathbb{T}} |G_m'|.$$

This proves the first inclusion, and the second follows by putting $s = \tau/(2\sqrt{N})$ and $R = Lm^{3/2}$. The assertion about uniform grids is the elementary mesh bound for M equally spaced points on a circle of arclength 2π, with the absolute constant C absorbing 2π. □

Lemma 3.6 (Uniform random points form a net)**.** *Let $U_1, \dots, U_M$ be independent Haar-uniform points of $\mathbb{T}$, and put $\mathcal{U}_M = \{U_1, \dots, U_M\}$. There are absolute constants $C, c > 0$ such that, for every $0 < h \le 1$,*

$$\mathbb{P}\{\mathcal{U}_M \text{ is not an } h\text{-net in arclength}\} \le \frac{C}{h}\exp(-cMh).$$

Proof. Choose a deterministic $h/2$-net $R_h \subset \mathbb{T}$ with $|R_h| \le C/h$. If every arclength ball $B(r, h/2)$, $r \in R_h$, contains at least one of the U_j, then $\mathcal{U}_M$ is an h-net. Each such ball has normalized arclength measure at least ch. Hence, by the union bound,

$$\mathbb{P}\{\exists r \in R_h : \mathcal{U}_M \cap B(r, h/2) = \emptyset\} \le \frac{C}{h}(1-ch)^M \le \frac{C}{h}\exp(-cMh).$$

□

Corollary 3.7 (Independent uniform random grids detect with one extra logarithm)**.** *Let $\mathcal{U}_m$ consist of M_m independent Haar-uniform points of $\mathbb{T}$, sampled independently of the coefficients of G_m, and put $N = m+1$. Fix $L \ge 1$ and $\tau > 0$. Then*

$$E_{m,L,\tau} := \Big\{\min_{z\in\mathbb{T}} |G_m(z)| \le \frac{\tau}{2\sqrt{N}}\Big\} \cap \Big\{\sup_{z\in\mathbb{T}} |G_m'(z)| \le Lm^{3/2}\Big\}.$$

With absolute constants $C, c > 0$,

$$\mathbb{P}\left(E_{m,L,\tau} \setminus \left\{\min_{q\in\mathcal{U}_m} |G_m(q)| \le \frac{\tau}{\sqrt{N}}\right\}\right) \le C\,\frac{Lm^{3/2}\sqrt{N}}{\tau}\exp\left\{-c\,\frac{\tau M_m}{Lm^{3/2}\sqrt{N}}\right\}.$$

Consequently, for fixed L and fixed $\tau > 0$, independent uniform random grids of size $M_m \ge C_{A,L,\tau}N^2\log N$ detect continuum $N^{-1/2}$-scale excursions under the speed cutoff with miss probability $O_A(N^{-A})$.

More generally, assume the polynomial speed-tail hypothesis in Corollary 3.68 for a given $A > 0$. Then independent uniform random grids with

$$M_m \ge C_{A,\tau,L_A}N^2(\log N)^{3/2}$$

satisfy

$$\mathbb{P}\left(\left\{\min_{z\in\mathbb{T}} |G_m(z)| \le \frac{\tau}{2\sqrt{N}}\right\} \setminus \left\{\min_{q\in\mathcal{U}_m} |G_m(q)| \le \frac{\tau}{\sqrt{N}}\right\}\right) \le C_{A,\tau,L_A}m^{-A}.$$

Proof. Set

$$h = \frac{\tau}{2Lm^{3/2}\sqrt{N}}.$$

On the event that $\mathcal{U}_m$ is an h-net, Lemma 3.5 shows that the displayed miss event under the speed cutoff is empty. The first bound is therefore Lemma 3.6. If L and τ are fixed and $M_m \ge C_{A,L,\tau}N^2\log N$, then $M_m h \ge C_A \log N$ while $h^{-1} = O_{L,\tau}(N^2)$, so the net-failure probability is $O_A(N^{-A})$.

For the speed-tail version, put $L = L_A\sqrt{\log m}$. The same argument gives an $O_A(m^{-A})$ net-failure probability once $M_m \ge C_{A,\tau,L_A}m^{3/2}\sqrt{N}(\log m)^{3/2}$, which is the displayed condition after changing constants. Add the assumed $O_A(m^{-A})$ probability of the complementary speed event. □

Corollary 3.8 (Stratified random grids keep the deterministic scale)**.** *Let $I_{1,m}, \dots, I_{M_m,m}$ be the equal-arclength partition of $\mathbb{T}$. Choose one point $q_{j,m} \in I_{j,m}$ for each j, deterministically or randomly, and put*

$$\mathcal{S}_m := \{q_{1,m}, \dots, q_{M_m,m}\}.$$

Let $N = m + 1$, fix $L \ge 1$ and $\tau > 0$, and let $E_{m,L,\tau}$ be the event from Corollary 3.7. If

$$M_m \ge C\,\frac{Lm^{3/2}\sqrt{N}}{\tau},$$

then, for every realization of the stratified grid,

$$E_{m,L,\tau} \subseteq \left\{\min_{q\in\mathcal{S}_m} |G_m(q)| \le \frac{\tau}{\sqrt{N}}\right\}.$$

Consequently, under the polynomial speed-tail hypothesis of Corollary 3.68, if

$$M_m \ge C_\tau L_A m^{3/2}\sqrt{N}\,\sqrt{\log m},$$

then

$$\mathbb{P}\left(\left\{\min_{z\in\mathbb{T}} |G_m(z)| \le \frac{\tau}{2\sqrt{N}}\right\} \setminus \left\{\min_{q\in\mathcal{S}_m} |G_m(q)| \le \frac{\tau}{\sqrt{N}}\right\}\right) \le C_A m^{-A}.$$

Thus stratified one-point-per-cell sampling has the same $O_{A,\tau}(L_A N^2\sqrt{\log N})$ detection complexity as a deterministic net and does not pay the coupon-collector logarithm of independent uniform sampling. The same conclusion holds for random shifts of a uniform grid.

Proof. The set $\mathcal{S}_m$ is a $2\pi/M_m$-net in arclength: every point of $\mathbb{T}$ lies in one of the intervals $I_{j,m}$, and that whole interval has arclength $2\pi/M_m$. If M_m is at least a sufficiently large absolute constant times $Lm^{3/2}\sqrt{N}/\tau$, then

$$\frac{2\pi}{M_m} \le \frac{\tau}{2Lm^{3/2}\sqrt{N}}.$$

Lemma 3.5 gives the first inclusion. For the speed-tail conclusion, put $L = L_A\sqrt{\log m}$; outside the displayed speed event the miss probability is at most $C_A m^{-A}$, and on the event the first inclusion applies. A random shift of a uniform grid satisfies the same mesh bound, so the same proof applies to it directly. □

Remark 3.9 (What the grid estimate does not prove). Proposition 3.2 proves the density-level delocalization at every fixed boundary evaluation, including deterministic grids. It does not prove Hypothesis 1.4: the latter is a continuum statement, and an excursion of $|G_m|$ below t can have angular width only $\asymp t\, m^{-3/2}$. Corollary 3.3 makes the sampling gap quantitative at the continuum scale $t \asymp m^{-1/2}$: away from the two parity endpoints, any deterministic grid with $o(m^2)$ points has vanishing detection probability, and Corollary 3.4 gives the same conclusion for independent randomized grids with $o(m^2)$ expected size. Lemma 3.5 gives the matching deterministic sufficiency statement: once the curve speed is bounded by $Lm^{3/2}$, a grid with spacing $O(t/(Lm^{3/2}))$, hence $O_L(m^2)$ points at $t \asymp m^{-1/2}$, detects every continuum excursion up to a harmless factor of two in the threshold. For Gaussian coefficients, Corollary 3.11 combines this deterministic sufficiency with the Salem–Zygmund speed tail to give detection by $O(N^2\sqrt{\log N})$ grid points outside a polynomially small exceptional event. Corollary 3.7 gives the independent uniform random-grid analogue, with the expected extra logarithmic covering overhead: $O(N^2 \log N)$ points under a fixed speed cutoff and $O(N^2(\log N)^{3/2})$ points under the natural $\sqrt{\log N}$ speed tail. Corollary 3.8 shows that stratified random grids and random shifts of a uniform lattice avoid this coupon-collector overhead and keep the deterministic $O(N^2\sqrt{\log N})$ natural-speed scale. Corollary 3.69 packages this as the natural-speed sampling bracket. Without such a natural speed tail, the variance-only fallback in Corollary 3.63 gives fixed powers N^β, $\beta > 5/2$, with miss rate $O_\beta(N^{5-2\beta})$. Passing from fixed points to arcs therefore still requires a curve-crossing, occupation-measure, or derivative-control argument. For Gaussian coefficients away from $z = \pm 1$ this is supplied sharply by Theorem 1.1; the endpoint arcs are controlled at polynomial scales by Corollary 1.3. For general bounded-density laws the corresponding continuum step remains the universality gap.

Lemma 3.10 (Salem–Zygmund derivative tail)**.** *Let ξ_j be independent standard real Gaussians. There are absolute constants $C_1, c_1 > 0$ such that*

$$\mathbb{P}\Big\{\sup_{z\in\mathbb{T}} |G_m'(z)| \ge \lambda\, m^{3/2}\sqrt{\log m}\Big\} \le C_1\, m^{-c_1\lambda^2} \qquad (\lambda \ge 2).$$

In particular, $\sup_{\mathbb{T}} |G_m'| \le m^{3/2}\log m$ off an event of probability $O(m^{-10})$.

Proof. $G_m'(z) = \sum_{j=1}^m j\xi_j z^{j-1}$ is a trigonometric polynomial of degree m with $\mathbb{E}\,|G_m'(z)|^2 = \sum_{j=1}^m j^2 \asymp m^3$ at each fixed $z \in \mathbb{T}$, so the standard deviation at a point is $\asymp m^{3/2}$. By Gaussian concentration $|G_m'(z)|$ has Gaussian tails on that scale at each point; the Salem–Zygmund maximal inequality for random trigonometric polynomials (as used in [4]) upgrades the pointwise bound to the supremum at the cost of the $\sqrt{\log m}$ factor, giving the stated tail. Taking λ a large constant yields the high-probability bound. □

Corollary 3.11 (Gaussian quadratic-log grids detect continuum excursions)**.** *Let $G_m(z) = \sum_{j=0}^m \xi_j z^j$ with independent standard real Gaussian coefficients, and put $N = m+1$. Fix $A > 0$ and $\tau > 0$. There are constants $C_{A,\tau}, C_A < \infty$ such that, if $Q_m \subset \mathbb{T}$ is an h_m-net with*

$$h_m \le \frac{1}{C_{A,\tau} m^{3/2}\sqrt{N}\,\sqrt{\log m}},$$

then

$$\mathbb{P}\left(\Big\{\min_{z\in\mathbb{T}} |G_m(z)| \le \frac{\tau}{2\sqrt{N}}\Big\} \setminus \Big\{\min_{q\in Q_m} |G_m(q)| \le \frac{\tau}{\sqrt{N}}\Big\}\right) \le C_A m^{-A}.$$

Equivalently, a uniform grid with

$$M \ge C_{A,\tau} m^{3/2}\sqrt{N}\,\sqrt{\log m}$$

points detects every continuum excursion at this scale except on an $O_A(m^{-A})$ exceptional event. For fixed τ, this is $O_A(N^2\sqrt{\log N})$ points.

Proof. Choose $\lambda = \lambda_A \ge 2$ so large that the derivative-tail bound in Lemma 3.10 is at most $C_A m^{-A}$. Let

$$E_\lambda = \Big\{\sup_{\mathbb{T}} |G_m'| \le \lambda m^{3/2}\sqrt{\log m}\Big\}.$$

If $h_m \le \tau/(2\lambda m^{3/2}\sqrt{N}\sqrt{\log m})$, then Lemma 3.5, with $L = \lambda\sqrt{\log m}$, gives

$$\Big\{\min_{\mathbb{T}} |G_m| \le \frac{\tau}{2\sqrt{N}}\Big\} \cap E_\lambda \subseteq \Big\{\min_{q\in Q_m} |G_m(q)| \le \frac{\tau}{\sqrt{N}}\Big\}.$$

Thus the miss event in the statement is contained in E_λ^c, whose probability is $O_A(m^{-A})$. The uniform-grid formulation is again the elementary mesh estimate on a circle of arclength 2π, with constants absorbing τ^{-1} and λ_A. □

Lemma 3.12 (Kac–Rice crossing count away from the real points)**.** *Let $G_m(z) = \sum_{j=0}^m \xi_j z^j$ with ξ_j independent standard real Gaussians, and put*

$$X_\theta = (\Re G_m(e^{i\theta}), \Im G_m(e^{i\theta})), \qquad Q(\theta) = |X_\theta|^2.$$

Fix $\gamma > 0$. There are constants $C_\gamma < \infty$ and $m_0(\gamma) < \infty$ such that, whenever $m \geq m_0(\gamma)$, J is an interval whose points have angular distance at least γ/m from 0 and π, and $0 < t \leq 1$,

$$\mathbb{E}\,\#\{\theta \in J : Q(\theta) = t^2\} \leq C_\gamma |J|\, t\sqrt{m}.$$

Proof. Write $\dot{X}_\theta = dX_\theta/d\theta$, and let $\Sigma_\theta = \operatorname{Cov}(X_\theta)$. By the covariance computation in Lemma 3.1,

$$\lambda_{\min}(\Sigma_\theta) \geq c_\gamma m, \qquad \|\Sigma_\theta\| \leq Cm$$

uniformly on J. Also

$$\|\operatorname{Cov}(\dot{X}_\theta)\| \leq \sum_{j=1}^{m} j^2 \leq Cm^3, \qquad \|\operatorname{Cov}(\dot{X}_\theta, X_\theta)\| \leq \|\operatorname{Cov}(\dot{X}_\theta)\|^{1/2} \|\Sigma_\theta\|^{1/2} \leq Cm^2.$$

Thus, conditionally on $X_\theta = x$,

$$\dot{X}_\theta = B_\theta x + Z_\theta, \qquad \|B_\theta\| \leq C_\gamma m, \qquad \|\operatorname{Cov}(Z_\theta)\| \leq Cm^3,$$

where Z_θ is a centered Gaussian vector. For $|x| = t \leq 1$,

$$\mathbb{E}[|2x \cdot \dot{X}_\theta| \mid X_\theta = x] \leq C_\gamma(mt^2 + m^{3/2}t) \leq C_\gamma m^{3/2} t.$$

The density p_θ of X_θ satisfies $p_\theta(x) \leq C_\gamma/m$. The first-moment Kac–Rice formula applied to the smooth scalar process $Q(\theta) - t^2$, followed by the coarea formula for $x \mapsto |x|^2$, gives

$$\begin{aligned}\mathbb{E}\,\#\{\theta \in J : Q(\theta) = t^2\} &= \int_J \int_{|x|=t} \frac{\mathbb{E}[|2x \cdot \dot{X}_\theta| \,|\, X_\theta = x]\, p_\theta(x)}{2|x|}\, dH(x)\, d\theta \\ &\leq C_\gamma |J|\, t\sqrt{m}.\end{aligned}$$

□

Lemma 3.13 (Regression variance away from the parity points)**.** *Fix $\gamma > 0$. There are constants $\kappa_\gamma > 0$ and $m_0(\gamma) < \infty$ such that, for $m \geq m_0(\gamma)$, $\operatorname{dist}(\theta, \pi\mathbb{Z}) \geq \gamma/m$, and $u \in S^1$,*

$$\operatorname{dist}\Big((j(-u_1 \sin j\theta + u_2 \cos j\theta))_{j=0}^m,\ E_\theta\Big)^2 \geq \kappa_\gamma m^3,$$

where

$$a_\theta = (\cos j\theta)_{j=0}^m, \qquad b_\theta = (\sin j\theta)_{j=0}^m, \qquad E_\theta = \operatorname{span}\{a_\theta, b_\theta\}.$$

Proof. Put $v = (j(-u_1 \sin j\theta + u_2 \cos j\theta))_{j=0}^m$ and let G_θ be the 2×2 Gram matrix of (a_θ, b_θ). The squared distance is the Schur complement

$$\|v\|_2^2 - \begin{pmatrix}\langle v, a_\theta\rangle & \langle v, b_\theta\rangle\end{pmatrix} G_\theta^{-1} \begin{pmatrix}\langle v, a_\theta\rangle \\ \langle v, b_\theta\rangle\end{pmatrix}.$$

First consider the endpoint scale $\theta = \alpha/m$, $\alpha \in [\gamma, C]$. After division by m^3, this Schur complement converges uniformly to the $L^2[0,1]$-distance squared of

$$x(-u_1 \sin \alpha x + u_2 \cos \alpha x)$$

from $\operatorname{span}\{\cos\alpha x, \sin\alpha x\}$. The limiting distance is positive: otherwise analytic linear independence would force $x(-u_1\sin\alpha x + u_2\cos\alpha x)$ to be a linear combination of $\cos\alpha x$ and $\sin\alpha x$ on an interval. Compactness in $(\alpha, u) \in [\gamma, C]\times S^1$ gives a uniform positive lower bound. The endpoint scale near π, namely $\theta = \pi \pm \alpha/m$ modulo 2π, reduces to the same case after multiplying the j-th coordinate by $(-1)^j$, an orthogonal change of coordinates that sends E_θ to the corresponding endpoint space and replaces u by another unit vector.

It remains to treat the complementary region $\operatorname{dist}(\theta, \pi\mathbb{Z}) \geq C/m$. Write $\beta = u_2 + iu_1$, so $v_j = j\Re(\beta e^{ij\theta})$. With $S_k = \sum_{j=0}^m j^k$, summation by parts gives, uniformly in this region,

$$\sum_{j=0}^{m} j^k e^{2ij\theta} = O_k(m^k/\operatorname{dist}(\theta,\pi\mathbb{Z})) = O_k(m^{k+1}/C) \qquad (k = 0, 1, 2).$$

Consequently

$$\|v\|_2^2 = \frac{1}{2}S_2 + O(m^3/C), \qquad G_\theta = \frac{1}{2}S_0 I_2 + O(m/C),$$

and

$$\begin{pmatrix}\langle v, a_\theta\rangle \\ \langle v, b_\theta\rangle\end{pmatrix} = \frac{1}{2}S_1\begin{pmatrix}u_2 \\ -u_1\end{pmatrix} + O(m^2/C).$$

Since $G_\theta^{-1} = 2S_0^{-1}I_2 + O(1/(Cm))$, the Schur complement is

$$\frac{1}{2}S_2 - \frac{S_1^2}{2S_0} + O(m^3/C) = \frac{m(m+1)(m+2)}{24} + O(m^3/C).$$

Choosing $C = C(\gamma)$ large and then $m_0(\gamma)$ large gives the claimed lower bound. □

Proposition 3.14 (Away-from-endpoint first moment)**.** *Let $G_m(z) = \sum_{j=0}^m \xi_j z^j$ with ξ_j independent standard real Gaussians, and put $Q(\theta) = |G_m(e^{i\theta})|^2$. Fix $\gamma > 0$. There are constants $0 < c_\gamma < C_\gamma < \infty$ and $m_0(\gamma) < \infty$ such that, whenever $m \geq m_0(\gamma)$, J is an interval whose points satisfy $\operatorname{dist}(\theta, \pi\mathbb{Z}) \geq \gamma/m$, and $0 < t \leq 1$,*

$$c_\gamma |J|\, t\sqrt{m} \leq \mathbb{E}\,\#\{\theta \in J : Q(\theta) = t^2,\ Q'(\theta) < 0\} \leq C_\gamma |J|\, t\sqrt{m}.$$

The same two-sided bound holds for up-crossings and hence for all level crossings. In particular, on an arc of length c/m away from $\{\pm 1\}$, the expected number of level-t down-crossings is $\asymp_{\gamma,c} t/\sqrt{m}$.

Proof. The upper bound follows from Lemma 3.12. For the lower bound, write

$$a_\theta = (\cos j\theta)_{j=0}^m, \qquad b_\theta = (\sin j\theta)_{j=0}^m, \qquad E_\theta = \operatorname{span}\{a_\theta, b_\theta\}.$$

If $u = (u_1, u_2) \in S^1$, then

$$u \cdot \dot{X}_\theta = \sum_{j=0}^{m} \xi_j\, j(-u_1 \sin j\theta + u_2 \cos j\theta).$$

By Lemma 3.13, taking $u = x/t$, conditionally on $X_\theta = x$, with $|x| = t$, the Gaussian regression of the normal velocity satisfies

$$\operatorname{Var}(Q'(\theta) \mid X_\theta = x) = 4 \operatorname{Var}(x \cdot \dot{X}_\theta \mid X_\theta = x) \geq c_\gamma m^3 t^2.$$

The regression mean is $O_\gamma(mt^2)$, hence is $o(m^{3/2}t)$ for $0 < t \leq 1$ after increasing $m_0(\gamma)$. Therefore

$$\mathbb{E}[(-Q'(\theta))_+ \mid X_\theta = x] \geq c_\gamma m^{3/2} t.$$

The covariance bounds in Lemma 3.1 imply that the density p_θ of X_θ satisfies $p_\theta(x) \geq c_\gamma/m$ for $|x| \leq 1$. The Kac–Rice formula for down-crossings and coarea on $|x| = t$ now give

$$\begin{aligned}\mathbb{E}N_t^{\downarrow}(J) &= \int_J \int_{|x|=t} \frac{\mathbb{E}[(-Q'(\theta))_+ \,|\, X_\theta = x] p_\theta(x)}{2t}\, dH(x)\, d\theta \\ &\geq c_\gamma \int_J t\sqrt{m}\, d\theta = c_\gamma |J| t\sqrt{m}.\end{aligned}$$

The up-crossing lower bound is identical, and the proposition follows. □

Proposition 3.15 (Bulk Kac–Rice crossing constant)**.** *Let $G_m(z) = \sum_{j=0}^m \xi_j z^j$ with ξ_j independent standard real Gaussians, put $N = m + 1$, and set $Q(\theta) = |G_m(e^{i\theta})|^2$. Fix $\delta > 0$, and let*

$$B_\delta := \{\theta \in [-\pi, \pi] : \operatorname{dist}(\theta, \pi\mathbb{Z}) \geq \delta\}.$$

If $J \subset B_\delta$ is an interval and

$$N_t^{\downarrow}(J) := \#\{\theta \in J : Q(\theta) = t^2,\ Q'(\theta) < 0\},$$

then, uniformly for $0 < t \leq 1$ and for all such intervals J,

$$\mathbb{E}N_t^{\downarrow}(J) = (1 + o_\delta(1)) \frac{|J|\, t\sqrt{N}}{\sqrt{12\pi}} \qquad (m \to \infty). \tag{3.1}$$

The same asymptotic holds for up-crossings, so the leading bulk all-crossing intensity is $2t\sqrt{N}/\sqrt{12\pi}$ per unit arclength.

Proof. Write

$$X_\theta = (\Re G_m(e^{i\theta}), \Im G_m(e^{i\theta})), \qquad \dot{X}_\theta = \frac{d}{d\theta} X_\theta.$$

For $k = 0, 1, 2$, summation by parts gives, uniformly for $\theta \in B_\delta$,

$$\sum_{j=0}^m j^k e^{2ij\theta} = O_{\delta,k}(m^k).$$

With $S_1 = \sum_{j=0}^m j = N(N-1)/2$ and $S_2 = \sum_{j=0}^m j^2 = N(N-1)(2N-1)/6$, the covariance matrices therefore satisfy

$$\Sigma_\theta := \operatorname{Cov}(X_\theta) = \frac{N}{2} I_2 + O_\delta(1),$$

$$\Gamma_\theta := \operatorname{Cov}(\dot{X}_\theta) = \frac{S_2}{2} I_2 + O_\delta(N^2),$$

and

$$C_\theta := \mathrm{Cov}(\dot X_\theta, X_\theta) = \frac{S_1}{2}\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} + O_\delta(N).$$

Consequently, the conditional covariance of $\dot X_\theta$ given X_θ is

$$V_\theta := \Gamma_\theta - C_\theta \Sigma_\theta^{-1} C_\theta^T = \alpha_N I_2 + O_\delta(N^2), \qquad \alpha_N := \frac{N(N^2-1)}{24}.$$

Also

$$C_\theta \Sigma_\theta^{-1} = \frac{S_1}{N}\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} + O_\delta(1).$$

Thus, for $x = tu$, $u \in S^1$, the conditional law of $u \cdot \dot X_\theta$ given $X_\theta = x$ is Gaussian with variance $\alpha_N(1 + O_\delta(N^{-1}))$ and mean $O_\delta(t)$, uniformly in θ, u, and $0 < t \le 1$. Hence

$$\mathbb{E}[(-u \cdot \dot X_\theta)_+ \mid X_\theta = tu] = (1 + o_\delta(1))\sqrt{\frac{\alpha_N}{2\pi}}.$$

The density of X_θ satisfies, uniformly for $|x| \le 1$,

$$p_\theta(x) = \frac{1 + o_\delta(1)}{\pi N}.$$

Kac–Rice for down-crossings of $Q(\theta) - t^2$, followed by coarea on the circle $|x| = t$, gives the pointwise down-crossing intensity

$$\begin{aligned} \mathcal{I}_m(\theta, t) &= \int_{|x|=t} \frac{\mathbb{E}[(-Q'(\theta))_+ \mid X_\theta = x] p_\theta(x)}{2t}\, dH(x) \\ &= \int_{|x|=t} \mathbb{E}[(-u \cdot \dot X_\theta)_+ \mid X_\theta = tu] p_\theta(tu)\, dH(x) \\ &= (1 + o_\delta(1))\, 2\pi t \sqrt{\frac{\alpha_N}{2\pi}}\, \frac{1}{\pi N} = (1 + o_\delta(1))\frac{t\sqrt{N}}{\sqrt{12\pi}}. \end{aligned}$$

The error is uniform on B_δ, so integration over J proves (3.1). Replacing Q' by $-Q'$ gives the identical up-crossing asymptotic. □

Corollary 3.16 (Fixed-bulk continuum-scale crossing limit)**.** *With the notation of Proposition 3.15, fix $0 < \delta < \pi/2$ and write*

$$B_\delta = \{\theta \in [-\pi, \pi] : \mathrm{dist}(\theta, \pi\mathbb{Z}) \ge \delta\}.$$

For $0 \le \tau < \infty$, set $t_N = \tau/\sqrt{N}$. Then, uniformly for τ in compact subsets of $[0, \infty)$,

$$\mathbb{E}N^{\downarrow}_{t_N}(B_\delta) \longrightarrow \frac{|B_\delta|\,\tau}{\sqrt{12\pi}} = \frac{(2\pi - 4\delta)\tau}{\sqrt{12\pi}},$$

and the same limit holds for up-crossings. Hence the fixed-bulk all-crossing first moment tends to

$$\frac{2(2\pi - 4\delta)\tau}{\sqrt{12\pi}}.$$

Letting $\delta \downarrow 0$ after this fixed-bulk limit gives the non-endpoint candidate constants $\sqrt{\pi/3}\,\tau$ for down-crossings and $2\sqrt{\pi/3}\,\tau$ for all crossings. This

last passage is only a first-moment benchmark; Corollary 3.29 below shows that the endpoint neighborhoods do not change the corresponding probability upper constant, but it does not prove the lower-tail or exponential minimum-modulus law.

Proof. The set B_δ is the disjoint union of the two intervals $[-\pi+\delta,-\delta]$ and $[\delta,\pi-\delta]$, whose total length is $2\pi-4\delta$. Apply Proposition 3.15 to each component and put $t=t_N=\tau/\sqrt{N}$. The multiplicative $o_\delta(1)$ error in (3.1) is uniform for $0<t\le 1$, so for τ in a fixed compact interval the absolute error is $o_\delta(1)\tau$. This proves the down-crossing limit, and the up-crossing limit follows in the same way. The displayed all-crossing limit is their sum. □

Corollary 3.17 (Fixed-bulk continuum-scale probability upper bound)**.** *With the notation of Corollary 3.16, put*

$$P_{N,\delta}(\tau) := \mathbb{P}\Big\{\min_{\theta\in B_\delta}|G_m(e^{i\theta})| \le \frac{\tau}{\sqrt{N}}\Big\}.$$

For every $K<\infty$,

$$\limsup_{N\to\infty}\sup_{0\le\tau\le K}\left(P_{N,\delta}(\tau)-\frac{|B_\delta|\,\tau}{\sqrt{12\pi}}\right)_+ = 0.$$

Equivalently, for each fixed τ,

$$\limsup_{N\to\infty}\mathbb{P}\Big\{\min_{\theta\in B_\delta}|G_m(e^{i\theta})| \le \frac{\tau}{\sqrt{N}}\Big\} \le \frac{(2\pi-4\delta)\tau}{\sqrt{12\pi}}.$$

This is only a first-moment upper bound on a fixed bulk. It does not supply the matching lower bound or the exponential minimum-modulus law, which require anti-clustering and decorrelation of crossings.

Proof. Write $B_\delta = I_1\cup I_2$, where $I_1=[-\pi+\delta,-\delta]$ and $I_2=[\delta,\pi-\delta]$. Let $t_N=\tau/\sqrt{N}$. If $\min_{\theta\in I_j}|G_m(e^{i\theta})|\le t_N$, then either one endpoint of I_j has modulus at most t_N, or the level $|G_m(e^{i\theta})|=t_N$ is crossed inside I_j. If both endpoints are above level, the first entry into the sublevel set is a down-crossing of $Q(\theta)-t_N^2$. The standard nondegeneracy used in the Kac–Rice formula makes tangential level contacts a null event, so the latter contribution is bounded by the down-crossing count.

The four interval endpoints are a fixed positive distance from $\{\pm1\}$, so Lemma 3.1 gives a total endpoint contribution

$$O_\delta(t_N^2/N) = O_{\delta,K}(N^{-2}) \qquad (0\le\tau\le K).$$

By Corollary 3.16, the expected down-crossing count in B_δ is

$$(1+o_\delta(1))\,\frac{|B_\delta|\,\tau}{\sqrt{12\pi}},$$

uniformly for $0\le\tau\le K$. Markov's inequality gives the displayed uniform limsup, and the pointwise bound follows by fixing τ. □

Proposition 3.18 (Second-moment transfer to a fixed-bulk lower tail)**.** *With the notation of Corollary 3.16, fix $0 < \delta < \pi/2$, $T < \infty$, and put*

$$\mu_\delta(\tau) := \frac{|B_\delta|\,\tau}{\sqrt{12\pi}} = \frac{(2\pi - 4\delta)\tau}{\sqrt{12\pi}}, \qquad 0 \le \tau \le T.$$

For $N = m + 1$ and $t_N = \tau/\sqrt{N}$, set

$$X_{N,\delta}(\tau) := N^{\downarrow}_{t_N}(B_\delta).$$

Assume that for some constant $L_{\delta,T} < \infty$,

$$\limsup_{N\to\infty} \mathbb{E}\big[X_{N,\delta}(\tau)(X_{N,\delta}(\tau) - 1)\big] \le L_{\delta,T}\,\mu_\delta(\tau)^2 \qquad (0 \le \tau \le T). \tag{3.2}$$

Then, for every $0 \le \tau \le T$,

$$\liminf_{N\to\infty} \mathbb{P}\Big\{\min_{\theta\in B_\delta} |G_m(e^{i\theta})| \le \frac{\tau}{\sqrt{N}}\Big\} \ge \frac{\mu_\delta(\tau)}{1 + L_{\delta,T}\mu_\delta(\tau)}. \tag{3.3}$$

In particular, if the Poisson-scale decorrelation estimate

$$\mathbb{E}\big[X_{N,\delta}(\tau)(X_{N,\delta}(\tau) - 1)\big] \le (1 + o(1))\,\mu_\delta(\tau)^2$$

holds for fixed δ and τ, then the fixed-bulk lower tail has the small-τ expansion

$$\liminf_{N\to\infty} \mathbb{P}\Big\{\min_{\theta\in B_\delta} |G_m(e^{i\theta})| \le \frac{\tau}{\sqrt{N}}\Big\} \ge \mu_\delta(\tau) - O_\delta(\tau^2) \qquad (\tau \downarrow 0).$$

Proof. The case $\tau = 0$ is trivial, so assume $\tau > 0$. By Corollary 3.16,

$$\mathbb{E}X_{N,\delta}(\tau) \longrightarrow \mu_\delta(\tau).$$

The event $\{X_{N,\delta}(\tau) > 0\}$ is contained in

$$\left\{\min_{\theta\in B_\delta} |G_m(e^{i\theta})| \le \frac{\tau}{\sqrt{N}}\right\}.$$

Paley–Zygmund for the nonnegative integer-valued variable $X_{N,\delta}(\tau)$ gives

$$\mathbb{P}\{X_{N,\delta}(\tau) > 0\} \ge \frac{(\mathbb{E}X_{N,\delta}(\tau))^2}{\mathbb{E}X_{N,\delta}(\tau)^2}.$$

Since

$$\mathbb{E}X_{N,\delta}(\tau)^2 = \mathbb{E}X_{N,\delta}(\tau) + \mathbb{E}[X_{N,\delta}(\tau)(X_{N,\delta}(\tau) - 1)]\,,$$

the convergence of the first moment and the assumed factorial second-moment bound imply

$$\liminf_{N\to\infty} \mathbb{P}\{X_{N,\delta}(\tau) > 0\} \ge \frac{\mu_\delta(\tau)^2}{\mu_\delta(\tau) + L_{\delta,T}\mu_\delta(\tau)^2},$$

which is (3.3). Under the Poisson-scale bound, put $L_{\delta,T} = 1 + o(1)$ at the fixed value of τ to get $\mu_\delta(\tau)/(1+\mu_\delta(\tau))$, which is $\mu_\delta(\tau) - O_\delta(\tau^2)$ as $\tau \downarrow 0$. □

Corollary 3.19 (Fixed-bulk first-order law from a factorial moment bound)**.** *With the notation and hypotheses of Proposition 3.18, and with $P_{N,\delta}(\tau)$ as in Corollary 3.17, one has, for every $0 < \tau \le T$,*

$$\frac{\mu_\delta(\tau)}{1 + L_{\delta,T}\mu_\delta(\tau)} \le \liminf_{N\to\infty} P_{N,\delta}(\tau) \le \limsup_{N\to\infty} P_{N,\delta}(\tau) \le \mu_\delta(\tau).$$

Consequently,

$$\lim_{\tau\downarrow 0} \frac{1}{\tau} \liminf_{N\to\infty} P_{N,\delta}(\tau) = \lim_{\tau\downarrow 0} \frac{1}{\tau} \limsup_{N\to\infty} P_{N,\delta}(\tau) = \frac{|B_\delta|}{\sqrt{12\pi}}.$$

Proof. The lower bound is (3.3), and the upper bound is Corollary 3.17. Since $\mu_\delta(\tau) = |B_\delta|\tau/\sqrt{12\pi}$, dividing the sandwich by τ and letting $\tau \downarrow 0$ gives the displayed first-order coefficient. □

Corollary 3.20 (Uniform bulk second moment implies a global lower tail)**.** *Fix $T < \infty$. Suppose there is $L_T < \infty$ such that, for every $0 < \delta < \pi/2$ and $0 \le \tau \le T$, the variables*

$$X_{N,\delta}(\tau) = N^{\downarrow}_{\tau/\sqrt{N}}(B_\delta)$$

satisfy

$$\limsup_{N\to\infty} \mathbb{E}\big[X_{N,\delta}(\tau)(X_{N,\delta}(\tau) - 1)\big] \le L_T\, \mu_\delta(\tau)^2, \qquad \mu_\delta(\tau) = \frac{(2\pi - 4\delta)\tau}{\sqrt{12\pi}}.$$

Then, for every $0 \le \tau \le T$,

$$\liminf_{N\to\infty} \mathbb{P}\Big\{ \min_{\theta\in[-\pi,\pi]} |G_m(e^{i\theta})| \le \frac{\tau}{\sqrt{N}} \Big\} \ge \frac{\sqrt{\pi/3}\,\tau}{1 + L_T\sqrt{\pi/3}\,\tau}. \tag{3.4}$$

In particular, a uniform Poisson-scale factorial second-moment estimate $L_T = 1$ would give the small-τ global lower-tail expansion

$$\liminf_{N\to\infty} \mathbb{P}\Big\{ \min_{\mathbb{T}} |G_m| \le \frac{\tau}{\sqrt{N}} \Big\} \ge \sqrt{\frac{\pi}{3}}\,\tau - \frac{\pi}{3}\tau^2 + O(\tau^3) \qquad (\tau \downarrow 0).$$

Proof. For fixed δ, Proposition 3.18 gives

$$\liminf_{N\to\infty} \mathbb{P}\Big\{ \min_{\theta\in B_\delta} |G_m(e^{i\theta})| \le \frac{\tau}{\sqrt{N}} \Big\} \ge \frac{\mu_\delta(\tau)}{1 + L_T\mu_\delta(\tau)}.$$

The event on B_δ is contained in the corresponding whole-circle event. Letting $\delta \downarrow 0$ and using

$$\lim_{\delta\downarrow 0} \mu_\delta(\tau) = \frac{2\pi\tau}{\sqrt{12\pi}} = \sqrt{\frac{\pi}{3}}\,\tau$$

proves (3.4). The final expansion is the Taylor series of $x/(1+x)$ at $x = \sqrt{\pi/3}\,\tau$. □

Proof of Theorem 1.1. Parametrize I by an interval $J = [a, b]$, and set $Q(\theta) = |G_m(e^{i\theta})|^2$. If $\min_J Q \le t^2$, then either $Q(a) \le t^2$, or $Q(b) \le t^2$, or $Q(\theta) = t^2$ for some $\theta \in (a, b)$. Therefore Lemmas 3.1 and 3.12 give

$$\mathbb{P}\{\min_J Q \le t^2\} \le \mathbb{P}\{Q(a) \le t^2\} + \mathbb{P}\{Q(b) \le t^2\} + \mathbb{E}\,\#\{\theta \in J : Q(\theta) = t^2\}$$
$$\le C_\gamma \frac{t^2}{m} + C_\gamma \frac{c}{m} t\sqrt{m} \le C_\gamma (1+c) \frac{t}{\sqrt{m}},$$

since $0 < t \le 1$. This is (1.7). □

Proposition 3.21 (Log-loss single-arc boundary small ball)**.** *Let $G_m(z) = \sum_{j=0}^m \xi_j z^j$ with ξ_j independent standard real Gaussians. For every fixed $\gamma > 0$ and $A > 0$ there are constants $C_{\gamma,A} < \infty$ and $m_0(\gamma, A)$ such that, for every $m \ge m_0(\gamma, A)$, every $c \ge 1$, every arc $I \subset \mathbb{T}$ of arclength $|I| = c/m$ whose angular distance from $z = \pm 1$ is at least γ/m, and every $t \in (0, 1]$,*

$$\mathbb{P}\Big\{\min_{z\in I} |G_m(z)| \le t\Big\} \le C_{\gamma,A} \left((1+c) \frac{\sqrt{\log m}\, t}{\sqrt{m}} + m^{-A} \right).$$

Proof. Parametrize I by an interval J of length c/m, and set

$$g(\theta) := \left|G_m(e^{i\theta})\right|, \qquad \Lambda(s) := \int_J \mathbf{1}\{g(\theta) \le s\}\, d\theta.$$

By Fubini and Lemma 3.1, since every point of J is γ/m away from 0 and π,

$$\mathbb{E}\Lambda(s) \le |J|\, \frac{C_\gamma s^2}{m+1} \le \frac{C_\gamma c s^2}{m^2}.$$

Fix $\lambda = \lambda(A) \ge 2$ large enough that the event

$$E_\lambda := \Big\{\sup_{z\in\mathbb{T}} |G_m'(z)| \le \lambda m^{3/2} \sqrt{\log m}\Big\}$$

has complement of probability at most $C_A m^{-A}$, by Lemma 3.10. On E_λ, if $g(\theta_0) \le t$ for some $\theta_0 \in J$, then

$$g(\theta) \le 2t \qquad \text{whenever } \theta \in J, \quad |\theta - \theta_0| \le \delta := \frac{t}{\lambda m^{3/2}\sqrt{\log m}},$$

because $\left|\frac{d}{d\theta} G_m(e^{i\theta})\right| = \left|e^{i\theta} G_m'(e^{i\theta})\right|$. For m large, $c \ge 1$ and $t \le 1$ give $|J| = c/m \ge \delta$, so the intersection of J with this δ-neighborhood has length at least a constant multiple of δ. Hence

$$\mathbf{1}\Big\{\min_{\theta\in J} g(\theta) \le t\Big\} \mathbf{1}_{E_\lambda} \le C\, \frac{\Lambda(2t)}{\delta}.$$

Taking expectations and using the preceding occupation bound with $s = 2t$ gives

$$\mathbb{P}\Big\{\min_I |G_m| \le t\Big\} \le C_A m^{-A} + C\, \frac{\lambda m^{3/2}\sqrt{\log m}}{t}\, \frac{C_\gamma c t^2}{m^2} \le C_{\gamma,A} \left(c\, \frac{\sqrt{\log m}\, t}{\sqrt{m}} + m^{-A} \right).$$

Replacing c by $1 + c$ gives the stated form. □

Lemma 3.22 (Endpoint Dirichlet deficit)**.** *Fix $C_0 > 0$, and put*

$$S_m(\theta) := \sum_{j=0}^{m} e^{2ij\theta}.$$

There are constants $c_{C_0} > 0$ and $m_0(C_0)$ such that, for all $m \geq m_0(C_0)$ and $0 < \vartheta := |\theta| \leq C_0/m$,

$$(m+1) - |S_m(\theta)| \geq c_{C_0} m^3 \vartheta^2.$$

Proof. The geometric-sum formula gives

$$|S_m(\theta)| = \left| \frac{\sin((m+1)\theta)}{\sin\theta} \right|.$$

Let $x = (m+1)\vartheta$. Since $\vartheta \leq C_0/m$, the variable x stays in a fixed compact interval $(0, 2C_0]$ for all large m. The function $x \mapsto |\sin x|/x$ is strictly less than one on every compact subinterval of $(0, \infty)$, and near zero its Taylor expansion is

$$\frac{\sin x}{x} = 1 - \frac{x^2}{6} + O(x^4).$$

Also

$$\frac{\vartheta}{\sin\vartheta} = 1 + O(\vartheta^2) = 1 + O\left(\frac{x^2}{m^2}\right)$$

uniformly in the present range. Splitting $x \in (0, 2C_0]$ into a small neighborhood of zero and its compact complement, these two estimates give a constant $c'_{C_0} > 0$ such that

$$1 - \frac{|S_m(\theta)|}{m+1} = 1 - \frac{|\sin x|}{x} \frac{\vartheta}{\sin\vartheta} \geq c'_{C_0} x^2$$

for all large m. Multiplying by $m+1$ and using $(m+1)^3 \asymp m^3$ proves the claim. □

Lemma 3.23 (Endpoint occupation bound for bounded-density laws)**.** *Let $\xi_0, \ldots, \xi_m$ be independent real random variables whose densities are bounded by K. Fix $C_0 > 0$ and $B > 0$. Let*

$$E_{C_0} := \{\theta \in [-\pi, \pi] : \operatorname{dist}(\theta, \{0, \pi, -\pi\}) \leq C_0/m\}.$$

For m sufficiently large and $m^{-B} \leq s \leq 2$,

$$\mathbb{E} \int_{E_{C_0}} \mathbf{1}\{\left|G_m(e^{i\theta})\right| \leq s\}\, d\theta \leq C_{B,C_0,K}\, \frac{s^2 \log m}{m^2}.$$

Proof. It is enough to work near 0; the neighborhoods of π and $-\pi$ are identical after replacing ξ_j by $(-1)^j \xi_j$. Put $\vartheta = |\theta| \leq C_0/m$. The real-part coefficient vector has squared norm $\sum_j \cos^2(j\theta) \geq c_{C_0} m$: indeed, with $a = m\theta \in [-C_0, C_0]$, the normalized sums converge uniformly to $\int_0^1 \cos^2(ax)\, dx$,

whose minimum on this compact interval is positive. Projection onto the real axis gives the one-dimensional bound

$$\mathbb{P}\{\Big|G_m(e^{i\theta})\Big| \le s\} \le C_{C_0,K}\frac{s}{\sqrt{m}},$$

by the one-dimensional linear-image density estimate. For $\vartheta > 0$, let A_θ be the two-row evaluation matrix from Proposition 3.2. The eigenvalues of $A_\theta A_\theta^\top$ are

$$\frac{1}{2}(m+1 \pm |S_m(\theta)|), \qquad S_m(\theta) = \sum_{j=0}^{m} e^{2ij\theta}.$$

Lemma 3.22 gives

$$(m+1) - |S_m(\theta)| \ge c_{C_0} m^3 \vartheta^2 \qquad (0 < \vartheta \le C_0/m),$$

while the larger eigenvalue is $\asymp m$. Hence $\sqrt{\det(A_\theta A_\theta^\top)} \ge c_{C_0} m^2 \vartheta$. The two-dimensional linear-image density estimate gives

$$\mathbb{P}\{\Big|G_m(e^{i\theta})\Big| \le s\} \le C_{C_0,K}\frac{s^2}{m^2\vartheta}.$$

Combining the two estimates and integrating,

$$\begin{aligned}\int_0^{C_0/m} \mathbb{P}\{\Big|G_m(e^{i\theta})\Big| \le s\}\, d\theta &\le C\int_0^{s/m^{3/2}} \frac{s}{\sqrt{m}}\, d\theta + C\int_{s/m^{3/2}}^{C_0/m} \frac{s^2}{m^2\theta}\, d\theta \\ &\le C_{B,C_0,K}\frac{s^2\log m}{m^2},\end{aligned}$$

because $s \le 2$ makes $s/m^{3/2} \le C_0/m$ for all large m, and $s \ge m^{-B}$ controls the logarithm. Multiplying by the constant number of endpoint neighborhoods gives the claim. □

Proposition 3.24 (Endpoint single-arc boundary small ball)**.** *Fix $C_0 < \infty$, and set*

$$E_+(C_0) = \{e^{i\theta} : |\theta| \le C_0/m\}, \qquad E_-(C_0) = \{e^{i\theta} : |\theta - \pi| \le C_0/m\}.$$

Let $G_m(z) = \sum_{j=0}^m \xi_j z^j$, where the ξ_j are independent standard real Gaussians. There are constants $C_{C_0} < \infty$ and $m_0(C_0) < \infty$ such that, for $m \ge m_0(C_0)$ and $0 < t \le 1$,

$$\mathbb{P}\Big\{\min_{z\in E_\pm(C_0)} |G_m(z)| \le t\Big\} \le C_{C_0}\frac{t}{\sqrt{m}}.$$

Consequently, any arc of length c/m which meets one of the two real points satisfies the same bound with right-hand side $C_{C_0}(1+c)t/\sqrt{m}$.

Proof. It is enough to treat $E_+(C_0)$, since $E_-(C_0)$ is obtained after replacing ξ_j by $(-1)^j\xi_j$. Put

$$F_m(u) := m^{-1/2}G_m(e^{iu/m}), \qquad X_m(u) := (\Re F_m(u), \Im F_m(u)), \qquad \varepsilon := t/\sqrt{m}.$$

The endpoint value gives

$$\mathbb{P}\{|F_m(0)| \le \varepsilon\} \le C\varepsilon.$$

Thus it remains to bound the expected number of crossings $\#\{u \in (0, C_0] : |F_m(u)| = \varepsilon\}$.

For $0 < u \le u_0$, with $u_0 > 0$ small depending only on C_0, write

$$r(u) = m^{-1/2}(\cos(ju/m))_{j=0}^{m}, \qquad s(u) = m^{-1/2}(\sin(ju/m))_{j=0}^{m},$$

$$\beta(u) = \frac{\langle r(u), s(u)\rangle}{\|r(u)\|^2}, \qquad w(u) = s(u) - \beta(u)r(u).$$

Uniformly for large m,

$$c \le \|r(u)\|^2 \le C, \qquad cu^2 \le \|w(u)\|^2 \le Cu^2, \qquad \|r'(u)\| \le Cu, \qquad \|w'(u) - u^{-1}w(u)\| \le Cu,$$

and $|\beta(u)| \le Cu$, $|\beta'(u)| \le C$. These follow from Taylor expansion and the Riemann-sum identities $m^{-1}\sum_{j=0}^{m}(j/m)^k = (k+1)^{-1} + O(m^{-1})$.

Let $y_1 = x_1$ and $y_2 = x_2 - \beta(u)x_1$. Since $r(u) \perp w(u)$, the density of $X_m(u)$ satisfies

$$p_u(x) \le \frac{C}{u}\exp\left(-c\,\frac{y_2^2}{u^2}\right).$$

The same normal form and Gaussian regression give, for $|x| = \varepsilon$,

$$\mathbb{E}\big[|x \cdot \dot{X}_m(u)| \;\big|\; X_m(u) = x\big] \le C\left(u\varepsilon + \frac{y_2^2}{u}\right).$$

Kac–Rice for $Q(u) = |X_m(u)|^2$, followed by coarea on the circle $|x| = \varepsilon$, therefore gives

$$\mathbb{E}\#\{0 < u \le u_0 : Q(u) = \varepsilon^2\} \le C\int_0^{u_0}\int_{|x|=\varepsilon} \frac{u\varepsilon + y_2^2/u}{\varepsilon}\,\frac{1}{u}e^{-cy_2^2/u^2}\,dH(x)\,du.$$

On the circle $|x| = \varepsilon$,

$$\int e^{-cy_2^2/u^2}\,dH(x) \le C\min\{\varepsilon, u\}, \qquad \int y_2^2 e^{-cy_2^2/u^2}\,dH(x) \le C\min\{\varepsilon^3, u^3\}.$$

Hence the last display is bounded by

$$C\int_0^{u_0}\left(\min\{\varepsilon, u\} + \frac{\min\{\varepsilon^3, u^3\}}{u^2\varepsilon}\right)du \le C\varepsilon.$$

On $u \in [u_0, C_0]$, equivalently $\theta \in [u_0/m, C_0/m]$, Lemma 3.12 applies with $\gamma = u_0$ and contributes at most $C_{C_0}\varepsilon$. Therefore

$$\mathbb{E}\#\{0 < u \le C_0 : |F_m(u)| = \varepsilon\} \le C_{C_0}\varepsilon.$$

If the minimum over $E_+(C_0)$ is at most t, then either the endpoint value is at most t, or the level t is crossed inside the arc. Markov's inequality for the crossing count proves the displayed bound. The final sentence follows by applying this endpoint estimate on a fixed $O(1/m)$ neighborhood of the real point and Theorem 1.1 to the remaining part of the arc. □

Proposition 3.25 (Endpoint lower bound)**.** *For every $C_0 > 0$ there are constants $c_{C_0}, C_{C_0} > 0$ and $m_0(C_0) < \infty$ such that, for $m \geq m_0(C_0)$ and $0 < t \leq 1$,*

$$c_{C_0} \frac{t}{\sqrt{m}} \leq \mathbb{P}\Big\{ \min_{z \in E_\pm(C_0)} |G_m(z)| \leq t \Big\} \leq C_{C_0} \frac{t}{\sqrt{m}}.$$

In particular the endpoint estimate of Proposition 3.24 has the correct order in both t and m.

Proof. The upper bound is Proposition 3.24. For the lower bound, $1 \in E_+(C_0)$ and $-1 \in E_-(C_0)$, while

$$G_m(1) = \sum_{j=0}^{m} \xi_j, \qquad G_m(-1) = \sum_{j=0}^{m} (-1)^j \xi_j$$

are each real Gaussians with variance $m+1$. Thus, for $0 < t \leq 1$,

$$\mathbb{P}\{|G_m(\pm 1)| \leq t\} = \frac{1}{\sqrt{2\pi(m+1)}} \int_{-t}^{t} e^{-x^2/(2(m+1))}\, dx \geq c\, \frac{t}{\sqrt{m}},$$

after decreasing c and taking m large enough. Since $\{|G_m(\pm 1)| \leq t\} \subseteq \{\min_{E_\pm(C_0)} |G_m| \leq t\}$, the lower bound follows. □

Corollary 3.26 (Exact two-endpoint Gaussian formula and sharp expansion)**.** *Let $G_m(z) = \sum_{j=0}^{m} \xi_j z^j$ have independent standard real Gaussian coefficients. Assume $m \geq 1$, and put $N = m+1$ and*

$$\rho_N := \sum_{j=0}^{N-1} (-1)^j \in \{0, 1\}.$$

For $t \geq 0$, set

$$p_N(t) := \frac{1}{\sqrt{2\pi N}} \int_{-t}^{t} e^{-x^2/(2N)}\, dx$$

and

$$q_N(t) := \frac{1}{2\pi\sqrt{N^2 - \rho_N^2}} \int_{-t}^{t} \int_{-t}^{t} \exp\left\{ -\frac{N(x^2+y^2) - 2\rho_N xy}{2(N^2 - \rho_N^2)} \right\} dx\, dy.$$

Then the endpoint event is exactly

$$\mathbb{P}\Big\{ \min(|G_m(1)|, |G_m(-1)|) \leq t \Big\} = 2p_N(t) - q_N(t). \tag{3.5}$$

In particular, if N is even then the two endpoint values are independent and $q_N(t) = p_N(t)^2$. Uniformly for $0 \leq t \leq 1$,

$$\mathbb{P}\Big\{ \min(|G_m(1)|, |G_m(-1)|) \leq t \Big\} = 2\sqrt{\frac{2}{\pi}}\, \frac{t}{\sqrt{N}} - \frac{2}{\pi} \frac{t^2}{\sqrt{N^2 - \rho_N^2}} + O\left(\frac{t^3}{N^{3/2}} \right) \qquad (m \to \infty).$$

Thus the two parity endpoints contribute a sharp global lower floor of order $t/\sqrt{m}$, with leading constant $2\sqrt{2/\pi}$ before the non-real arcs are counted.

Proof. Set

$$X = G_m(1), \qquad Y = G_m(-1), \qquad \rho_N = \sum_{j=0}^{N-1} (-1)^j \in \{0, 1\}.$$

Then X and Y are centered Gaussians with

$$\operatorname{Var} X = \operatorname{Var} Y = N, \qquad \operatorname{Cov}(X, Y) = \rho_N.$$

Therefore $p_N(t) = \mathbb{P}\{|X| \le t\} = \mathbb{P}\{|Y| \le t\}$, and $q_N(t)$ is the bivariate Gaussian probability $\mathbb{P}\{|X| \le t,\ |Y| \le t\}$. Inclusion–exclusion proves (3.5). If N is even then $\rho_N = 0$, so the centered Gaussian variables X and Y are independent and $q_N(t) = p_N(t)^2$.

For the asymptotic expansion, one endpoint gives

$$p_N(t) = \frac{1}{\sqrt{2\pi N}} \int_{-t}^{t} e^{-x^2/(2N)}\, dx = \sqrt{\frac{2}{\pi}}\, \frac{t}{\sqrt{N}} + O\!\left(\frac{t^3}{N^{3/2}}\right),$$

uniformly for $0 \le t \le 1$. The joint density of (X, Y) is bounded by C/N for all $m \ge 1$, since $\det\left(\begin{smallmatrix} N & \rho_N \\ \rho_N & N \end{smallmatrix}\right) = N^2 - \rho_N^2 \asymp N^2$. More precisely, on the square $[-t, t]^2$,

$$\exp\left\{-\frac{N(x^2 + y^2) - 2\rho_N xy}{2(N^2 - \rho_N^2)}\right\} = 1 + O\!\left(\frac{t^2}{N}\right),$$

uniformly for $0 \le t \le 1$. Therefore

$$q_N(t) = \frac{2}{\pi} \frac{t^2}{\sqrt{N^2 - \rho_N^2}} + O\!\left(\frac{t^4}{N^2}\right).$$

Combining this with (3.5) and the one-point expansion gives the displayed formula, since $t^4/N^2 \le t^3/N^{3/2}$ on $0 \le t \le 1$ and $N \ge 2$. □

Corollary 3.27 (Endpoint scale separation at $t = \tau/\sqrt{N}$)**.** *Let $G_m(z) = \sum_{j=0}^{m} \xi_j z^j$ have independent standard real Gaussian coefficients, and put $N = m + 1$. For every $K < \infty$, uniformly for $0 \le \tau \le K$,*

$$\begin{aligned} &\mathbb{P}\Big\{\min(|G_m(1)|, |G_m(-1)|) \le \frac{\tau}{\sqrt{N}}\Big\} \\ &\quad = 2\sqrt{\frac{2}{\pi}}\, \frac{\tau}{N} - \frac{2}{\pi} \frac{\tau^2}{N\sqrt{N^2 - \rho_N^2}} + O_K(N^{-3}) = 2\sqrt{\frac{2}{\pi}}\, \frac{\tau}{N} + O_K(N^{-2}), \end{aligned}$$

where $\rho_N = \sum_{j=0}^{N-1} (-1)^j \in \{0, 1\}$. Consequently

$$N\, \mathbb{P}\Big\{\min(|G_m(1)|, |G_m(-1)|) \le \frac{\tau}{\sqrt{N}}\Big\} \longrightarrow 2\sqrt{\frac{2}{\pi}}\, \tau$$

uniformly for τ in compact subsets of $[0, \infty)$. Thus the two parity endpoints have vanishing unscaled probability at the continuum boundary-minimum

threshold $t = \tau/\sqrt{N}$; the $O(1)$ candidate global law must come from the non-real arcs.

Proof. Substitute $t = \tau/\sqrt{N}$ in the expansion of Corollary 3.26. For $0 \le \tau \le K$, the condition $t \le 1$ holds for all sufficiently large N, and

$$\frac{t^2}{\sqrt{N^2-\rho_N^2}} = \frac{\tau^2}{N\sqrt{N^2-\rho_N^2}}, \qquad \frac{t^3}{N^{3/2}} = O_K(N^{-3}).$$

This gives the displayed asymptotic. Multiplication by N and $\rho_N \in \{0,1\}$ give the uniform compact limit. □

Corollary 3.28 (Endpoint neighborhoods are negligible at continuum scale)**.** *Fix $C_0 < \infty$, and put*

$$E_{C_0} := E_+(C_0) \cup E_-(C_0).$$

Let $G_m(z) = \sum_{j=0}^m \xi_j z^j$ have independent standard real Gaussian coefficients, and put $N = m+1$. There are constants $C_{C_0} < \infty$ and $m_0(C_0) < \infty$ such that, for $m \ge m_0(C_0)$ and $0 < \tau \le \sqrt{N}$,

$$\mathbb{P}\Big\{ \min_{z \in E_{C_0}} |G_m(z)| \le \frac{\tau}{\sqrt{N}} \Big\} \le C_{C_0} \frac{\tau}{N}.$$

In particular, for every fixed τ the natural $O(1/m)$ neighborhoods of the two real endpoints have vanishing probability at the continuum boundary-minimum threshold. They can contribute only on the N^{-1} probability scale, not to an $O(1)$ whole-circle limiting law.

Proof. Apply Proposition 3.24 to $E_+(C_0)$ and $E_-(C_0)$, with $t = \tau/\sqrt{N}$. Since $0 < \tau \le \sqrt{N}$, this threshold satisfies $0 < t \le 1$, and the union bound gives

$$\mathbb{P}\Big\{ \min_{z \in E_{C_0}} |G_m(z)| \le \frac{\tau}{\sqrt{N}} \Big\} \le 2C_{C_0} \frac{\tau}{\sqrt{mN}}.$$

Because $m = N-1$, the last quantity is at most $C'_{C_0}\tau/N$ after increasing $m_0(C_0)$ and the constant. □

Corollary 3.29 (Sharp global continuum-scale probability upper bound)**.** *Let $G_m(z) = \sum_{j=0}^m \xi_j z^j$ have independent standard real Gaussian coefficients, put $N = m+1$, and define*

$$P_N^{\mathbb{T}}(\tau) := \mathbb{P}\Big\{ \min_{\theta \in [-\pi,\pi]} |G_m(e^{i\theta})| \le \frac{\tau}{\sqrt{N}} \Big\}.$$

Then, for every $K < \infty$,

$$\limsup_{N\to\infty} \sup_{0 \le \tau \le K} \left(P_N^{\mathbb{T}}(\tau) - \sqrt{\frac{\pi}{3}}\,\tau \right)_+ = 0.$$

Equivalently, for every fixed $\tau \ge 0$,

$$\limsup_{N\to\infty} P_N^{\mathbb{T}}(\tau) \le \sqrt{\frac{\pi}{3}}\,\tau.$$

Proof. Fix $K < \infty$, $0 < \eta < \pi/4$, and a fixed $C_0 \geq 4$. Let

$$A_\eta := \{\theta \in [-\pi, \pi] : \operatorname{dist}(\theta, \pi\mathbb{Z}) < \eta\}.$$

The complement of A_η is B_η. Corollary 3.17 gives, uniformly for $0 \leq \tau \leq K$,

$$\limsup_{N\to\infty} \sup_{0\leq\tau\leq K} \left(\mathbb{P}\Big\{ \min_{\theta\in B_\eta} |G_m(e^{i\theta})| \leq \frac{\tau}{\sqrt{N}} \Big\} - \frac{|B_\eta|\,\tau}{\sqrt{12\pi}} \right)_+ = 0.$$

It remains to bound A_η. Remove from A_η the natural endpoint neighborhoods E_{C_0}. For all large m, the remaining set is covered by at most $C\eta m$ arcs J_ℓ of arclength c_ℓ/m, with $1 \leq c_\ell \leq 2$, each at angular distance at least $(C_0/2)/m$ from $\{\pm 1\}$. Theorem 1.1, with $\gamma = C_0/2$ and $t = \tau/\sqrt{N}$, gives

$$\mathbb{P}\Big\{ \min_{e^{i\theta}\in J_\ell} |G_m(e^{i\theta})| \leq \frac{\tau}{\sqrt{N}} \Big\} \leq C_{C_0,K} \frac{\tau}{N} \qquad (0 \leq \tau \leq K),$$

after increasing m if necessary. Summing over the $O(\eta m)$ arcs gives

$$\limsup_{N\to\infty} \sup_{0\leq\tau\leq K} \mathbb{P}\Big\{ \min_{\theta\in A_\eta,\ e^{i\theta}\notin E_{C_0}} |G_m(e^{i\theta})| \leq \frac{\tau}{\sqrt{N}} \Big\} \leq C_{C_0}\eta K.$$

By Corollary 3.28, the natural endpoint neighborhoods E_{C_0} contribute $o_N(1)$, uniformly for $0 \leq \tau \leq K$. Therefore

$$\limsup_{N\to\infty} \sup_{0\leq\tau\leq K} \left(P_N^{\mathbb{T}}(\tau) - \frac{|B_\eta|\,\tau}{\sqrt{12\pi}} \right)_+ \leq C_{C_0}\eta K.$$

Since $|B_\eta| = 2\pi - 4\eta$, the coefficient $|B_\eta|/\sqrt{12\pi}$ is at most $2\pi/\sqrt{12\pi} = \sqrt{\pi/3}$. Thus the same bound holds with $\sqrt{\pi/3}\,\tau$ subtracted in place of $|B_\eta|\tau/\sqrt{12\pi}$. Letting $\eta \downarrow 0$ proves the uniform upper bound, and the pointwise bound follows by fixing τ. □

Corollary 3.30 (Global first-order law from a uniform factorial moment bound)**.** *Under the hypotheses of Corollary 3.20, with $P_N^{\mathbb{T}}$ as in Corollary 3.29, one has, for every $0 < \tau \leq T$,*

$$\frac{\sqrt{\pi/3}\,\tau}{1 + L_T\sqrt{\pi/3}\,\tau} \leq \liminf_{N\to\infty} P_N^{\mathbb{T}}(\tau) \leq \limsup_{N\to\infty} P_N^{\mathbb{T}}(\tau) \leq \sqrt{\frac{\pi}{3}}\,\tau.$$

Consequently,

$$\lim_{\tau\downarrow 0} \frac{1}{\tau} \liminf_{N\to\infty} P_N^{\mathbb{T}}(\tau) = \lim_{\tau\downarrow 0} \frac{1}{\tau} \limsup_{N\to\infty} P_N^{\mathbb{T}}(\tau) = \sqrt{\frac{\pi}{3}}.$$

Proof. The lower bound is (3.4), and the upper bound is Corollary 3.29. Divide by τ and let $\tau \downarrow 0$. □

Corollary 3.31 (Negative moments below the boundary exponent)**.** *Let*

$$M_N := \sqrt{N} \min_{z\in\mathbb{T}} |G_m(z)|, \qquad N = m + 1,$$

for the standard real Gaussian polynomial G_m. For every $0 < q < 1$,

$$\sup_{N\geq 2} \mathbb{E} M_N^{-q} < \infty.$$

More sharply, for every $a > 0$,

$$\limsup_{N\to\infty} \mathbb{E}\Big[M_N^{-q}\mathbf{1}\{M_N \le a\}\Big] \le \frac{\sqrt{\pi/3}}{1-q}\, a^{1-q}.$$

Proof. First $M_N > 0$ almost surely. Indeed, a unit-circle zero of G_m forces G_m and the reflected polynomial $z^m G_m(1/z)$ to have a common root, so the coefficient vector lies in the zero set of a non-trivial resultant; the Gaussian coefficient law gives that algebraic hypersurface probability zero.

Put $F_N(s) = \mathbb{P}\{M_N \le s\}$. Theorem 1.2, with $t = s/\sqrt{N}$, gives $F_N(s) \le Cs$ for $0 < s \le 1$, after increasing C to cover the finitely many small N. Therefore, for $0 < q < 1$,

$$\mathbb{E}M_N^{-q} = q\int_0^\infty u^{q-1}\mathbb{P}\{M_N^{-1} > u\}\,du \le 1 + qC\int_1^\infty u^{q-2}\,du,$$

uniformly in N.

For the sharp small-window estimate, fix $0 < \rho < a$. Stieltjes integration by parts gives

$$\mathbb{E}\Big[M_N^{-q}\mathbf{1}\{M_N \le a\}\Big] = a^{-q}F_N(a) + q\int_0^a s^{-q-1}F_N(s)\,ds.$$

On $(0,\rho)$ use the uniform bound $F_N(s) \le Cs$. On $[\rho, a]$ use Corollary 3.29, which gives

$$\limsup_{N\to\infty} \sup_{\rho\le s\le a} \big(F_N(s) - \sqrt{\pi/3}\, s\big)_+ = 0.$$

Taking the limsup in N yields

$$\limsup_{N\to\infty} \mathbb{E}\Big[M_N^{-q}\mathbf{1}\{M_N \le a\}\Big] \le \sqrt{\frac{\pi}{3}}\, a^{1-q} + \frac{qC}{1-q}\rho^{1-q} + \frac{q\sqrt{\pi/3}}{1-q}\left(a^{1-q} - \rho^{1-q}\right).$$

Letting $\rho \downarrow 0$ gives the claimed bound. □

Corollary 3.32 (Weak endpoint bound for the inverse scaled minimum)**.** *With M_N as in Corollary 3.31, put $Y_N = M_N^{-1}$. Then*

$$\sup_{N\ge 2}\sup_{u\ge 1} u\,\mathbb{P}\{Y_N > u\} < \infty.$$

Moreover, for every $1 \le R < \infty$,

$$\limsup_{N\to\infty} \sup_{1\le u\le R} \left(u\,\mathbb{P}\{Y_N > u\} - \sqrt{\frac{\pi}{3}}\right)_+ = 0.$$

Thus the endpoint exponent for Y_N is weak L^1, and the sharp global upper constant controls every fixed inverse scale.

Proof. Let $F_N(s) = \mathbb{P}\{M_N \le s\}$. The proof of Corollary 3.31, using Theorem 1.2, gives a constant C such that $F_N(s) \le Cs$ for all $N \ge 2$ and $0 < s \le 1$. Therefore, for $u \ge 1$,

$$\mathbb{P}\{Y_N > u\} = F_N(1/u) \le \frac{C}{u},$$

which proves the uniform weak bound.

For the sharp finite-window constant, Corollary 3.29 with $K = 1$ gives

$$\varepsilon_N := \sup_{0\le s\le 1} \left(F_N(s) - \sqrt{\frac{\pi}{3}}\, s \right)_+ \longrightarrow 0.$$

If $1 \le u \le R$, then

$$u\,\mathbb{P}\{Y_N > u\} = uF_N(1/u) \le \sqrt{\frac{\pi}{3}} + R\varepsilon_N.$$

Taking the supremum over $1 \le u \le R$ and then the limsup in N proves the claim. □

Corollary 3.33 (Quantiles of the inverse scaled minimum)**.** *With M_N and $Y_N = M_N^{-1}$ as in Corollary 3.32, define, for $0 < \alpha \le 1$,*

$$Q_N(\alpha) := \inf\{u \ge 1 : \mathbb{P}\{Y_N > u\} \le \alpha\}.$$

Then there is an absolute constant $C < \infty$ such that

$$Q_N(\alpha) \le \frac{C}{\alpha} \qquad (N \ge 2,\ 0 < \alpha \le 1).$$

More sharply, for every fixed $0 < \alpha \le 1$,

$$\limsup_{N\to\infty} Q_N(\alpha) \le \frac{1}{\alpha}\sqrt{\frac{\pi}{3}}.$$

In lower-quantile form, if

$$q_N(p) := \inf\{s \ge 0 : \mathbb{P}\{M_N \le s\} \ge p\},$$

then there is $c > 0$ such that

$$q_N(p) \ge cp \qquad (N \ge 2,\ 0 < p < 1),$$

and, for every fixed $0 < p < 1$,

$$\liminf_{N\to\infty} q_N(p) \ge p\sqrt{\frac{3}{\pi}}.$$

Thus target tail probabilities for the inverse minimum translate into linear clearance thresholds for the scaled minimum, with the same sharp global upper constant in the limit.

Proof. Enlarge the constant in the uniform weak bound of Corollary 3.32 so that $C \ge 1$. Then

$$\mathbb{P}\{Y_N > C/\alpha\} \le \alpha,$$

and hence $Q_N(\alpha) \le C/\alpha$.

For the sharp limsup, put $K_0 = \sqrt{\pi/3}$, fix $\varepsilon > 0$, and set $u_\varepsilon = (K_0 + \varepsilon)/\alpha$. The finite-window part of Corollary 3.32, applied with $R = u_\varepsilon$, gives

$$\limsup_{N\to\infty} \mathbb{P}\{Y_N > u_\varepsilon\} \le \frac{K_0}{u_\varepsilon} < \alpha.$$

Thus $Q_N(\alpha) \le u_\varepsilon$ for all sufficiently large N, and $\varepsilon \downarrow 0$ gives the displayed bound.

For the lower quantiles of M_N, let $F_N(s) = \mathbb{P}\{M_N \le s\}$. The proof of Corollary 3.32 gives $F_N(s) \le C_0 s$ for all $N \ge 2$ and $0 < s \le 1$. Choosing $c \le (2C_0)^{-1}$ gives $F_N(cp) < p$, hence $q_N(p) \ge cp$. For the sharp asymptotic statement, fix $a < p\sqrt{3/\pi}$. Corollary 3.29 gives

$$\limsup_{N\to\infty} F_N(a) \le \sqrt{\frac{\pi}{3}}\, a < p.$$

Therefore $q_N(p) \ge a$ for all sufficiently large N. Letting $a \uparrow p\sqrt{3/\pi}$ proves the last claim. □

Corollary 3.34 (Finite-family weak endpoint bound)**.** *Fix $L \ge 1$. For each N, let $Y_{N,1}, \dots, Y_{N,L}$ be positive random variables whose marginals have the same law as Y_N in Corollary 3.32; no independence between them is required. Put*

$$Y^*_{N,L} := \max_{1\le \ell \le L} Y_{N,\ell}.$$

Then

$$\sup_{N\ge 2} \sup_{u\ge 1} u\, \mathbb{P}\{Y^*_{N,L} > u\} \le CL$$

with an absolute constant $C < \infty$. More sharply, for every fixed $L \ge 1$ and $1 \le R < \infty$,

$$\limsup_{N\to\infty} \sup_{1\le u\le R} \left(u\, \mathbb{P}\{Y^*_{N,L} > u\} - L\sqrt{\frac{\pi}{3}} \right)_+ = 0.$$

Thus finite simultaneous testing preserves the weak L^1 endpoint, paying only the linear union-bound factor in the sharp fixed-window constant.

Proof. Let C_1 be the uniform weak constant in Corollary 3.32. For $u \ge 1$, the union bound gives

$$\mathbb{P}\{Y^*_{N,L} > u\} \le \sum_{\ell=1}^{L} \mathbb{P}\{Y_{N,\ell} > u\} \le \frac{LC_1}{u},$$

which proves the uniform estimate.

For the sharp fixed-window statement, put $K_0 = \sqrt{\pi/3}$. The finite-window part of Corollary 3.32 gives

$$\varepsilon_N(R) := \sup_{1\le u\le R} (u\mathbb{P}\{Y_N > u\} - K_0)_+ \longrightarrow 0.$$

Hence, uniformly for $1 \le u \le R$,

$$u\, \mathbb{P}\{Y^*_{N,L} > u\} \le L\,(K_0 + \varepsilon_N(R))\,.$$

Taking the supremum in u and the limsup in N proves the claim. □

Corollary 3.35 (Finite-family inverse-minimum quantiles)**.** *Fix $L \ge 1$. For each N, let $Y_{N,1}, \dots, Y_{N,L}$ be positive random variables whose marginals have the same law as Y_N in Corollary 3.32; no independence between them is required. Put*

$$Y^*_{N,L} := \max_{1\le \ell \le L} Y_{N,\ell}$$

and, for $0 < \alpha \le 1$,

$$Q_{N,L}(\alpha) := \inf\{u \ge 1 : \mathbb{P}\{Y^*_{N,L} > u\} \le \alpha\}.$$

Then there is an absolute constant $C < \infty$ *such that*

$$Q_{N,L}(\alpha) \le \frac{CL}{\alpha} \qquad (N \ge 2,\ L \ge 1,\ 0 < \alpha \le 1).$$

Moreover, for every fixed L *and fixed* $0 < \alpha \le 1$,

$$\limsup_{N\to\infty} Q_{N,L}(\alpha) \le \frac{L}{\alpha}\sqrt{\frac{\pi}{3}}.$$

Equivalently, if $M_{N,\ell} := Y^{-1}_{N,\ell}$,

$$M_{N,L,*} := \min_{1\le \ell\le L} M_{N,\ell}, \qquad q_{N,L}(p) := \inf\{s \ge 0 : \mathbb{P}\{M_{N,L,*} \le s\} \ge p\},$$

then there is $c > 0$ *such that*

$$q_{N,L}(p) \ge c\,\frac{p}{L} \qquad (N \ge 2,\ L \ge 1,\ 0 < p < 1),$$

and, for every fixed L *and fixed* $0 < p < 1$,

$$\liminf_{N\to\infty} q_{N,L}(p) \ge \frac{p}{L}\sqrt{\frac{3}{\pi}}.$$

Thus a simultaneous clearance requirement for L *Gaussian symbols costs only the union-bound factor* L *at the inverse-minimum endpoint, with the sharp single-symbol constant preserved for fixed* L.

Proof. Let C_1 be the absolute constant in Corollary 3.34, enlarged so that $C_1 \ge 1$. That corollary gives $Q_{N,L}(\alpha) \le C_1 L/\alpha$.

For the sharp fixed-L bound, put $K_0 = \sqrt{\pi/3}$, fix $\varepsilon > 0$, and set $u_\varepsilon = (LK_0 + \varepsilon)/\alpha$. The finite-window part of Corollary 3.34, applied with $R = u_\varepsilon$, gives

$$\limsup_{N\to\infty} \mathbb{P}\{Y^*_{N,L} > u_\varepsilon\} \le L\,\frac{K_0}{u_\varepsilon} < \alpha.$$

Hence $Q_{N,L}(\alpha) \le u_\varepsilon$ for all sufficiently large N, and $\varepsilon \downarrow 0$ proves the displayed limsup estimate.

For the lower-quantile form, write $F_N(s) = \mathbb{P}\{Y_N^{-1} \le s\}$. The proof of Corollary 3.32 gives $F_N(s) \le C_0 s$ for all $N \ge 2$ and $0 < s \le 1$. Therefore

$$\mathbb{P}\{M_{N,L,*} \le s\} \le LF_N(s) \le LC_0 s.$$

Taking $s = cp/L$ with $c \le (2C_0)^{-1}$ proves the uniform lower bound. For the sharp bound, if $a < p\sqrt{3/\pi}/L$, then Corollary 3.29 and the union bound give

$$\limsup_{N\to\infty} \mathbb{P}\{M_{N,L,*} \le a\} \le L\sqrt{\frac{\pi}{3}}\,a < p.$$

Thus $q_{N,L}(p) \ge a$ eventually, and $a \uparrow p\sqrt{3/\pi}/L$ completes the proof. □

Corollary 3.36 (Finite-family bad-clearance counts)**.** *In the setting of Corollary 3.35, define, for* $s \geq 0$,

$$Z_{N,L}(s) := \sum_{\ell=1}^{L} \mathbf{1}\{M_{N,\ell} \leq s\}.$$

Then

$$\mathbb{E}Z_{N,L}(s) \leq CLs \qquad (N \geq 2,\ L \geq 1,\ s \geq 0)$$

with an absolute constant $C < \infty$*. More sharply, for every fixed* $L \geq 1$ *and* $K < \infty$,

$$\limsup_{N\to\infty} \sup_{0\leq s\leq K} \left(\mathbb{E}Z_{N,L}(s) - L\sqrt{\frac{\pi}{3}}\, s\right)_+ = 0.$$

Consequently, for every integer $r \geq 1$,

$$\mathbb{P}\{Z_{N,L}(s) \geq r\} \leq \frac{CLs}{r},$$

and, for fixed L, K, r,

$$\limsup_{N\to\infty} \sup_{0\leq s\leq K} \left(r\,\mathbb{P}\{Z_{N,L}(s) \geq r\} - L\sqrt{\frac{\pi}{3}}\, s\right)_+ = 0.$$

Thus finite simultaneous testing not only controls the first bad clearance but also the expected number of bad clearances, with the same sharp linear coefficient and no independence assumption.

Proof. Let $F_N(s) = \mathbb{P}\{Y_N^{-1} \leq s\}$. Since each $M_{N,\ell}$ has the same law as Y_N^{-1},

$$\mathbb{E}Z_{N,L}(s) = \sum_{\ell=1}^{L} \mathbb{P}\{M_{N,\ell} \leq s\} = LF_N(s).$$

The proof of Corollary 3.32 gives $F_N(s) \leq C_0 s$ for $0 < s \leq 1$, and the case $s = 0$ is trivial since $F_N(0) = 0$. Also $F_N(s) \leq 1 \leq s$ for $s \geq 1$. This proves the uniform bound after increasing C.

For the sharp fixed-window statement, Corollary 3.29 gives, for every $K < \infty$,

$$\limsup_{N\to\infty} \sup_{0\leq s\leq K} \left(F_N(s) - \sqrt{\frac{\pi}{3}}\, s\right)_+ = 0.$$

Multiplying by L gives the expectation estimate. The two probability bounds follow from Markov's inequality $r\mathbb{P}\{Z_{N,L}(s) \geq r\} \leq \mathbb{E}Z_{N,L}(s)$. □

Corollary 3.37 (Finite-family clearance order statistics)**.** *In the setting of Corollary 3.35, let*

$$M_{N,L}^{(1)} \leq M_{N,L}^{(2)} \leq \cdots \leq M_{N,L}^{(L)}$$

be the increasing rearrangement of $M_{N,1}, \ldots, M_{N,L}$*. For* $1 \leq r \leq L$ *and* $0 < p < 1$*, define*

$$q_{N,L,r}(p) := \inf\{s \geq 0 : \mathbb{P}\{M_{N,L}^{(r)} \leq s\} \geq p\}.$$

Then there is an absolute constant $c > 0$ such that

$$q_{N,L,r}(p) \geq c\,\frac{pr}{L} \qquad (N \geq 2,\ 1 \leq r \leq L,\ 0 < p < 1).$$

Moreover, for every fixed $L \geq 1$, $1 \leq r \leq L$, and $0 < p < 1$,

$$\liminf_{N\to\infty} q_{N,L,r}(p) \geq \frac{pr}{L}\sqrt{\frac{3}{\pi}}.$$

Equivalently, the threshold at which r out of L simultaneous symbols can have bad boundary clearance is at least r/L times the one-symbol continuum scale, up to the same fixed-window constant, without any independence assumption among the L symbols.

Proof. For every $s \geq 0$,

$$\{M_{N,L}^{(r)} \leq s\} = \{Z_{N,L}(s) \geq r\}.$$

Corollary 3.36 gives

$$\mathbb{P}\{M_{N,L}^{(r)} \leq s\} \leq \frac{CLs}{r}.$$

Taking $s = cpr/L$, with $c \leq (2C)^{-1}$, proves the uniform bound.

For the fixed-L, r asymptotic statement, put $K_0 = \sqrt{\pi/3}$. If $0 < a < pr/(LK_0)$, then Corollary 3.36, used with $K = a$, gives

$$\limsup_{N\to\infty} \mathbb{P}\{M_{N,L}^{(r)} \leq a\} \leq \frac{LK_0 a}{r} < p.$$

Thus $q_{N,L,r}(p) \geq a$ eventually, and $a \uparrow pr/(LK_0)$ completes the proof. □

Corollary 3.38 (Finite-family truncated and logarithmic inverse moments)**.** *In the setting of Corollary 3.35, for every $R \geq 1$,*

$$\sup_{N\geq 2} \mathbb{E}\big[(Y_{N,L}^{*} \wedge R) - 1\big]_{+} \leq CL \log R$$

with an absolute constant $C < \infty$. More sharply, for every fixed $L \geq 1$ and fixed $1 \leq R < \infty$,

$$\limsup_{N\to\infty} \mathbb{E}\big[(Y_{N,L}^{*} \wedge R) - 1\big]_{+} \leq L\sqrt{\frac{\pi}{3}}\,\log R.$$

Also, for every $p > 0$,

$$\sup_{N\geq 2} \mathbb{E}\big[(\log_{+} Y_{N,L}^{*})^{p}\big] \leq C_p L,$$

and, for every fixed $L \geq 1$,

$$\limsup_{N\to\infty} \mathbb{E}\big[(\log_{+} Y_{N,L}^{*})^{p}\big] \leq L\sqrt{\frac{\pi}{3}}\,\Gamma(p+1).$$

Thus finite simultaneous testing preserves the endpoint $q = 1$ obstruction: the price of L marginals is only the linear union-bound factor in the critical truncation and logarithmic-integrability constants.

Proof. For any nonnegative random variable Y,

$$\mathbb{E}\big[(Y \wedge R) - 1\big]_+ = \int_1^R \mathbb{P}\{Y > u\}\, du.$$

The union-bound tail estimate from Corollary 3.35,

$$\mathbb{P}\{Y^*_{N,L} > u\} \le LC_1/u \qquad (u \ge 1),$$

gives the uniform truncated bound. For the sharp fixed-window coefficient, the finite-window part of Corollary 3.32 gives, for fixed R,

$$\varepsilon_N(R) := \sup_{1\le u\le R} \left(u\mathbb{P}\{Y_N > u\} - \sqrt{\frac{\pi}{3}} \right)_+ \longrightarrow 0.$$

Hence, uniformly for $1 \le u \le R$,

$$\mathbb{P}\{Y^*_{N,L} > u\} \le L\left(\sqrt{\frac{\pi}{3}} + \varepsilon_N(R)\right) u^{-1}.$$

Integrating over $1 \le u \le R$ and taking the limsup proves the truncated estimate.

For logarithmic moments, use layer cake:

$$\mathbb{E}\big[(\log_+ Y^*_{N,L})^p\big] = p\int_0^\infty t^{p-1}\mathbb{P}\{Y^*_{N,L} > e^t\}\, dt.$$

The uniform tail bound gives

$$\mathbb{P}\{Y^*_{N,L} > e^t\} \le LC_1 e^{-t},$$

so the uniform C_pL estimate follows. For the sharp limsup, fix $T < \infty$. The preceding finite-window bound with $R = e^T$ yields

$$\limsup_{N\to\infty} p\int_0^T t^{p-1}\mathbb{P}\{Y^*_{N,L} > e^t\}\, dt \le L\sqrt{\frac{\pi}{3}}\, p\int_0^T t^{p-1}e^{-t}\, dt.$$

The tail $t \ge T$ is at most

$$LC_1 p\int_T^\infty t^{p-1}e^{-t}\, dt$$

uniformly in N. Letting $T \to \infty$ gives $L\sqrt{\pi/3}\,\Gamma(p+1)$. □

Corollary 3.39 (Finite-family subcritical inverse powers)**.** *In the setting of Corollary 3.35, for every $0 < q < 1$,*

$$\sup_{N\ge 2} \mathbb{E}\big[(Y^*_{N,L})^q\big] = \sup_{N\ge 2} \mathbb{E}M^{-q}_{N,L,*} \le C_qL,$$

where $C_q < \infty$ depends only on q. More sharply, for every fixed $L \ge 1$ and every $a > 0$,

$$\limsup_{N\to\infty} \mathbb{E}\Big[M^{-q}_{N,L,*}\mathbf{1}\{M_{N,L,*} \le a\}\Big] \le \frac{L\sqrt{\pi/3}}{1-q}\, a^{1-q}.$$

Thus finite simultaneous clearance preserves the same subcritical endpoint exponent; L marginals only multiply the sharp small-clearance coefficient by L.

Proof. Put

$$F_{N,L}(s) = \mathbb{P}\{M_{N,L,*} \le s\}, \qquad F_N(s) = \mathbb{P}\{M_N \le s\}.$$

The proof of Corollary 3.35 gives a constant C_0 such that $F_N(s) \le C_0 s$ for all $N \ge 2$ and $0 < s \le 1$. Hence, by the union bound,

$$F_{N,L}(s) \le L F_N(s) \le L C_0 s \qquad (0 < s \le 1).$$

For $u \ge 1$, this gives

$$\mathbb{P}\{Y^*_{N,L} > u\} \le F_{N,L}(1/u) \le L C_0/u.$$

Therefore

$$\mathbb{E}\big[(Y^*_{N,L})^q\big] = q \int_0^\infty u^{q-1} \mathbb{P}\{Y^*_{N,L} > u\}\, du \le 1 + qLC_0 \int_1^\infty u^{q-2}\, du \le C_q L.$$

For the sharp small-window estimate, fix $0 < \rho < \min\{a, 1\}$. Stieltjes integration by parts gives

$$\mathbb{E}\Big[M^{-q}_{N,L,*} \mathbf{1}\{M_{N,L,*} \le a\}\Big] = a^{-q} F_{N,L}(a) + q \int_0^a s^{-q-1} F_{N,L}(s)\, ds.$$

On $(0, \rho)$ use $F_{N,L}(s) \le LC_0 s$. On $[\rho, a]$, the union bound and Corollary 3.29 give

$$\limsup_{N\to\infty} \sup_{\rho \le s \le a} \big(F_{N,L}(s) - L\sqrt{\pi/3}\, s\big)_+ = 0.$$

Taking the limsup in N yields

$$\limsup_{N\to\infty} \mathbb{E}\Big[M^{-q}_{N,L,*} \mathbf{1}\{M_{N,L,*} \le a\}\Big] \le L\sqrt{\frac{\pi}{3}}\, a^{1-q} + \frac{qLC_0}{1-q} \rho^{1-q} + \frac{qL\sqrt{\pi/3}}{1-q} \left(a^{1-q} - \rho^{1-q}\right).$$

Letting $\rho \downarrow 0$ proves the claim. □

Corollary 3.40 (Summable subcritical inverse-clearance mass)**.** *For each $N \ge 2$, let $L_N \ge 1$, and let $Y_{N,1}, \dots, Y_{N,L_N}$ be defined on a common probability space, each with the same marginal law as Y_N from Corollary 3.32. Put $M_{N,\ell} = Y^{-1}_{N,\ell}$. For $s \ge 0$, put*

$$Z_N(s) := \sum_{\ell=1}^{L_N} \mathbf{1}\{M_{N,\ell} \le s\}.$$

For $0 < q < 1$ and $a \ge 0$, define the subcritical inverse-clearance mass

$$S^{(q)}_N(a) := \sum_{\ell=1}^{L_N} M^{-q}_{N,\ell} \mathbf{1}\{M_{N,\ell} \le a\}.$$

Then, for $0 \le a \le 1$,

$$\mathbb{E} S^{(q)}_N(a) \le C_q L_N a^{1-q},$$

where $C_q < \infty$ depends only on q. More sharply, if $L_N \equiv L$, then for every fixed $L \geq 1$ and $0 < a < \infty$,

$$\limsup_{N\to\infty} \mathbb{E}S_N^{(q)}(a) \leq \frac{L\sqrt{\pi/3}}{1-q}\, a^{1-q}.$$

Consequently, for any deterministic $b_N > 0$, if $0 \leq a_N \leq 1$ and

$$\sum_{N=2}^{\infty} \frac{L_N a_N^{1-q}}{b_N} < \infty,$$

then

$$\mathbb{E}\sum_{N=2}^{\infty} \frac{S_N^{(q)}(a_N)}{b_N} < \infty, \qquad \sum_{N=2}^{\infty} \frac{S_N^{(q)}(a_N)}{b_N} < \infty \quad \text{almost surely.}$$

In particular $S_N^{(q)}(a_N) = o(b_N)$ almost surely. If $L_N \leq C_L N^{\beta}$ and $b_N \geq c_b N^{-\delta}$, then for every $\varepsilon > 0$,

$$S_N^{(q)}\Big(N^{-(\beta+\delta+1)/(1-q)-\varepsilon}\Big) = o\Big(N^{-\delta}\Big) \qquad \text{almost surely.}$$

For actual Gaussian symbols this says that the total subcritical inverse mass of symbols with

$$\min_{z\in\mathbb{T}} |G_m^{(\ell)}(z)| \leq N^{-(\beta+\delta+1)/(1-q)-1/2-\varepsilon}$$

is $o(N^{-\delta})$ after the natural $N^{-q/2}$ scaling. No independence is required within a batch or across N.

Proof. The case $a = 0$ is trivial. For $x > 0$ and $a > 0$,

$$x^{-q}\mathbf{1}\{x \leq a\} = a^{-q}\mathbf{1}\{x \leq a\} + q\int_0^a s^{-q-1}\mathbf{1}\{x \leq s\}\, ds,$$

with the evident zero value when $a = 0$. Therefore Tonelli's theorem gives

$$\mathbb{E}S_N^{(q)}(a) = a^{-q}\mathbb{E}Z_N(a) + q\int_0^a s^{-q-1}\mathbb{E}Z_N(s)\, ds.$$

For $a \leq 1$, Corollary 3.36 gives $\mathbb{E}Z_N(s) \leq CL_N s$, hence

$$\mathbb{E}S_N^{(q)}(a) \leq CL_N a^{1-q} + CqL_N\int_0^a s^{-q}\, ds \leq C_q L_N a^{1-q}.$$

For the sharp fixed-L bound, put $K_0 = \sqrt{\pi/3}$ and fix $0 < \rho < \min\{a, 1\}$. On $(0, \rho)$, use the uniform bound $\mathbb{E}Z_N(s) \leq CLs$. On $[\rho, a]$, Corollary 3.36 gives

$$\limsup_{N\to\infty} \sup_{\rho\leq s\leq a} \left(\mathbb{E}Z_N(s) - LK_0 s\right)_+ = 0.$$

Taking the limsup in N in the preceding layer-cake formula yields

$$\limsup_{N\to\infty} \mathbb{E}S_N^{(q)}(a) \leq LK_0 a^{1-q} + \frac{qCL}{1-q}\rho^{1-q} + \frac{qLK_0}{1-q}\left(a^{1-q} - \rho^{1-q}\right).$$

Letting $\rho \downarrow 0$ gives the sharp coefficient.

The normalized summability assertion follows by monotone convergence:

$$\mathbb{E}\sum_{N=2}^{\infty}\frac{S_N^{(q)}(a_N)}{b_N}=\sum_{N=2}^{\infty}\frac{\mathbb{E}S_N^{(q)}(a_N)}{b_N}\le C_q\sum_{N=2}^{\infty}\frac{L_N a_N^{1-q}}{b_N}<\infty.$$

Thus the nonnegative normalized sum is finite almost surely, so its terms tend to zero. The polynomial specialization takes

$$a_N=N^{-(\beta+\delta+1)/(1-q)-\varepsilon}.$$

Finally $M_{N,\ell}=\sqrt{N}\min_{\mathbb{T}}|G_m^{(\ell)}|$ gives the stated unscaled interpretation. □

Corollary 3.41 (Summable Stieltjes inverse-clearance gauge mass)**.** *For each $N\ge 2$, let $L_N\ge 1$, and let $Y_{N,1},\dots,Y_{N,L_N}$ be defined on a common probability space, each with the same marginal law as Y_N from Corollary 3.32. Put $M_{N,\ell}=Y_{N,\ell}^{-1}$. Let $a>0$, and let $\psi:(0,a]\to[0,\infty)$ be nonincreasing and right-continuous. Let ν_ψ be the positive Stieltjes measure on $(0,a]$ determined by*

$$\nu_\psi((u,v])=\psi(u)-\psi(v)\qquad(0<u<v\le a),$$

and suppose

$$B_\psi(a):=a\psi(a)+\int_{(0,a]}s\,d\nu_\psi(s)<\infty.$$

Define

$$T_N^{\psi}(a):=\sum_{\ell=1}^{L_N}\psi(M_{N,\ell})\mathbf{1}\{M_{N,\ell}\le a\}.$$

Then

$$\mathbb{E}T_N^{\psi}(a)\le CL_NB_\psi(a),$$

with an absolute constant $C<\infty$. More sharply, if $L_N\equiv L$, then for every fixed $L\ge 1$, $a<\infty$, and such ψ,

$$\limsup_{N\to\infty}\mathbb{E}T_N^{\psi}(a)\le L\sqrt{\frac{\pi}{3}}\,B_\psi(a).$$

Consequently, for deterministic $b_N>0$, if $\psi_N:(0,a_N]\to[0,\infty)$ are nonincreasing right-continuous gauges, with associated Stieltjes measures ν_{ψ_N} and $B_{\psi_N}(a_N)<\infty$, and

$$\sum_{N=2}^{\infty}\frac{L_NB_{\psi_N}(a_N)}{b_N}<\infty,$$

then

$$\mathbb{E}\sum_{N=2}^{\infty}\frac{T_N^{\psi_N}(a_N)}{b_N}<\infty,\qquad\sum_{N=2}^{\infty}\frac{T_N^{\psi_N}(a_N)}{b_N}<\infty\quad\text{almost surely.}$$

In particular $T_N^{\psi_N}(a_N)=o(b_N)$ almost surely. In the absolutely continuous case $d\nu_\psi(s)=(-\psi'(s))\,ds$. The choice $\psi(s)=s^{-q}$, $0<q<1$, gives $B_\psi(a)=a^{1-q}/(1-q)$ and recovers the subcritical power scale. The choices $\psi(s)=s^{-q}(\log(e/s))^{\gamma}$, $0<q<1$, $\gamma\ge 0$, and $\psi(s)=(\log(e/s))^{\gamma}$, $\gamma>0$, give

the corresponding logarithmic refinements with $B_\psi(a) \asymp_{q,\gamma} a^{1-q}(\log(e/a))^\gamma$ *and* $B_\psi(a) \asymp_\gamma a(\log(e/a))^\gamma$*, respectively, for* $0 < a \le 1$.

Proof. For $s \ge 0$, put

$$Z_N(s) := \sum_{\ell=1}^{L_N} \mathbf{1}\{M_{N,\ell} \le s\}.$$

For $x > 0$,

$$\psi(x)\mathbf{1}\{x \le a\} = \psi(a)\mathbf{1}\{x \le a\} + \int_{(0,a]} \mathbf{1}\{x < s\}\, d\nu_\psi(s).$$

Tonelli's theorem therefore gives

$$\mathbb{E}T_N^\psi(a) \le \psi(a)\mathbb{E}Z_N(a) + \int_{(0,a]} \mathbb{E}Z_N(s)\, d\nu_\psi(s).$$

The uniform count estimate in Corollary 3.36 gives $\mathbb{E}Z_N(s) \le CL_N s$, hence

$$\mathbb{E}T_N^\psi(a) \le CL_N \left(a\psi(a) + \int_{(0,a]} s\, d\nu_\psi(s) \right).$$

For the sharp fixed-L estimate, set $K_0 = \sqrt{\pi/3}$ and fix $0 < \rho < a$. Corollary 3.36 gives

$$\varepsilon_N(\rho, a) := \sup_{\rho \le s \le a} (\mathbb{E}Z_N(s) - LK_0 s)_+ \longrightarrow 0.$$

Since $\nu_\psi([\rho, a]) < \infty$, the error term contributes at most $\varepsilon_N(\rho, a)\nu_\psi([\rho, a])$, and the preceding bound yields

$$\limsup_{N\to\infty} \mathbb{E}T_N^\psi(a) \le LK_0 a\psi(a) + CL \int_{(0,\rho)} s\, d\nu_\psi(s) + LK_0 \int_{[\rho,a]} s\, d\nu_\psi(s).$$

Letting $\rho \downarrow 0$ and using $B_\psi(a) < \infty$ proves the sharp bound. Monotone convergence gives the normalized summability statement exactly as in Corollary 3.40. The displayed examples follow by direct one-variable integration. □

Corollary 3.42 (Inverse-gauge confidence budgets)**.** *In the setting of Corollary 3.41, let*

$$Q_N^\psi(\alpha) := \inf\{r \ge 0 : \mathbb{P}\{T_N^\psi(a) > r\} \le \alpha\}, \qquad 0 < \alpha \le 1.$$

Then

$$Q_N^\psi(\alpha) \le \frac{CL_N B_\psi(a)}{\alpha} \qquad (N \ge 2,\ 0 < \alpha \le 1),$$

with an absolute constant $C < \infty$*. More sharply, if* $L_N \equiv L$*, then for every fixed* $L \ge 1$*,* $a < \infty$*, gauge* ψ*, and* $0 < \alpha \le 1$*,*

$$\limsup_{N\to\infty} Q_N^\psi(\alpha) \le \frac{L\sqrt{\pi/3}\, B_\psi(a)}{\alpha}.$$

Consequently, for a triangular family ψ_N, a_N as in Corollary 3.41 and any $0 < \eta_N \le 1$,

$$\mathbb{P}\left\{T_N^{\psi_N}(a_N) > \frac{CL_N B_{\psi_N}(a_N)}{\eta_N}\right\} \le \eta_N.$$

For every finite index set I, the preceding bounds hold simultaneously for all $N \in I$ outside an event of probability at most $\sum_{N\in I}\eta_N$. If $\sum_N \eta_N < \infty$, then almost surely, for all sufficiently large N,

$$T_N^{\psi_N}(a_N) \le \frac{CL_N B_{\psi_N}(a_N)}{\eta_N}.$$

No independence is required in the batch or across N.

Proof. Markov's inequality and Corollary 3.41 give

$$\mathbb{P}\{T_N^{\psi}(a) > r\} \le \frac{CL_N B_{\psi}(a)}{r},$$

which proves the uniform quantile estimate after taking $r = CL_N B_\psi(a)/\alpha$, with C enlarged if necessary. If $L_N \equiv L$, the sharp expectation bound in Corollary 3.41 gives, for every $\varepsilon > 0$,

$$\limsup_{N\to\infty} \mathbb{P}\left\{T_N^{\psi}(a) > \frac{L\sqrt{\pi/3}\, B_\psi(a) + \varepsilon}{\alpha}\right\} < \alpha.$$

Thus $Q_N^{\psi}(\alpha) \le (L\sqrt{\pi/3}\, B_\psi(a)+\varepsilon)/\alpha$ for all sufficiently large N; letting $\varepsilon \downarrow 0$ proves the fixed-L quantile coefficient.

The displayed one-step confidence bound is the same Markov estimate with $r = CL_N B_{\psi_N}(a_N)/\eta_N$. The finite-horizon statement is the union bound, and the almost-sure eventual statement is the first Borel–Cantelli lemma. □

Corollary 3.43 (Simultaneous inverse-gauge menu budgets)**.** *For each $N \ge 2$, let $L_N \ge 1$, and let $Y_{N,1}, \dots, Y_{N,L_N}$ have the common marginal law from Corollary 3.32, with no independence assumption. Put $M_{N,\ell} = Y_{N,\ell}^{-1}$. Let J_N be a finite or countable deterministic index set. For each $j \in J_N$, let $\psi_{N,j} : (0, a_{N,j}] \to [0,\infty)$ be a nonincreasing right-continuous gauge with Stieltjes budget $B_{N,j} := B_{\psi_{N,j}}(a_{N,j}) < \infty$, and put*

$$T_{N,j} := \sum_{\ell=1}^{L_N} \psi_{N,j}(M_{N,\ell})\mathbf{1}\{M_{N,\ell} \le a_{N,j}\}.$$

There is an absolute constant $C < \infty$ such that, for any finite set $\mathcal{I} \subset \{(N,j) : N \ge 2,\ j \in J_N\}$ and any $0 < \eta_{N,j} \le 1$,

$$\mathbb{P}\left\{\exists (N,j) \in \mathcal{I} : T_{N,j} > \frac{CL_N B_{N,j}}{\eta_{N,j}}\right\} \le \sum_{(N,j)\in\mathcal{I}} \eta_{N,j}.$$

If

$$\sum_{N=2}^{\infty}\sum_{j\in J_N} \eta_{N,j} < \infty,$$

then almost surely, for all but finitely many pairs (N, j),

$$T_{N,j} \le \frac{CL_N B_{N,j}}{\eta_{N,j}}.$$

In particular, for a single N, *a finite menu* J_N, *and* $0 < \alpha \le 1$, *choosing* $\eta_{N,j} = \alpha/|J_N|$ *gives simultaneous control of the whole menu with failure probability at most* α*:*

$$T_{N,j} \le \frac{C|J_N|L_N B_{N,j}}{\alpha} \qquad (j \in J_N).$$

Thus power, logarithmic, and jump gauges may be tested on the same batch with only the explicit union-budget cost.

Proof. Apply Corollary 3.42 to each (N, j), with confidence parameter $\eta_{N,j}$, and sum the resulting probability bounds over $\mathcal{I}$. This proves the finite-menu assertion. For the countable almost-sure statement, enumerate the countable set of pairs (N, j) with $\eta_{N,j} > 0$. The displayed summability makes the expected number of failures finite, equivalently the sum of their probabilities finite; the first Borel–Cantelli lemma gives only finitely many failures almost surely. The final finite-menu specialization is the same bound with $\eta_{N,j} = \alpha/|J_N|$. □

Corollary 3.44 (Aggregate inverse-gauge menu mass)**.** *In the setting of Corollary 3.43, set*

$$\mathcal{T}_N := \sum_{j \in J_N} T_{N,j}, \qquad \mathcal{B}_N := \sum_{j \in J_N} B_{N,j}.$$

If $\mathcal{B}_N < \infty$, *then*

$$\mathbb{E}\mathcal{T}_N \le CL_N\mathcal{B}_N$$

with an absolute constant $C < \infty$. *More sharply, if* $L_N \equiv L$ *and the menu* J, *gauges* ψ_j, *windows* a_j, *and budgets* B_j *are fixed with* $\sum_{j \in J} B_j < \infty$, *then*

$$\limsup_{N \to \infty} \mathbb{E} \sum_{j \in J} T_{N,j} \le L\sqrt{\frac{\pi}{3}} \sum_{j \in J} B_j.$$

Consequently, for any deterministic $b_N > 0$, *if*

$$\sum_{N=2}^{\infty} \frac{L_N \mathcal{B}_N}{b_N} < \infty,$$

then

$$\mathbb{E} \sum_{N=2}^{\infty} \frac{\mathcal{T}_N}{b_N} < \infty, \qquad \sum_{N=2}^{\infty} \frac{\mathcal{T}_N}{b_N} < \infty \quad \textit{almost surely.}$$

In particular $\mathcal{T}_N = o(b_N)$ *almost surely. Thus a countable menu of power, logarithmic, and jump gauges may be aggregated as a single endpoint diagnostic whenever its Stieltjes budgets are summable.*

Proof. Apply Corollary 3.41 to each j and sum. Monotone convergence justifies the countable sum:

$$\mathbb{E}\mathcal{T}_N = \sum_{j\in J_N} \mathbb{E}T_{N,j} \le CL_N \sum_{j\in J_N} B_{N,j}.$$

For the sharp fixed-L statement, enumerate J as $\{j_1, j_2, \dots\}$, with the evident finite truncation when J is finite. For a finite initial segment, the sharp bound in Corollary 3.41 gives

$$\limsup_{N\to\infty} \mathbb{E} \sum_{m=1}^{M} T_{N,j_m} \le L\sqrt{\frac{\pi}{3}} \sum_{m=1}^{M} B_{j_m}.$$

The remaining tail has expectation at most

$$CL \sum_{m>M} B_{j_m}$$

uniformly in N. Letting $M \to \infty$ proves the sharp countable-menu coefficient. The normalized summability assertion follows from the first display and monotone convergence, and finiteness of the nonnegative series implies its terms tend to zero. □

Corollary 3.45 (Summable finite-family clearance schedules)**.** *For each $N \ge 2$, let $L_N \ge 1$, and let $Y_{N,1}, \dots, Y_{N,L_N}$ be defined on a common probability space, each with the same marginal law as Y_N from Corollary 3.32. Put $M_{N,\ell} = Y_{N,\ell}^{-1}$,*

$$M_{N,L_N,*} := \min_{1\le\ell\le L_N} M_{N,\ell}, \qquad Y^*_{N,L_N} := \max_{1\le\ell\le L_N} Y_{N,\ell}.$$

If $0 < a_N \le 1$ and

$$\sum_{N=2}^{\infty} L_N a_N < \infty,$$

then

$$\mathbb{P}\{M_{N,L_N,*} \le a_N \text{ for infinitely many } N\} = 0.$$

Equivalently, almost surely $Y^*_{N,L_N} < a_N^{-1}$ for all sufficiently large N. In particular, if $L_N \le C_L N^\beta$, then for every $\varepsilon > 0$,

$$M_{N,L_N,*} > N^{-\beta-1-\varepsilon}$$

eventually almost surely. For actual Gaussian symbols this means

$$\min_{1\le\ell\le L_N} \min_{z\in\mathbb{T}} |G_m^{(\ell)}(z)| > N^{-\beta-3/2-\varepsilon}$$

eventually almost surely, with no independence required among the symbols or across N.

Proof. As in Corollary 3.39, there is an absolute C_0 such that, for $0 < s \le 1$,

$$\mathbb{P}\{M_{N,L_N,*} \le s\} \le C_0 L_N s.$$

Thus

$$\sum_{N=2}^{\infty} \mathbb{P}\{M_{N,L_N,*} \le a_N\} \le C_0 \sum_{N=2}^{\infty} L_N a_N < \infty.$$

The first Borel–Cantelli lemma gives the summable-schedule statement. The polynomial example follows by taking $a_N = N^{-\beta-1-\varepsilon}$, since then $\sum_N L_N a_N \le C_L \sum_N N^{-1-\varepsilon} < \infty$. Dividing the scaled clearance $M_{N,L_N,*}$ by $\sqrt{N}$ gives the final unscaled form. □

Corollary 3.46 (Summable total bad-clearance mass)**.** *For each $N \ge 2$, let $L_N \ge 1$, and let $Y_{N,1}, \dots, Y_{N,L_N}$ be defined on a common probability space, each with the same marginal law as Y_N from Corollary 3.32. Put $M_{N,\ell} = Y_{N,\ell}^{-1}$ and*

$$Z_N(s) := \sum_{\ell=1}^{L_N} \mathbf{1}\{M_{N,\ell} \le s\}.$$

If $0 < a_N \le 1$ and

$$\sum_{N=2}^{\infty} L_N a_N < \infty,$$

then

$$\mathbb{E} \sum_{N=2}^{\infty} Z_N(a_N) < \infty \qquad \text{and hence} \qquad \sum_{N=2}^{\infty} Z_N(a_N) < \infty \quad \text{almost surely.}$$

Equivalently, only finitely many individual pairs (N, ℓ) have $M_{N,\ell} \le a_N$, almost surely. In particular, if $L_N \le C_L N^{\beta}$, then for every $\varepsilon > 0$,

$$\sum_{N=2}^{\infty} \sum_{\ell=1}^{L_N} \mathbf{1}\{M_{N,\ell} \le N^{-\beta-1-\varepsilon}\} < \infty \qquad \text{almost surely.}$$

For actual Gaussian symbols, this says that only finitely many individual (N, ℓ) satisfy

$$\min_{z \in \mathbb{T}} |G_m^{(\ell)}(z)| \le N^{-\beta-3/2-\varepsilon}.$$

No independence is required within a batch or across N.

Proof. Corollary 3.36 gives an absolute constant C such that, for $0 < s \le 1$,

$$\mathbb{E} Z_N(s) \le C L_N s.$$

Therefore

$$\mathbb{E} \sum_{N=2}^{\infty} Z_N(a_N) = \sum_{N=2}^{\infty} \mathbb{E} Z_N(a_N) \le C \sum_{N=2}^{\infty} L_N a_N < \infty,$$

where monotone convergence justifies the exchange. A nonnegative random variable with finite expectation is finite almost surely. The polynomial specialization takes $a_N = N^{-\beta-1-\varepsilon}$, and the unscaled Gaussian form divides the scaled clearance by $\sqrt{N}$. □

Corollary 3.47 (Summable clearance-deficit mass)**.** *In the setting of Corollary 3.46, define the total deficit below level $a \geq 0$ by*

$$D_N(a) := \sum_{\ell=1}^{L_N} (a - M_{N,\ell})_+.$$

For $0 \leq a \leq 1$,

$$\mathbb{E}D_N(a) \leq CL_N a^2$$

with an absolute constant $C < \infty$. More sharply, for every fixed $L \geq 1$ and $K < \infty$, if $L_N \equiv L$, then

$$\limsup_{N\to\infty} \sup_{0\leq a\leq K} \left(\mathbb{E}D_N(a) - \frac{L}{2}\sqrt{\frac{\pi}{3}}\, a^2\right)_+ = 0.$$

Consequently, if $0 \leq a_N \leq 1$ and

$$\sum_{N=2}^{\infty} L_N a_N^2 < \infty,$$

then

$$\mathbb{E}\sum_{N=2}^{\infty} D_N(a_N) < \infty \qquad \text{and hence} \qquad \sum_{N=2}^{\infty} D_N(a_N) < \infty \quad \text{almost surely.}$$

In particular, if $L_N \leq C_L N^\beta$, then for every $\varepsilon > 0$,

$$\sum_{N=2}^{\infty}\sum_{\ell=1}^{L_N} \left(N^{-(\beta+1)/2-\varepsilon} - M_{N,\ell}\right)_+ < \infty \qquad \text{almost surely.}$$

For actual Gaussian symbols this is equivalently

$$\sum_{N=2}^{\infty}\sum_{\ell=1}^{L_N} \sqrt{N}\left(N^{-(\beta+2)/2-\varepsilon} - \min_{z\in\mathbb{T}} |G_m^{(\ell)}(z)|\right)_+ < \infty \qquad \text{almost surely.}$$

Thus one can sum shortfall sizes below a much larger clearance threshold than the threshold for summing the bad-clearance indicators themselves. No independence is required within a batch or across N.

Proof. For each $a \geq 0$,

$$(a - M_{N,\ell})_+ = \int_0^a \mathbf{1}\{M_{N,\ell} \leq s\}\, ds,$$

and hence Tonelli's theorem gives

$$\mathbb{E}D_N(a) = \int_0^a \mathbb{E}Z_N(s)\, ds.$$

Corollary 3.36 gives $\mathbb{E}Z_N(s) \leq CL_N s$ for $0 \leq s \leq 1$, which proves $\mathbb{E}D_N(a) \leq CL_N a^2$ for $0 \leq a \leq 1$, after changing C.

For the sharp fixed-L statement, put $K_0 = \sqrt{\pi/3}$. By Corollary 3.36,

$$\varepsilon_N(K) := \sup_{0\leq s\leq K} (\mathbb{E}Z_N(s) - LK_0 s)_+ \longrightarrow 0.$$

Therefore, uniformly for $0 \le a \le K$,

$$\mathbb{E}D_N(a) \le \int_0^a (LK_0 s + \varepsilon_N(K))\, ds \le \frac{LK_0}{2} a^2 + K\varepsilon_N(K).$$

Taking the limsup proves the displayed sharp bound.

If $\sum_N L_N a_N^2 < \infty$, then monotone convergence and the uniform deficit estimate give

$$\mathbb{E}\sum_{N=2}^{\infty} D_N(a_N) = \sum_{N=2}^{\infty} \mathbb{E}D_N(a_N) \le C \sum_{N=2}^{\infty} L_N a_N^2 < \infty.$$

Thus the nonnegative sum is finite almost surely. The polynomial example takes $a_N = N^{-(\beta+1)/2-\varepsilon}$, so $L_N a_N^2 \le C_L N^{-1-2\varepsilon}$. Finally

$$M_{N,\ell} = \sqrt{N} \min_{z\in\mathbb{T}} |G_m^{(\ell)}(z)|$$

turns the scaled deficit into the displayed unscaled Gaussian form. □

Corollary 3.48 (Normalized clearance-deficit budgets)**.** *In the setting of Corollary 3.47, let $b_N > 0$ be deterministic. If $0 \le a_N \le 1$ and*

$$\sum_{N=2}^{\infty} \frac{L_N a_N^2}{b_N} < \infty,$$

then

$$\mathbb{E}\sum_{N=2}^{\infty} \frac{D_N(a_N)}{b_N} < \infty, \qquad \sum_{N=2}^{\infty} \frac{D_N(a_N)}{b_N} < \infty \quad \text{almost surely.}$$

In particular,

$$D_N(a_N) = o(b_N) \qquad \text{almost surely.}$$

Equivalently, the total shortfall below a_N is eventually negligible relative to any deterministic budget b_N whose reciprocal still leaves $\sum_N L_N a_N^2/b_N$ summable. If $L_N \le C_L N^{\beta}$ and $b_N \ge c_b N^{-\delta}$ with $\delta \ge 0$, then for every $\varepsilon > 0$,

$$D_N\Big(N^{-(\beta+\delta+1)/2-\varepsilon}\Big) = o\Big(N^{-\delta}\Big) \qquad \text{almost surely.}$$

For actual Gaussian symbols this becomes

$$\sum_{\ell=1}^{L_N} \sqrt{N}\left(N^{-(\beta+\delta+2)/2-\varepsilon} - \min_{z\in\mathbb{T}} |G_m^{(\ell)}(z)|\right)_+ = o\Big(N^{-\delta}\Big) \qquad \text{almost surely.}$$

No independence is required within a batch or across N.

Proof. Corollary 3.47 gives $\mathbb{E}D_N(a_N) \le CL_N a_N^2$. Hence, by monotone convergence,

$$\mathbb{E}\sum_{N=2}^{\infty} \frac{D_N(a_N)}{b_N} = \sum_{N=2}^{\infty} \frac{\mathbb{E}D_N(a_N)}{b_N} \le C \sum_{N=2}^{\infty} \frac{L_N a_N^2}{b_N} < \infty.$$

The nonnegative normalized sum is therefore finite almost surely, which implies $D_N(a_N)/b_N \to 0$. For the polynomial specialization take

$$a_N = N^{-(\beta+\delta+1)/2-\varepsilon}.$$

Then

$$\frac{L_N a_N^2}{b_N} \le \frac{C_L}{c_b} N^{-1-2\varepsilon},$$

which is summable. Finally

$$M_{N,\ell} = \sqrt{N} \min_{z\in\mathbb{T}} |G_m^{(\ell)}(z)|$$

gives the unscaled Gaussian form. □

Corollary 3.49 (Powered clearance-deficit budgets)**.** *In the setting of Corollary 3.47, let $p > 0$ and define*

$$D_N^{(p)}(a) := \sum_{\ell=1}^{L_N} (a - M_{N,\ell})_+^p.$$

For $0 \le a \le 1$,

$$\mathbb{E} D_N^{(p)}(a) \le C_p L_N a^{p+1}$$

with a constant $C_p < \infty$ depending only on p. More sharply, for every fixed $L \ge 1$ and $K < \infty$, if $L_N \equiv L$, then

$$\limsup_{N\to\infty} \sup_{0\le a\le K} \left(\mathbb{E} D_N^{(p)}(a) - \frac{L}{p+1}\sqrt{\frac{\pi}{3}}\, a^{p+1} \right)_+ = 0.$$

Consequently, for any deterministic $b_N > 0$, if $0 \le a_N \le 1$ and

$$\sum_{N=2}^{\infty} \frac{L_N a_N^{p+1}}{b_N} < \infty,$$

then

$$\mathbb{E} \sum_{N=2}^{\infty} \frac{D_N^{(p)}(a_N)}{b_N} < \infty, \qquad \sum_{N=2}^{\infty} \frac{D_N^{(p)}(a_N)}{b_N} < \infty \quad \text{almost surely,}$$

and hence $D_N^{(p)}(a_N) = o(b_N)$ almost surely. If $L_N \le C_L N^{\beta}$ and $b_N \ge c_b N^{-\delta}$, then for every $\varepsilon > 0$,

$$D_N^{(p)}\left(N^{-(\beta+\delta+1)/(p+1)-\varepsilon}\right) = o\left(N^{-\delta}\right) \qquad \text{almost surely.}$$

For actual Gaussian symbols this is equivalently

$$\sum_{\ell=1}^{L_N} N^{p/2} \left(N^{-(\beta+\delta+1)/(p+1)-1/2-\varepsilon} - \min_{z\in\mathbb{T}} |G_m^{(\ell)}(z)| \right)_+^p = o\left(N^{-\delta}\right) \qquad \text{almost surely.}$$

The case $p = 1$ recovers Corollary 3.48. No independence is required within a batch or across N.

Proof. For $p > 0$ and $a \ge 0$,

$$(a - M_{N,\ell})_+^p = p \int_0^a (a-s)^{p-1} \mathbf{1}\{M_{N,\ell} \le s\}\, ds.$$

Thus Tonelli's theorem gives

$$\mathbb{E} D_N^{(p)}(a) = p \int_0^a (a-s)^{p-1} \mathbb{E} Z_N(s)\, ds.$$

For $0 \le a \le 1$, Corollary 3.36 gives $\mathbb{E}Z_N(s) \le CL_N s$. Therefore

$$\mathbb{E}D_N^{(p)}(a) \le CL_N p \int_0^a (a-s)^{p-1} s\, ds = \frac{CL_N}{p+1} a^{p+1},$$

which is the uniform estimate after absorbing constants into C_p.

For the sharp fixed-L statement, put $K_0 = \sqrt{\pi/3}$. As in Corollary 3.47, let

$$\varepsilon_N(K) := \sup_{0 \le s \le K} (\mathbb{E}Z_N(s) - LK_0 s)_+ \longrightarrow 0.$$

Then, uniformly for $0 \le a \le K$,

$$\begin{aligned}\mathbb{E}D_N^{(p)}(a) &\le p \int_0^a (a-s)^{p-1} (LK_0 s + \varepsilon_N(K))\, ds \\ &= \frac{LK_0}{p+1} a^{p+1} + \varepsilon_N(K) a^p \le \frac{LK_0}{p+1} a^{p+1} + \varepsilon_N(K) K^p.\end{aligned}$$

Taking the limsup proves the sharp coefficient.

The normalized summability assertion follows by monotone convergence:

$$\mathbb{E} \sum_{N=2}^{\infty} \frac{D_N^{(p)}(a_N)}{b_N} = \sum_{N=2}^{\infty} \frac{\mathbb{E}D_N^{(p)}(a_N)}{b_N} \le C_p \sum_{N=2}^{\infty} \frac{L_N a_N^{p+1}}{b_N} < \infty.$$

Hence the nonnegative normalized sum is finite almost surely, so its terms tend to zero. For the polynomial specialization take

$$a_N = N^{-(\beta+\delta+1)/(p+1)-\varepsilon}.$$

Then

$$\frac{L_N a_N^{p+1}}{b_N} \le \frac{C_L}{c_b} N^{-1-(p+1)\varepsilon},$$

which is summable. Finally

$$M_{N,\ell} = \sqrt{N} \min_{z \in \mathbb{T}} |G_m^{(\ell)}(z)|$$

gives the unscaled Gaussian form. □

Corollary 3.50 (Jump-weighted clearance-deficit functionals)**.** *In the setting of Corollary 3.47, let $\Phi : [0, K] \to [0, \infty)$ be nondecreasing, right-continuous on $(0, K]$, and satisfy $\Phi(0) = 0$. Put*

$$\Delta_0 \Phi := \lim_{u \downarrow 0} \Phi(u),$$

and let μ_Φ be the Stieltjes measure on $(0, K]$ determined by $\mu_\Phi((0, v]) = \Phi(v) - \Delta_0\Phi$ and $\mu_\Phi((u, v]) = \Phi(v) - \Phi(u)$ for $0 < u < v$. For $0 \le a \le K$, define

$$W_N^\Phi(a) := \sum_{\ell=1}^{L_N} \Phi((a - M_{N,\ell})_+)$$

and

$$H_\Phi(a) := a\, \Delta_0\Phi + \int_{(0,a]} (a-u)\, d\mu_\Phi(u).$$

If $0 \le a \le \min\{K, 1\}$, *then*

$$\mathbb{E}W_N^{\Phi}(a) \le CL_N H_{\Phi}(a)$$

with an absolute constant $C < \infty$. *More sharply, for every fixed* $L \ge 1$ *and* $K < \infty$, *if* $L_N \equiv L$, *then*

$$\limsup_{N\to\infty} \sup_{0\le a\le K} \left(\mathbb{E}W_N^{\Phi}(a) - L\sqrt{\frac{\pi}{3}}\, H_{\Phi}(a)\right)_+ = 0.$$

Consequently, let $0 < a_N \le 1$, $b_N > 0$, *and let* $\Phi_N : [0, a_N] \to [0, \infty)$ *be nondecreasing, right-continuous on* $(0, a_N]$, *and satisfy* $\Phi_N(0) = 0$. *Let* $\Delta_N = \lim_{u\downarrow 0} \Phi_N(u)$, *let* μ_N *be the corresponding Stieltjes measure on* $(0, a_N]$, *and put*

$$H_N := a_N\Delta_N + \int_{(0,a_N]} (a_N - u)\, d\mu_N(u).$$

If

$$\sum_{N=2}^{\infty} \frac{L_N H_N}{b_N} < \infty,$$

then

$$\sum_{N=2}^{\infty} \frac{1}{b_N} \sum_{\ell=1}^{L_N} \Phi_N\big((a_N - M_{N,\ell})_+\big) < \infty \qquad \textit{almost surely.}$$

Thus any summable weighted shortfall budget follows from the same one-dimensional occupation integral, now allowing an entrance jump at zero as well as later jump penalties and smooth weights. If $\Delta_0\Phi = 0$ *and* Φ *is absolutely continuous, then*

$$H_{\Phi}(a) = \int_0^a \Phi'(a - s)s\, ds.$$

Taking $\Phi(u) = u^p$ *recovers Corollary 3.49; taking* $\Phi(u) = u$ *recovers the first-power deficit estimates; and taking* $\Phi(u) = \mathbf{1}_{\{u>0\}}$ *recovers the strict bad-clearance count scale.*

Proof. For $0 \le x \le K$,

$$\Phi(x) = \Delta_0\Phi\, \mathbf{1}\{x > 0\} + \int_{(0,K]} \mathbf{1}\{u \le x\}\, d\mu_{\Phi}(u).$$

Therefore, for $0 \le a \le K$,

$$\Phi\big((a - M_{N,\ell})_+\big) = \Delta_0\Phi\, \mathbf{1}\{M_{N,\ell} < a\} + \int_{(0,a]} \mathbf{1}\{M_{N,\ell} \le a - u\}\, d\mu_{\Phi}(u).$$

Summing over ℓ and applying Tonelli's theorem gives

$$\mathbb{E}W_N^{\Phi}(a) \le \Delta_0\Phi\, \mathbb{E}Z_N(a) + \int_{(0,a]} \mathbb{E}Z_N(a - u)\, d\mu_{\Phi}(u).$$

For $a \le 1$, Corollary 3.36 gives $\mathbb{E}Z_N(s) \le CL_N s$, which proves the uniform estimate.

For the sharp fixed-L statement, put $K_0 = \sqrt{\pi/3}$ and

$$\varepsilon_N(K) := \sup_{0\le s\le K} (\mathbb{E}Z_N(s) - LK_0 s)_+ \longrightarrow 0.$$

Then, uniformly for $0 \le a \le K$,

$$\begin{aligned}\mathbb{E}W_N^{\Phi}(a) &\le LK_0 H_\Phi(a) + \varepsilon_N(K)(\Delta_0\Phi + \mu_\Phi((0,a])) \\ &\le LK_0 H_\Phi(a) + \varepsilon_N(K)\Phi(K).\end{aligned}$$

Taking the limsup proves the sharp bound.

The schedule statement follows by applying the uniform estimate with $\Phi = \Phi_N$ and $a = a_N$, then using monotone convergence:

$$\begin{aligned}\mathbb{E}\sum_{N=2}^{\infty}\frac{1}{b_N}\sum_{\ell=1}^{L_N}\Phi_N((a_N - M_{N,\ell})_+) &\le C\sum_{N=2}^{\infty}\frac{L_N H_N}{b_N} \\ &< \infty.\end{aligned}$$

The nonnegative sum is therefore finite almost surely. The smooth specializations follow by evaluating H_Φ; the indicator specialization has $\Delta_0\Phi = 1$ and $\mu_\Phi = 0$. □

Corollary 3.51 (Summable bad-count schedules)**.** *For each $N \ge 2$, let $L_N \ge 1$, and let $Y_{N,1}, \dots, Y_{N,L_N}$ be defined on a common probability space, each with the same marginal law as Y_N from Corollary 3.32. Put $M_{N,\ell} = Y_{N,\ell}^{-1}$ and*

$$Z_N(s) := \sum_{\ell=1}^{L_N}\mathbf{1}\{M_{N,\ell} \le s\}.$$

Let $1 \le r_N \le L_N$ be deterministic integers. If $0 < a_N \le 1$ and

$$\sum_{N=2}^{\infty}\frac{L_N a_N}{r_N} < \infty,$$

then

$$\mathbb{P}\{Z_N(a_N) \ge r_N \text{ for infinitely many } N\} = 0.$$

Equivalently, the r_N-th smallest clearance among the L_N symbols is larger than a_N for all sufficiently large N, almost surely. In particular, if $L_N \le C_L N^\beta$ and $r_N \ge c_r N^\gamma$ with $\gamma \ge 0$, then for every $\varepsilon > 0$,

$$Z_N\left(N^{-\beta+\gamma-1-\varepsilon}\right) < r_N$$

eventually almost surely. Thus a polynomially growing allowed bad-count budget raises the eventual simultaneous clearance threshold by the factor r_N/L_N, again without any independence assumption.

Proof. By Corollary 3.36, there is an absolute constant C such that, for $0 < s \le 1$,

$$\mathbb{P}\{Z_N(s) \ge r_N\} \le \frac{CL_N s}{r_N}.$$

Therefore

$$\sum_{N=2}^{\infty} \mathbb{P}\{Z_N(a_N) \geq r_N\} \leq C \sum_{N=2}^{\infty} \frac{L_N a_N}{r_N} < \infty.$$

The first Borel–Cantelli lemma gives the general assertion. For the polynomial consequence, take $a_N = N^{-\beta+\gamma-1-\varepsilon}$. Then

$$\frac{L_N a_N}{r_N} \leq \frac{C_L}{c_r} N^{-1-\varepsilon},$$

which is summable. □

Corollary 3.52 (Critical truncated inverse moment)**.** *With $Y_N = M_N^{-1}$ as in Corollary 3.32, for every $R \geq 1$,*

$$\sup_{N \geq 2} \mathbb{E}\big[(Y_N \wedge R) - 1\big]_+ \leq C \log R$$

with an absolute constant $C < \infty$. More sharply, for every fixed $1 \leq R < \infty$,

$$\limsup_{N \to \infty} \mathbb{E}\big[(Y_N \wedge R) - 1\big]_+ \leq \sqrt{\frac{\pi}{3}} \log R.$$

Equivalently, the critical $q = 1$ inverse moment grows at most logarithmically under truncation, while Corollary 3.31 gives strong L^q control for every $q < 1$.

Proof. For a nonnegative random variable Y,

$$\mathbb{E}\big[(Y \wedge R) - 1\big]_+ = \int_1^R \mathbb{P}\{Y > u\}\, du.$$

The uniform weak bound in Corollary 3.32 gives the first assertion. For the sharp finite-window coefficient, the second assertion of Corollary 3.32 implies that, for fixed R,

$$\varepsilon_N(R) := \sup_{1 \leq u \leq R} \left(u\, \mathbb{P}\{Y_N > u\} - \sqrt{\frac{\pi}{3}} \right)_+ \longrightarrow 0.$$

Hence

$$\mathbb{E}\big[(Y_N \wedge R) - 1\big]_+ \leq \left(\sqrt{\frac{\pi}{3}} + \varepsilon_N(R) \right) \log R,$$

and the limsup in N gives the claim. □

Corollary 3.53 (Logarithmic moments at the inverse endpoint)**.** *With $Y_N = M_N^{-1}$ as in Corollary 3.32, for every $p > 0$,*

$$\sup_{N \geq 2} \mathbb{E}\big[(\log_+ Y_N)^p\big] < \infty,$$

where $\log_+ u := \max\{\log u, 0\}$. More sharply,

$$\limsup_{N \to \infty} \mathbb{E}\big[(\log_+ Y_N)^p\big] \leq \sqrt{\frac{\pi}{3}}\, \Gamma(p+1).$$

Thus the weak-L^1 endpoint still gives full logarithmic integrability of the inverse scaled minimum, with the same sharp upper-tail constant.

Proof. For any positive random variable Y,

$$\mathbb{E}\big[(\log_+ Y)^p\big] = p\int_0^\infty t^{p-1}\mathbb{P}\{Y > e^t\}\,dt.$$

The uniform weak bound in Corollary 3.32 gives $\mathbb{P}\{Y_N > e^t\} \le Ce^{-t}$ for all $t \ge 0$, uniformly in N, which proves the first assertion.

For the sharp limsup, fix $T < \infty$. The finite-window part of Corollary 3.32, applied with $R = e^T$, gives

$$\limsup_{N\to\infty} p\int_0^T t^{p-1}\mathbb{P}\{Y_N > e^t\}\,dt \le \sqrt{\frac{\pi}{3}}\,p\int_0^T t^{p-1}e^{-t}\,dt.$$

The tail $t \ge T$ is bounded uniformly in N by

$$Cp\int_T^\infty t^{p-1}e^{-t}\,dt,$$

again by the weak bound. Letting $T \to \infty$ gives the displayed estimate, since $p\int_0^\infty t^{p-1}e^{-t}\,dt = \Gamma(p+1)$. □

Corollary 3.54 (Endpoint floor for the global minimum)**.** *For standard real Gaussian coefficients there are constants $c > 0$ and $m_0 < \infty$ such that, for $m \ge m_0$ and $0 < t \le 1$,*

$$\mathbb{P}\Big\{\min_{z\in\mathbb{T}} |G_m(z)| \le t\Big\} \ge c\,\frac{t}{\sqrt{m}}.$$

Equivalently,

$$\mathbb{P}\Big\{\min_{z\in\mathbb{T}} |G_m(z)| \le \frac{\tau}{\sqrt{m}}\Big\} \ge c\,\frac{\tau}{m} \qquad (0 < \tau \le \sqrt{m}).$$

Thus the two parity endpoints already force a nonzero global lower floor, but only at order $t/\sqrt{m}$; Corollaries 3.26, 3.27, and 3.28 give the exact bivariate formula, sharp two-endpoint expansion, and continuum-scale $O_\tau(N^{-1})$ contribution of both the endpoint values and their natural neighborhoods. The conjectural whole-circle lower tail at order $\sqrt{m}\,t$ must come from the $O(m)$ non-real arcs and their decorrelation.

Proof. The event in Corollary 3.26 is contained in $\{\min_{\mathbb{T}} |G_m| \le t\}$, and its lower asymptotic is $\gg t/\sqrt{m}$. The rescaled statement is the same estimate with $t = \tau/\sqrt{m}$. □

Proof of Theorem 1.2. Fix endpoint arcs E_{C_0} with C_0 large enough that the complement can be partitioned into $O(m)$ arcs I_ℓ of lengths c_ℓ/m, with $1 \le c_\ell \le 2$, each at angular distance at least $1/m$ from $\{\pm 1\}$. Applying Theorem 1.1, then summing over the $O(m)$ arcs, gives

$$\mathbb{P}\Big\{\min_{\theta\notin E_{C_0}} \big|G_m(e^{i\theta})\big| \le t\Big\} \le C\sqrt{m}\,t.$$

By Proposition 3.24,

$$\mathbb{P}\Big\{\min_{\theta\in E_{C_0}} \big|G_m(e^{i\theta})\big| \le t\Big\} \le C\,\frac{t}{\sqrt{m}}.$$

Combining the endpoint and non-endpoint estimates proves (1.8). □

Proof of Corollary 1.3. This is immediate from Theorem 1.2, after increasing $C_{A,B}$ and $m_0(A,B)$ if necessary. □

Lemma 3.55 (Arc sampling under a speed cutoff)**.** *Let $I \subset \mathbb{T}$ be an arc of arclength $|I| = c/m$, sampled at $L := \lceil 4c \rceil$ equally spaced nodes $z_1, \dots, z_L \in I$. For every $z \in I$ there is a node z_ℓ with*

$$\big|\,|G_m(z)| - |G_m(z_\ell)|\,\big| \;\le\; |G_m(z) - G_m(z_\ell)| \;\le\; \frac{|I|}{L} \sup_{w \in I} |G_m'(w)|\,.$$

Proof. This is the fundamental theorem of calculus along I, since each $z \in I$ lies within arclength $|I|/L$ of a node and $\big|\,|a| - |b|\,\big| \le |a-b|$. □

Proposition 3.56 (Bounded-density local bounds under a speed cutoff)**.** *Let $\xi_0, \dots, \xi_m$ be independent real random variables with densities bounded by K, and let $G_m(z) = \sum_{j=0}^m \xi_j z^j$.*

(i) Away from the parity points. *Fix $\gamma > 0$. If $I \subset \mathbb{T}$ is an arc of arclength c/m, $c \ge 1$, with $\operatorname{dist}_{\mathbb{T}}(I, \{\pm 1\}) \ge \gamma/m$, then for $L \ge 1$ and $0 < t \le 1$,*

$$\mathbb{P}\Big(\Big\{\min_{z \in I} |G_m(z)| \le t\Big\} \cap \Big\{\sup_{z \in I} |G_m'(z)| \le L m^{3/2}\Big\}\Big) \le C_{\gamma,K}(1+c)\frac{Lt}{\sqrt{m}}.$$

(ii) Endpoint arcs. *Fix $C_0, B > 0$, and put*

$$E_{C_0} := \{\theta \in [-\pi, \pi] : \operatorname{dist}(\theta, \{0, \pi, -\pi\}) \le C_0/m\}.$$

For $m^{-B} \le t \le 1$ and $L \ge 1$,

$$\mathbb{P}\Big(\Big\{\min_{\theta \in E_{C_0}} \Big|G_m(e^{i\theta})\Big| \le t\Big\} \cap \Big\{\sup_{\mathbb{T}} |G_m'| \le L m^{3/2}\Big\}\Big) \le C_{B,C_0,K}\frac{Lt \log m}{\sqrt{m}}.$$

Proof. For (i), parametrize I by an interval J of length c/m and set $\Lambda(s) = \int_J \mathbf{1}\{\Big|G_m(e^{i\theta})\Big| \le s\}\, d\theta$. By Fubini and Proposition 3.2,

$$\mathbb{E}\Lambda(s) \le \frac{C_{\gamma,K} c s^2}{m^2}.$$

On the speed event, one visit to $\{|G_m| \le t\}$ forces $|G_m| \le 2t$ on an interval of length at least a constant multiple of $\delta = t/(Lm^{3/2})$, truncated only by a harmless endpoint factor because $|J| = c/m \ge \delta$ for large m. Hence

$$\mathbf{1}\{\min_I |G_m| \le t\}\mathbf{1}_{\{\sup_I |G_m'| \le L m^{3/2}\}} \le C\,\frac{\Lambda(2t)}{\delta}.$$

Taking expectations gives $C_{\gamma,K} c L t/\sqrt{m}$, and $c \le 1 + c$.

For (ii), the same argument uses the endpoint occupation estimate Lemma 3.23 instead of the away-from-endpoint occupation estimate. On the speed event a visit contributes length $\delta = t/(Lm^{3/2})$ inside E_{C_0}, up to a constant one-sided loss at the ends of the endpoint arcs, and Lemma 3.23 with $s = 2t$ gives

$$C\,\frac{Lm^{3/2}}{t} \cdot C_{B,C_0,K}\frac{t^2 \log m}{m^2} \le C_{B,C_0,K}\frac{Lt \log m}{\sqrt{m}}.$$

□

Corollary 3.57 (Uniform bounded-density boundary bound under a speed cutoff)**.** *Let $\xi_0, \dots, \xi_m$ be independent real random variables with densities bounded by K, and let $G_m(z) = \sum_{j=0}^m \xi_j z^j$. For every $B > 0$ there are constants $C_{B,K} < \infty$ and $m_0(B,K)$ such that, for $m \ge m_0(B,K)$, $m^{-B} \le t \le 1$, and $L \ge 1$,*

$$\mathbb{P}\Big(\Big\{\min_{z\in\mathbb{T}} |G_m(z)| \le t\Big\} \cap \Big\{\sup_{z\in\mathbb{T}} |G_m'(z)| \le Lm^{3/2}\Big\}\Big) \le C_{B,K} L\sqrt{m}\, t.$$

Proof. Choose endpoint arcs E_{C_0} with fixed C_0 large enough that the complement can be partitioned into $O(m)$ arcs of lengths c_ℓ/m, with $1 \le c_\ell \le 2$, all at angular distance at least $1/m$ from $\{\pm 1\}$. Sum Proposition 3.56(i) over those arcs, using $\gamma = 1$, to obtain $C_K L\sqrt{m}\, t$ on the complement. On the endpoint arcs, Proposition 3.56(ii) gives

$$C_{B,C_0,K} \frac{Lt \log m}{\sqrt{m}} \le C_{B,K} L\sqrt{m}\, t$$

for m large. Combining the two estimates proves the claim. □

Lemma 3.58 (Polynomial derivative cutoff from variance)**.** *Let $\xi_0, \dots, \xi_m$ be independent mean-zero real random variables with $\mathbb{E}|\xi_j|^2 = 1$, and let $G_m(z) = \sum_{j=0}^m \xi_j z^j$. For every $\delta > 1/2$, there is a constant $C_\delta < \infty$ such that*

$$\mathbb{P}\Big\{\sup_{z\in\mathbb{T}} |G_m'(z)| > m^{3/2+\delta}\Big\} \le C_\delta m^{1-2\delta}.$$

Proof. Put

$$H_m(\theta) := G_m'(e^{i\theta}) = \sum_{j=1}^m j\xi_j e^{i(j-1)\theta}.$$

Bernstein's inequality for trigonometric polynomials gives a grid $\{\theta_\ell\}_{\ell=1}^N$ with $N \le Cm$ and mesh at most $1/(4m)$ such that

$$\sup_\theta |H_m(\theta)| \le 2 \max_{1\le\ell\le N} |H_m(\theta_\ell)|.$$

Indeed, at a maximum point choose the nearest grid point and use $\|H_m'\|_\infty \le m\|H_m\|_\infty$. For each fixed grid point, independence and centering give

$$\mathbb{E}|H_m(\theta_\ell)|^2 = \sum_{j=1}^m j^2 \mathbb{E}|\xi_j|^2 \le Cm^3.$$

Chebyshev's inequality therefore gives

$$\mathbb{P}\{|H_m(\theta_\ell)| > \tfrac{1}{2} m^{3/2+\delta}\} \le Cm^{-2\delta}.$$

The union bound over $N \le Cm$ grid points yields

$$\mathbb{P}\{\sup_\theta |H_m(\theta)| > m^{3/2+\delta}\} \le Cm^{1-2\delta} = o(1),$$

because $\delta > 1/2$. □

Corollary 3.59 (Finite-confidence derivative cutoff from variance)**.** *Let $\xi_0, \dots, \xi_m$ be independent mean-zero real random variables with $\mathbb{E}|\xi_j|^2 = 1$, and let $G_m(z) = \sum_{j=0}^m \xi_j z^j$. There is an absolute constant $C < \infty$ such that, for every $R > 0$,*

$$\mathbb{P}\Big\{\sup_{z\in\mathbb{T}} |G_m'(z)| > R\Big\} \le C \min\left\{1, \frac{m^4}{R^2}\right\}.$$

Consequently, for every $0 < \eta \le 1$,

$$\mathbb{P}\Big\{\sup_{z\in\mathbb{T}} |G_m'(z)| > Cm^2\eta^{-1/2}\Big\} \le \eta.$$

Proof. The proof of Lemma 3.58, before the final substitution $R = m^{3/2+\delta}$, gives the quantitative estimate. Indeed, Bernstein's inequality for trigonometric polynomials reduces the supremum of

$$H_m(\theta) = \frac{d}{d\theta} G_m(e^{i\theta})$$

to a grid of $O(m)$ points, and for each grid point $\mathbb{E}|H_m(\theta_\ell)|^2 \le Cm^3$. Chebyshev's inequality and the union bound therefore give

$$\mathbb{P}\{\sup_\theta |H_m(\theta)| > R\} \le \frac{Cm^4}{R^2}.$$

Since $|H_m(\theta)| = |G_m'(e^{i\theta})|$ on $\mathbb{T}$, this is the first display, after taking the minimum with the trivial bound 1. The finite-confidence form follows by taking $R = Cm^2\eta^{-1/2}$ with C large enough. □

Corollary 3.60 (Tail-free polynomial boundary separation)**.** *Let $\xi_0, \dots, \xi_m$ be independent mean-zero real random variables whose densities are bounded by K and satisfy $\mathbb{E}|\xi_j|^2 = 1$. If $G_m(z) = \sum_{j=0}^m \xi_j z^j$, then, for every $A > 1$,*

$$\mathbb{P}\Big\{\min_{z\in\mathbb{T}} |G_m(z)| \le m^{-A}\Big\} \le C_{A,K}\, m^{-2(A-1)/3}.$$

Equivalently, every bounded-density variance-one law has a polynomial continuum separation floor m^{-A}, for any prescribed exponent $A > 1$, without any tail assumption, and with an explicit polynomial exceptional probability.

Proof. Choose

$$\delta = \frac{A + 1/2}{3}.$$

Since $A > 1$, we have $1/2 < \delta < A - 1/2$. Let

$$E_\delta = \Big\{\sup_{\mathbb{T}} |G_m'| \le m^{3/2+\delta}\Big\}.$$

By Lemma 3.58,

$$\mathbb{P}(E_\delta^c) \le C_A m^{1-2\delta}.$$

On E_δ, apply Corollary 3.57 with $L = m^\delta$, $B = A$, and $t = m^{-A}$. This gives

$$\mathbb{P}\Big(\Big\{\min_{\mathbb{T}} |G_m| \le m^{-A}\Big\} \cap E_\delta\Big) \le C_{A,K} m^{\delta+1/2-A},$$

and the chosen δ makes

$$1-2\delta=\delta+\frac{1}{2}-A=-\frac{2(A-1)}{3}.$$

Adding the two estimates gives the displayed bound. □

Corollary 3.61 (Finite-confidence boundary separation from variance)**.** *Let $\xi_0,\dots,\xi_m$ be independent mean-zero real random variables whose densities are bounded by K and satisfy $\mathbb{E}|\xi_j|^2=1$, and let $G_m(z)=\sum_{j=0}^m \xi_j z^j$. Fix $B>0$. There are constants $C_{B,K}<\infty$ and $m_0(B,K)$ such that, for all $m\ge m_0(B,K)$, all $0<\eta\le 1$, and all $m^{-B}\le t\le 1$,*

$$\mathbb{P}\Big\{\min_{z\in\mathbb{T}}|G_m(z)|\le t\Big\}\le \eta+C_{B,K}\frac{mt}{\sqrt{\eta}}.$$

In particular, if $m^{-B}\le t\le c_{B,K}\eta^{3/2}/m$, then

$$\mathbb{P}\Big\{\min_{z\in\mathbb{T}}|G_m(z)|\le t\Big\}\le 2\eta.$$

Proof. By Corollary 3.59, outside an event of probability at most η,

$$\sup_{\mathbb{T}}|G_m'|\le Cm^2\eta^{-1/2}=Lm^{3/2},\qquad L=Cm^{1/2}\eta^{-1/2}.$$

Let

$$E_\eta:=\left\{\sup_{\mathbb{T}}|G_m'|\le Cm^2\eta^{-1/2}\right\}.$$

On this speed event, Corollary 3.57 gives

$$\mathbb{P}\left(\left\{\min_{\mathbb{T}}|G_m|\le t\right\}\cap E_\eta\right)\le C_{B,K}Lm^{1/2}t\le C_{B,K}\frac{mt}{\sqrt{\eta}}.$$

Adding the speed-exception probability proves the first display. The second display follows by choosing $c_{B,K}$ so that $C_{B,K}mt/\sqrt{\eta}\le\eta$. □

Corollary 3.62 (Variance-only polynomial grids detect continuum excursions)**.** *Let $\xi_0,\dots,\xi_m$ be independent mean-zero real random variables with $\mathbb{E}|\xi_j|^2=1$, let $G_m(z)=\sum_{j=0}^m\xi_jz^j$, and put $N=m+1$. Fix $\delta>1/2$ and $\tau>0$. If $Q_m\subset\mathbb{T}$ is an h_m-net in arclength with*

$$h_m\le\frac{\tau}{2m^{3/2+\delta}\sqrt{N}},$$

then there is a constant $C_\delta<\infty$ such that

$$\mathbb{P}\left(\left\{\min_{z\in\mathbb{T}}|G_m(z)|\le\frac{\tau}{2\sqrt{N}}\right\}\setminus\left\{\min_{q\in Q_m}|G_m(q)|\le\frac{\tau}{\sqrt{N}}\right\}\right)\le C_\delta m^{1-2\delta}.$$

Thus a uniform grid with $M\ge Cm^{3/2+\delta}\sqrt{N}/\tau$ points, in particular $M=O_\tau(N^{2+\delta})$, has this detection property with the same exceptional-probability bound.

Proof. Let

$$E_m = \Big\{ \sup_{\mathbb{T}} |G_m'| \le m^{3/2+\delta} \Big\}.$$

On E_m, Lemma 3.5, with $L = m^\delta$, gives

$$\Big\{ \min_{\mathbb{T}} |G_m| \le \frac{\tau}{2\sqrt{N}} \Big\} \subseteq \Big\{ \min_{q \in Q_m} |G_m(q)| \le \frac{\tau}{\sqrt{N}} \Big\}.$$

The miss event is therefore contained in E_m^c, whose probability is at most $C_\delta m^{1-2\delta}$ by Lemma 3.58. The uniform-grid assertion is the same elementary mesh estimate as in Lemma 3.5. □

Corollary 3.63 (Fixed-power variance-only grids)**.** *Under the hypotheses of Corollary 3.62, fix $\beta > 5/2$ and $\tau > 0$. There are constants $C_{\beta,\tau}, C_\beta < \infty$ such that, if Q_m is a uniform grid on $\mathbb{T}$ with*

$$|Q_m| \ge C_{\beta,\tau} N^\beta,$$

then

$$\mathbb{P}\left(\Big\{ \min_{z \in \mathbb{T}} |G_m(z)| \le \frac{\tau}{2\sqrt{N}} \Big\} \setminus \Big\{ \min_{q \in Q_m} |G_m(q)| \le \frac{\tau}{\sqrt{N}} \Big\} \right) \le C_\beta N^{5-2\beta}.$$

Thus every fixed power strictly above $N^{5/2}$ gives a tail-free variance-only detector with an explicit polynomial miss rate.

Proof. Apply Corollary 3.62 with $\delta = \beta - 2 > 1/2$. A uniform grid with M points has mesh at most C/M, so $M \ge C_{\beta,\tau} N^\beta$ implies

$$h_m \le \frac{\tau}{2m^{3/2+\delta}\sqrt{N}}$$

after increasing $C_{\beta,\tau}$, because $N = m + 1$. The miss probability is at most

$$C_\delta m^{1-2\delta} = C_\beta m^{5-2\beta} \le C_\beta N^{5-2\beta}.$$

□

Corollary 3.64 (Tail-free polynomial-confidence grids)**.** *Under the hypotheses of Corollary 3.62, fix $A > 0$ and $\tau > 0$. If $Q_m \subset \mathbb{T}$ is an h_m-net in arclength with*

$$h_m \le \frac{\tau}{2m^{2+A/2}\sqrt{N}},$$

then there is a constant $C_A < \infty$ such that

$$\mathbb{P}\left(\Big\{ \min_{z \in \mathbb{T}} |G_m(z)| \le \frac{\tau}{2\sqrt{N}} \Big\} \setminus \Big\{ \min_{q \in Q_m} |G_m(q)| \le \frac{\tau}{\sqrt{N}} \Big\} \right) \le C_A m^{-A}.$$

Equivalently, a uniform grid of size

$$M = O_{A,\tau}(N^{(A+5)/2})$$

detects continuum $N^{-1/2}$-scale excursions with failure probability $O_A(m^{-A})$, using only centering and unit variance.

Proof. Apply Corollary 3.62 with $\delta = (A+1)/2$. Then $1-2\delta = -A$, and

$$m^{3/2+\delta} = m^{2+A/2}.$$

The uniform-grid size follows from the same mesh estimate as in Corollary 3.62, using $N = m+1$. □

Corollary 3.65 (Finite-confidence variance-only grids)**.** *Let $\xi_0, \dots, \xi_m$ be independent mean-zero real random variables with $\mathbb{E}|\xi_j|^2 = 1$, let $G_m(z) = \sum_{j=0}^m \xi_j z^j$, and put $N = m+1$. Fix $\tau > 0$ and $0 < \eta \le 1$. If $Q_m \subset \mathbb{T}$ is an h_m-net in arclength with*

$$h_m \le c\,\frac{\tau\sqrt{\eta}}{m^2\sqrt{N}},$$

then

$$\mathbb{P}\left(\left\{\min_{z\in\mathbb{T}} |G_m(z)| \le \frac{\tau}{2\sqrt{N}}\right\} \setminus \left\{\min_{q\in Q_m} |G_m(q)| \le \frac{\tau}{\sqrt{N}}\right\}\right) \le \eta.$$

Equivalently, a uniform grid of size

$$M \ge C_\tau m^2 \sqrt{N}\,\eta^{-1/2}$$

detects continuum $N^{-1/2}$-scale excursions with miss probability at most η, using only centering and unit variance. The same conclusion holds for stratified one-point-per-equal-arc grids and for random shifts of a uniform grid with the same mesh bound.

Proof. By Corollary 3.59, with probability at least $1-\eta$,

$$\sup_{\mathbb{T}} |G_m'| \le R_\eta := C m^2 \eta^{-1/2}.$$

If $h_m \le \tau/(2R_\eta\sqrt{N})$, then Lemma 3.5 gives

$$\left\{\min_{\mathbb{T}} |G_m| \le \frac{\tau}{2\sqrt{N}}\right\} \cap \left\{\sup_{\mathbb{T}} |G_m'| \le R_\eta\right\} \subseteq \left\{\min_{q\in Q_m} |G_m(q)| \le \frac{\tau}{\sqrt{N}}\right\}.$$

Thus the miss event is contained in the derivative-speed exceptional event. The uniform-grid and stratified-grid assertions are the elementary circle mesh bound. □

Corollary 3.66 (Continuum-scale sampling bracket)**.** *Let $\xi_0, \dots, \xi_m$ be independent mean-zero real random variables whose densities are bounded by K and satisfy $\mathbb{E}|\xi_j|^2 = 1$, and put $N = m+1$.*

(i) *Fix $\gamma > 0$, $T < \infty$, and deterministic finite sets $Q_m^- \subset \mathbb{T}$ such that every point of $Q_m^- \setminus \{\pm 1\}$ has angular distance at least γ/m from $\{\pm 1\}$. If $|Q_m^-| = o(N^2)$, then*

$$\sup_{0\le\tau\le T} \mathbb{P}\Big\{\min_{q\in Q_m^-} |G_m(q)| \le \frac{\tau}{\sqrt{N}}\Big\} \to 0.$$

The same conclusion holds for independent random grids $\mathcal{Q}_m^-$ satisfying the same endpoint-separation condition almost surely and $\mathbb{E}|\mathcal{Q}_m^-| = o(N^2)$.

(ii) *Fix $A > 0$ and $\tau > 0$. If $Q_m^+ \subset \mathbb{T}$ is an h_m-net with*

$$h_m \le \frac{\tau}{2m^{2+A/2}\sqrt{N}},$$

then

$$\mathbb{P}\left(\left\{\min_{z\in\mathbb{T}} |G_m(z)| \le \frac{\tau}{2\sqrt{N}}\right\} \setminus \left\{\min_{q\in Q_m^+} |G_m(q)| \le \frac{\tau}{\sqrt{N}}\right\}\right) \le C_A m^{-A}.$$

Consequently, sampling at the continuum threshold has a genuine scale separation: deterministic and independent-random subquadratic grids see only the pointwise small-ball scale, whereas any fixed power above $N^{5/2}$ is complete with a polynomial miss rate under variance alone, and sufficiently higher polynomial powers give arbitrary polynomial failure.

Proof. Part (i) is Corollaries 3.3 and 3.4. Part (ii) is Corollary 3.64. The final sentence only restates these two estimates at the common threshold $N^{-1/2}$. □

Corollary 3.67 (Boundary law from a polynomial speed tail)**.** *Assume the hypotheses of Corollary 3.57. Suppose in addition that, for this fixed coefficient law, for every $A > 0$ there are constants $L_A, C_A < \infty$ and $m_0(A)$ such that*

$$\mathbb{P}\Big\{\sup_{z\in\mathbb{T}} |G_m'(z)| > L_A m^{3/2}\sqrt{\log m}\Big\} \le C_A m^{-A} \qquad (m \ge m_0(A)).$$

Then Hypothesis 1.4 holds for this coefficient law, with constants depending on K and on the displayed speed-tail constants.

Proof. Fix $A, B > 0$, let

$$E_A := \Big\{\sup_{\mathbb{T}} |G_m'| \le L_A m^{3/2}\sqrt{\log m}\Big\},$$

and put $L = L_A\sqrt{\log m}$ in Corollary 3.57. For $m^{-B} \le t \le 1$,

$$\begin{aligned}\mathbb{P}\{\min_{\mathbb{T}} |G_m| \le t\} &\le \mathbb{P}(\{\min_{\mathbb{T}} |G_m| \le t\} \cap E_A) + \mathbb{P}(E_A^c)\\ &\le C_{B,K} L_A \sqrt{m\log m}\, t + C_A m^{-A}.\end{aligned}$$

Absorbing L_A, C_A, and the threshold into the constants in Hypothesis 1.4 gives the claim. The conclusion is therefore uniform over any class of coefficient laws for which the speed-tail constants are uniform. □

Corollary 3.68 (Grid detection from a polynomial speed tail)**.** *Let $G_m(z) = \sum_{j=0}^m \xi_j z^j$, put $N = m+1$, and suppose that for a given coefficient law and a given $A > 0$ there are constants $L_A, C_A < \infty$ and $m_0(A)$ such that*

$$\mathbb{P}\Big\{\sup_{z\in\mathbb{T}} |G_m'(z)| > L_A m^{3/2}\sqrt{\log m}\Big\} \le C_A m^{-A} \qquad (m \ge m_0(A)).$$

Fix $\tau > 0$. If $Q_m \subset \mathbb{T}$ is an h_m-net with

$$h_m \le \frac{\tau}{2L_A m^{3/2}\sqrt{N}\,\sqrt{\log m}},$$

then, for $m \ge m_0(A)$,

$$\mathbb{P}\left(\left\{\min_{z\in\mathbb{T}}|G_m(z)| \le \frac{\tau}{2\sqrt{N}}\right\} \setminus \left\{\min_{q\in Q_m}|G_m(q)| \le \frac{\tau}{\sqrt{N}}\right\}\right) \le C_A m^{-A}.$$

Thus a uniform grid with $M \ge CL_A m^{3/2}\sqrt{N}\sqrt{\log m}/\tau$ points has the same property. In particular, the conclusion holds for the Gaussian law, for uniformly subexponential bounded-density laws, and for bounded-density laws with a finite p-moment satisfying $p > 2A + 4$, using the speed-tail inputs proved in Lemma 3.10 and Corollaries 3.70–3.71.

Proof. Let

$$E_A = \left\{\sup_{\mathbb{T}}|G_m'| \le L_A m^{3/2}\sqrt{\log m}\right\}.$$

On E_A, Lemma 3.5, with $L = L_A\sqrt{\log m}$, gives

$$\left\{\min_{\mathbb{T}}|G_m| \le \frac{\tau}{2\sqrt{N}}\right\} \subseteq \left\{\min_{q\in Q_m}|G_m(q)| \le \frac{\tau}{\sqrt{N}}\right\}.$$

Therefore the miss event in the statement is contained in E_A^c, whose probability is bounded by the assumed speed tail. The uniform-grid assertion is again the elementary mesh estimate on a circle of arclength 2π. The listed coefficient classes have the required speed tails by the cited results, with the polynomial-moment case choosing p according to the requested exponent A. □

Corollary 3.69 (Natural-speed sampling bracket)**.** *Let $\xi_0, \dots, \xi_m$ be independent real random variables whose densities are bounded by K, let $G_m(z) = \sum_{j=0}^m \xi_j z^j$, and put $N = m + 1$.*

(i) *Fix $\gamma > 0$, $T < \infty$, and deterministic finite sets $Q_m^- \subset \mathbb{T}$ such that every point of $Q_m^- \setminus \{\pm 1\}$ has angular distance at least γ/m from $\{\pm 1\}$. If $|Q_m^-| = o(N^2)$, then*

$$\sup_{0\le\tau\le T}\mathbb{P}\Big\{\min_{q\in Q_m^-}|G_m(q)| \le \frac{\tau}{\sqrt{N}}\Big\} \to 0.$$

The same conclusion holds for independent random grids $\mathcal{Q}_m^-$ satisfying the same endpoint-separation condition almost surely and $\mathbb{E}|\mathcal{Q}_m^-| = o(N^2)$.

(ii) *Suppose, in addition, that for a given $A > 0$ there are constants $L_A, C_A < \infty$ and $m_0(A)$ such that*

$$\mathbb{P}\Big\{\sup_{z\in\mathbb{T}}|G_m'(z)| > L_A m^{3/2}\sqrt{\log m}\Big\} \le C_A m^{-A} \qquad (m \ge m_0(A)).$$

Fix $\tau > 0$. If $Q_m^+ \subset \mathbb{T}$ is an h_m-net with

$$h_m \le \frac{\tau}{2L_A m^{3/2}\sqrt{N}\,\sqrt{\log m}},$$

then, for $m \ge m_0(A)$,

$$\mathbb{P}\left(\left\{\min_{z\in\mathbb{T}} |G_m(z)| \le \frac{\tau}{2\sqrt{N}}\right\} \setminus \left\{\min_{q\in Q_m^+} |G_m(q)| \le \frac{\tau}{\sqrt{N}}\right\}\right) \le C_A m^{-A}.$$

Thus, under a natural polynomial derivative-speed tail, sampling at the continuum threshold is squeezed between deterministic or independent random subquadratic grids, which see only the pointwise small-ball scale, and grids of size

$$O_{A,\tau}(L_A N^2 \sqrt{\log N}),$$

which detect every continuum excursion up to $O_A(m^{-A})$ miss probability. Independent uniform random grids also detect, with the coupon-collector overhead

$$O_{A,\tau,L_A}(N^2 (\log N)^{3/2}).$$

Stratified one-point-per-equal-arc grids, including random shifts of a uniform lattice, detect at the deterministic

$$O_{A,\tau}(L_A N^2 \sqrt{\log N})$$

scale without that extra logarithm. The upper side applies to Gaussian coefficients, uniformly subexponential bounded-density laws, and bounded-density laws with a finite p-moment satisfying $p > 2A + 4$.

Proof. Part (i) is Corollaries 3.3 and 3.4. Part (ii) is Corollary 3.68. A uniform grid of M points has mesh $O(1/M)$, so the displayed mesh condition is satisfied once $M \ge C_\tau L_A m^{3/2}\sqrt{N}\sqrt{\log m}$, which is $O_\tau(L_A N^2 \sqrt{\log N})$. The independent-uniform statement is Corollary 3.7, and the stratified statement is Corollary 3.8. The listed coefficient classes have the required speed tails by Lemma 3.10 and Corollaries 3.70–3.71, choosing p according to the requested exponent in the polynomial-moment case. □

Corollary 3.70 (Polynomial-moment bounded-density boundary law)**.** *Let $\xi_0, \dots, \xi_m$ be independent, mean-zero, variance-one real random variables whose densities are bounded by K. Fix $A_* > 0$, and suppose that for some $p > 2A_* + 4$,*

$$M_p := \sup_{m\ge 1} \max_{0\le j\le m} \mathbb{E}\,|\xi_j|^p < \infty.$$

Then there are constants $L_{A_*,p,M_p}, C_{A_*,p,M_p} < \infty$ and $m_0(A_*, p, M_p)$ such that

$$\mathbb{P}\Big\{\sup_{z\in\mathbb{T}} |G_m'(z)| > L_{A_*,p,M_p} m^{3/2}\sqrt{\log m}\Big\} \le C_{A_*,p,M_p} m^{-A_*} \qquad (m \ge m_0).$$

Consequently, if the displayed moment bound holds for every $p < \infty$, then Hypothesis 1.4 holds for this coefficient law, with constants depending on K and on the moment bounds used for the requested error exponent.

Proof. By Corollary 3.67, it suffices to prove the displayed speed tail. Put

$$H_m(\theta) := G_m'(e^{i\theta}) = \sum_{j=1}^{m} j\xi_j e^{i(j-1)\theta}.$$

As in the proof of Corollary 3.71, Bernstein's inequality for trigonometric polynomials reduces the supremum to a grid $\{\theta_\ell\}_{\ell=1}^N$ with $N \le Cm$:

$$\sup_\theta |H_m(\theta)| \le 2 \max_{1\le \ell\le N} |H_m(\theta_\ell)|\,.$$

Fix a grid point and write either the real or imaginary part as a sum

$$S_\ell = \sum_{j=1}^m a_{j,\ell}\xi_j, \qquad |a_{j,\ell}| \le j.$$

The Fuk–Nagaev inequality [8] gives, for $u > 0$,

$$\mathbb{P}\{|S_\ell| > u\} \le C_p u^{-p} \sum_{j=1}^m \mathbb{E}\,|a_{j,\ell}\xi_j|^p + 2\exp\left\{-c\frac{u^2}{\sum_{j=1}^m \mathbb{E}(a_{j,\ell}^2\xi_j^2)}\right\}.$$

Since $\sum_j j^p \le C_p m^{p+1}$ and $\sum_j j^2 \le Cm^3$, this implies

$$\mathbb{P}\{|H_m(\theta_\ell)| > u\} \le C_{p,M_p} u^{-p} m^{p+1} + 4\exp\{-cu^2/m^3\}.$$

With $u = Lm^{3/2}\sqrt{\log m}$, the first term is at most

$$C_{p,M_p,L} m^{1-p/2}(\log m)^{-p/2},$$

and the exponential term is at most $4m^{-cL^2}$. After the union bound over the $O(m)$-point grid, the polynomial term is $O(m^{2-p/2})$, which is $O(m^{-A_*})$ because $p > 2A_* + 4$. Choosing L large enough in terms of A_* controls the exponential term as well. This proves the speed-tail estimate, and the final assertion follows from Corollary 3.67, choosing p separately for each requested error exponent. □

Corollary 3.71 (Subexponential bounded-density boundary law)**.** *Let $\xi_0, \dots, \xi_m$ be independent, mean-zero, variance-one real random variables whose densities are bounded by K, and assume in addition that $\|\xi_j\|_{\psi_1} \le \kappa$ uniformly in j and m. For every $A > 0$ there are constants $L_{A,\kappa}, C_{A,\kappa} < \infty$ and $m_0(A,\kappa)$ such that*

$$\mathbb{P}\Big\{\sup_{z\in\mathbb{T}} |G_m'(z)| > L_{A,\kappa} m^{3/2}\sqrt{\log m}\Big\} \le C_{A,\kappa} m^{-A} \qquad (m \ge m_0(A,\kappa)).$$

Hence Hypothesis 1.4 holds, with constants depending also on κ. Consequently, if Hypothesis 1.5 holds for this coefficient law, the conclusion of Theorem 1.6 holds.

Proof. We first verify the displayed polynomial speed tail. Put

$$H_m(\theta) := G_m'(e^{i\theta}) = \sum_{j=1}^m j\xi_j e^{i(j-1)\theta}.$$

We first reduce the supremum to a grid. If h is a trigonometric polynomial of degree at most m, Bernstein's inequality and a grid $\{\theta_\ell\}_{\ell=1}^N$ with $N \asymp m$ and mesh at most $1/(4m)$ give

$$\sup_\theta |h(\theta)| \le 2 \max_{1\le \ell\le N} |h(\theta_\ell)|\,.$$

Indeed, at a maximum point choose the nearest grid point and use $\|h'\|_\infty \le m\,\|h\|_\infty$.

For each grid point, the real and imaginary parts of $H_m(\theta_\ell)$ are centered subexponential linear forms. The Bernstein inequality for subexponential sums [9, Thm. 2.8.1] gives

$$\mathbb{P}\{|H_m(\theta_\ell)| > u\} \le 4\exp\left\{-c_\kappa \min\left(\frac{u^2}{m^3}, \frac{u}{m}\right)\right\}.$$

Here $\sum_{j=1}^m j^2 = O(m^3)$ and $\max_j j \le m$ are the only coefficient norms used. Taking $u = Lm^{3/2}\sqrt{\log m}$, the quadratic term controls the minimum for all large m, and therefore

$$\mathbb{P}\{|H_m(\theta_\ell)| > Lm^{3/2}\sqrt{\log m}\} \le 4m^{-c_\kappa L^2}.$$

A union bound over the $O(m)$ grid points gives, after choosing $L = L_A$ large in terms of A and κ, constants $L_A, C_A < \infty$ such that

$$\mathbb{P}\Big\{\sup_{z\in\mathbb{T}} |G_m'(z)| > L_A m^{3/2}\sqrt{\log m}\Big\} \le C_A m^{-A}.$$

Corollary 3.67 now gives Hypothesis 1.4. The final assertion is Theorem 1.6. □

Remark 3.72 (Speed cutoffs and the full boundary law). Corollary 3.57 should not be read as an unconditional bounded-density boundary law under the assumptions of Theorem 1.6. Even for standard Gaussian coefficients, no fixed constant L makes $\{\sup_{\mathbb{T}} |G_m'| \le Lm^{3/2}\}$ a high-probability event, since

$$G_m'(1) = \sum_{j=1}^m j\xi_j, \qquad m^{-3/2}G_m'(1) \Longrightarrow N(0, 1/3).$$

Moreover, bounded density and unit variance alone do not give the polynomially small exceptional probabilities needed to combine a growing speed cutoff with Hypothesis 1.4: heavy-tailed absolutely continuous laws with bounded density can make the complement of $\{\sup_{\mathbb{T}} |G_m'| \le L_m m^{3/2}\}$ only logarithmically small at the natural Gaussian scale $L_m \asymp \sqrt{\log m}$. Removing the cutoff therefore requires a genuine continuum crossing or local-limit argument, not just a derivative tail estimate. Theorem 1.6 uses only the weaker tail-free separation in Corollary 3.60: a polynomially small exceptional probability at a deliberately polynomial speed cutoff is enough for the single boundary-minimum threshold needed by finite-section conditioning.

3.2. **Occupation-measure route to the single-arc bound.**

Remark 3.73 (The occupation-measure proof and its logarithmic loss). Fix an arc I of arclength $|I| = c/m$, $c \ge 1$, whose closure is at angular distance $\ge \gamma_0/m$ from $\{\pm 1\}$ for a fixed $\gamma_0 > 0$. Parametrize I by $\theta \in J$, an interval of length c/m, and write $g(\theta) := \left|G_m(e^{i\theta})\right|$. The event $\{\min_I g \le t\}$ is the

event that the planar curve $\theta \mapsto G_m(e^{i\theta})$ enters the disk $B(0,t)$ for some $\theta \in J$.

Occupation measure. Let $\Lambda(s) := \int_J \mathbf{1}\{g(\theta) \le s\}\, d\theta$ be the time the curve spends in $B(0,s)$. By Fubini and Lemma 3.1 (the away-from-± 1 branch applies, since J is γ_0/m away from $0, \pi$),

$$\mathbb{E}\Lambda(s) = \int_J \mathbb{P}\{g(\theta) \le s\}\, d\theta \le |J| \cdot \frac{C_{\gamma_0} s^2}{m+1} = \frac{C_{\gamma_0}\, c\, s^2}{m(m+1)}. \tag{3.6}$$

Sojourn lower bound per visit. If one had the localized speed event

$$G_I := \{\sup_I |G_m'| \le C_2\, m^{3/2}\}$$

with probability $1 - o(t/\sqrt{m})$, then the curve speed would be $|\frac{d}{d\theta} G_m(e^{i\theta})| = |iz\, G_m'(z)| \le C_2 m^{3/2}$. Hence whenever $g(\theta_0) \le t$, the curve would stay in $B(0,2t)$ for all θ with $|\theta - \theta_0| \le \delta$, where $\delta := \frac{t}{C_2 m^{3/2}}$, because g changes by at most $\delta\, C_2 m^{3/2} = t$. Since $|J| = c/m \ge \delta$ for $t \le 1$ and m large, each visit of $\{g \le t\}$ would contribute at least δ to $\Lambda(2t)$, so on G_I

$$\mathbf{1}\Big\{\min_J g \le t\Big\} \le \frac{\Lambda(2t)}{\delta}.$$

Combine. Taking expectations and using (3.6) with $s = 2t$ would give

$$\mathbb{P}\Big\{\min_I g \le t\Big\} \le \mathbb{P}(G_I^c) + \frac{1}{\delta}\,\mathbb{E}\Lambda(2t) \le \mathbb{P}(G_I^c) + \frac{C_2 m^{3/2}}{t} \cdot \frac{4C_{\gamma_0}\, c\, t^2}{m(m+1)} \le \frac{C\, c\, t}{\sqrt{m}}.$$

This is the desired rate (1.7). The obstruction to proving it by this elementary occupation route is the speed/level-crossing step: the available global Salem–Zygmund tail in Lemma 3.10 carries the logarithmic loss recorded in Proposition 3.21. Lemma 3.12 removes that loss in the Gaussian case away from $z = \pm 1$ by counting actual level crossings instead of enforcing a deterministic speed cutoff.

Proposition 3.74 (Exact Gaussian root-grid law)**.** *Let $N = m + 1 \ge 3$, let $\omega = \exp(2\pi i/N)$, and let*

$$G_m(z) = \sum_{j=0}^{N-1} \xi_j z^j$$

with ξ_j independent standard real Gaussians. Put

$$\mathcal{R}_N := \{\omega^k : 1 \le k \le \lfloor (N-1)/2 \rfloor\}.$$

Then the random variables $N^{-1/2} G_m(\omega^k)$, $1 \le k \le \lfloor (N-1)/2 \rfloor$, are independent standard complex Gaussians $(X + iY)/\sqrt{2}$, with X, Y independent $N(0,1)$. Consequently, for every $t \ge 0$,

$$\mathbb{P}\Big\{\min_{z \in \mathcal{R}_N} |G_m(z)| \le t\Big\} = 1 - \exp\Big(-\frac{\lfloor (N-1)/2 \rfloor}{N}\, t^2\Big). \tag{3.7}$$

In particular, for $N \geq 3$ and $0 \leq t \leq 1$, this probability is $\asymp t^2$, with absolute constants. If $\mathcal{U}_N := \{\omega^k : 0 \leq k < N\}$ is the full N-th-root grid, set

$$s_N := 1 + \mathbf{1}_{\{2|N\}}, \qquad \alpha(x) := \mathbb{P}\{|Z| \leq x\}, \quad Z \sim N(0,1).$$

Then the full grid has the exact law

$$\mathbb{P}\Big\{\min_{z\in\mathcal{U}_N} |G_m(z)| > t\Big\} = \big(1 - \alpha(t/\sqrt{N})\big)^{s_N} \exp\Big(-\frac{\lfloor (N-1)/2 \rfloor}{N}\, t^2\Big). \tag{3.8}$$

Consequently,

$$\mathbb{P}\Big\{\min_{z\in\mathcal{U}_N} |G_m(z)| \leq t\Big\} \leq C\left(\frac{t}{\sqrt{N}} + t^2\right) \qquad (0 \leq t \leq 1), \tag{3.9}$$

with an absolute constant C.

Proof. Let $M = \lfloor (N-1)/2 \rfloor$. For $1 \leq k \leq M$, set

$$A_k = \sqrt{\frac{2}{N}} \sum_{j=0}^{N-1} \xi_j \cos(2\pi kj/N), \qquad B_k = \sqrt{\frac{2}{N}} \sum_{j=0}^{N-1} \xi_j \sin(2\pi kj/N).$$

The vectors

$$\sqrt{2/N}\,(\cos(2\pi kj/N))_{j=0}^{N-1}, \qquad \sqrt{2/N}\,(\sin(2\pi kj/N))_{j=0}^{N-1}, \qquad 1 \leq k \leq M,$$

are orthonormal in $\mathbb{R}^N$. Since $(\xi_0, \dots, \xi_{N-1})$ is a standard Gaussian vector in $\mathbb{R}^N$, the coordinates A_k, B_k are independent standard real Gaussians. Moreover

$$N^{-1/2} G_m(\omega^k) = \frac{A_k + iB_k}{\sqrt{2}}.$$

Thus the displayed variables are independent standard complex Gaussians, and $\left|N^{-1/2} G_m(\omega^k)\right|^2$ are independent exponential variables of mean 1. This gives (3.7).

For the full root grid, the conjugate roots do not add new moduli: $G_m(\omega^{N-k}) = \overline{G_m(\omega^k)}$. The only additional points are the real DFT coordinates 1, and also -1 when N is even. The corresponding normalized real coordinates are

$$C_0 = N^{-1/2} \sum_{j=0}^{N-1} \xi_j, \qquad C_{N/2} = N^{-1/2} \sum_{j=0}^{N-1} (-1)^j \xi_j \quad (N \text{ even}).$$

Together with the cosine and sine vectors above, the relevant real vectors form an orthonormal family in $\mathbb{R}^N$. Hence these s_N real $N(0,1)$ coordinates are independent of the non-real complex coordinates. The event $\min_{\mathcal{U}_N} |G_m| > t$ is therefore the intersection of the non-real event from (3.7) with the s_N independent events $\{|C| > t/\sqrt{N}\}$, proving (3.8). The upper bound (3.9) follows from $\alpha(x) \leq Cx$, $1 - e^{-x} \leq x$, and a union bound applied to the exact survival formula. □

Corollary 3.75 (The DFT root-grid minimum has pointwise scale)**.** *Under the hypotheses and notation of Proposition 3.74, let*

$$M_N^{\mathrm{nr}} := \min_{z\in\mathcal{R}_N} |G_m(z)|, \qquad M_N^{\mathrm{full}} := \min_{z\in\mathcal{U}_N} |G_m(z)|.$$

For every fixed $T<\infty$ there is $C_T<\infty$ such that

$$\sup_{0\le t\le T}\left|\mathbb{P}\{M_N^{\mathrm{nr}}>t\}-e^{-t^2/2}\right| \le \frac{C_T}{N}$$

and

$$\sup_{0\le t\le T}\left|\mathbb{P}\{M_N^{\mathrm{full}}>t\}-e^{-t^2/2}\right| \le \frac{C_T}{\sqrt{N}}.$$

Thus both root-grid minima converge in distribution to the Rayleigh law with survival function $e^{-t^2/2}$. In particular, the DFT grid minimum is typically of constant pointwise size, not of the continuum scale $N^{-1/2}$. Moreover, for every fixed $p>0$ and $N\ge 3$, with $M=\lfloor (N-1)/2\rfloor$, $s_N=1+\mathbf{1}_{\{2|N\}}$, and $R_p:=2^{p/2}\Gamma(1+p/2)$,

$$\mathbb{E}(M_N^{\mathrm{nr}})^p = \Gamma\!\left(1+\frac{p}{2}\right)\left(\frac{N}{M}\right)^{p/2} = R_p + \frac{s_N p}{2N}R_p + O_p(N^{-2}),$$

and

$$\mathbb{E}(M_N^{\mathrm{full}})^p = R_p - \frac{s_N p\, 2^{p/2}\Gamma((p+1)/2)}{\sqrt{\pi N}} + \frac{pR_p}{N}\left(\frac{s_N}{2}+\frac{s_N(s_N-1)}{\pi}\right) + O_p(N^{-3/2}).$$

In particular,

$$\mathbb{E}M_N^{\mathrm{nr}} = \frac{\sqrt{\pi N}}{2\sqrt{M}} = \sqrt{\frac{\pi}{2}} + \frac{s_N}{2N}\sqrt{\frac{\pi}{2}} + O(N^{-2}),$$

and

$$\mathbb{E}M_N^{\mathrm{full}} = \sqrt{\frac{\pi}{2}} - s_N\sqrt{\frac{2}{\pi N}} + \frac{1}{N}\sqrt{\frac{\pi}{2}}\left(\frac{s_N}{2}+\frac{s_N(s_N-1)}{\pi}\right) + O(N^{-3/2}).$$

Proof. Put $M=\lfloor (N-1)/2\rfloor$. From (3.7),

$$\mathbb{P}\{M_N^{\mathrm{nr}}>t\} = \exp\!\left(-\frac{M}{N}t^2\right), \qquad \left|\frac{M}{N}-\frac{1}{2}\right| \le \frac{1}{N}.$$

The first bound follows uniformly for $0\le t\le T$ by the mean-value theorem applied to $u\mapsto e^{-ut^2}$. For the full grid, use (3.8). Uniformly for $0\le t\le T$,

$$0\le \alpha(t/\sqrt{N}) \le C_T N^{-1/2}, \qquad \left|(1-\alpha(t/\sqrt{N}))^{s_N}-1\right| \le C_T N^{-1/2},$$

while the non-real exponential factor differs from $e^{-t^2/2}$ by $O_T(N^{-1})$. Multiplying the two factors gives the second bound.

For $p > 0$, the non-real moment follows by integrating the exact survival function:

$$\mathbb{E}(M_N^{\mathrm{nr}})^p = p \int_0^\infty t^{p-1} \exp\left(-\frac{M}{N} t^2\right) dt = \Gamma\left(1 + \frac{p}{2}\right) \left(\frac{N}{M}\right)^{p/2}.$$

Since

$$\frac{M}{N} = \frac{1}{2} - \frac{s_N}{2N},$$

this gives the displayed non-real expansion. For the full grid, use the exact survival formula (3.8). Split the moment integral at $t = \sqrt{N}$; the tail $t \geq \sqrt{N}$ is exponentially small. On $0 \leq t \leq \sqrt{N}$, with $c_0 := \sqrt{2/\pi}$,

$$\alpha(t/\sqrt{N}) = c_0 \frac{t}{\sqrt{N}} + O\left(\frac{t^3}{N^{3/2}}\right),$$

and

$$(1 - \alpha(t/\sqrt{N}))^{s_N} = 1 - s_N c_0 \frac{t}{\sqrt{N}} + \binom{s_N}{2} c_0^2 \frac{t^2}{N} + O\left(\frac{t^3 + t^4}{N^{3/2}}\right).$$

Also,

$$e^{-Mt^2/N} = e^{-t^2/2} \left(1 + \frac{s_N t^2}{2N} + O\left(\frac{t^4}{N^2}\right)\right).$$

Multiplying and integrating against $pt^{p-1}\,dt$ gives

$$\begin{aligned} \mathbb{E}(M_N^{\mathrm{full}})^p = p &\int_0^\infty t^{p-1} e^{-t^2/2}\,dt \\ &- \frac{p s_N c_0}{\sqrt{N}} \int_0^\infty t^p e^{-t^2/2}\,dt \\ &+ \frac{p}{N}\left(\frac{s_N}{2} + \frac{s_N(s_N-1)}{\pi}\right) \int_0^\infty t^{p+1} e^{-t^2/2}\,dt + O_p(N^{-3/2}). \end{aligned}$$

The three displayed integrals are

$$\frac{R_p}{p}, \qquad 2^{(p-1)/2}\Gamma\left(\frac{p+1}{2}\right), \qquad R_p,$$

respectively. Substituting them gives the full-grid moment expansion. The expectation statement is the case $p = 1$. □

Corollary 3.76 (The DFT grid misses the continuum scale)**.** *Under the hypotheses and notation of Proposition 3.74, for every fixed $T < \infty$ there is a constant $C_T < \infty$ such that*

$$\sup_{0 \leq \tau \leq T} \left| N\mathbb{P}\Big\{ \min_{z \in \mathcal{U}_N} |G_m(z)| \leq \frac{\tau}{\sqrt{N}} \Big\} - \left(s_N \sqrt{\frac{2}{\pi}}\,\tau + \frac{\tau^2}{2} \right) \right| \leq \frac{C_T}{N}.$$

In particular, along odd N the limiting coefficient is $\sqrt{2/\pi}\,\tau + \tau^2/2$, while along even N it is $2\sqrt{2/\pi}\,\tau + \tau^2/2$. Thus for $0 \leq \tau \leq T$,

$$\mathbb{P}\Big\{ \min_{z \in \mathcal{U}_N} |G_m(z)| \leq \frac{\tau}{\sqrt{N}} \Big\} \leq \frac{C_T}{N}.$$

Thus the natural N-point DFT grid has vanishing detection probability at the continuum boundary-minimum scale $t \asymp N^{-1/2}$.

Proof. Put $t = \tau/\sqrt{N}$ in the exact formula (3.8). Set

$$a_N(\tau) := \alpha(\tau/N), \qquad b_N(\tau) := \frac{\lfloor (N-1)/2 \rfloor}{N^2}\,\tau^2.$$

Uniformly for $0 \le \tau \le T$,

$$a_N(\tau) = \sqrt{\frac{2}{\pi}}\,\frac{\tau}{N} + O_T(N^{-3}), \qquad b_N(\tau) = \frac{\tau^2}{2N} + O_T(N^{-2}).$$

Since $s_N \le 2$ and $a_N + b_N = O_T(N^{-1})$, Taylor expansion gives

$$\begin{aligned}\mathbb{P}\Big\{\min_{z\in\mathcal{U}_N} |G_m(z)| \le \frac{\tau}{\sqrt{N}}\Big\} &= 1 - (1-a_N(\tau))^{s_N} e^{-b_N(\tau)} \\ &= s_N a_N(\tau) + b_N(\tau) + O_T(N^{-2}).\end{aligned}$$

Multiplying by N proves the displayed estimate, and the final $O_T(N^{-1})$ bound follows by increasing C_T. □

Remark 3.77 (The t^2 point bound is a grid-resolution artifact). The excursion of $\{g \le t\}$ predicted above has angular length of order $\delta \asymp t\,m^{-3/2}$: at curve speed $\asymp m^{3/2}$, dropping the modulus below t and recovering takes only $\Theta(t\,m^{-3/2})$ radians. A numerical scan of $\mathbb{T}$ on a grid of spacing h therefore *misses* such an excursion unless $h \ll t\,m^{-3/2}$; a coarse grid sees only the pointwise event $\{g(z_\ell) \le t\}$, of probability $\asymp t^2/m$ per non-real node. Proposition 3.74 makes this exact on the $N = m+1$ root grid: Corollary 3.75 shows that the root-grid minimum has a non-degenerate Rayleigh limit at fixed t, while the non-real small-t grid minimum has probability $\asymp t^2$, and the real endpoint coordinates multiply the survival probability by the exact one-dimensional factors in (3.8). Corollary 3.76 shows in particular that at the continuum scale $t = \tau/\sqrt{N}$, fixed τ, this root grid sees the event with the exact first-order coefficient $(s_N\sqrt{2/\pi}\,\tau + \tau^2/2)/N$, up to $O_\tau(N^{-2})$. The sharp continuum law is $\sqrt{m}\,t$ (Theorem 1.2): the t^2 pointwise rate is a sampling artifact, not the true small-ball exponent. We record this so the linear law is not re-broken by under-resolved numerics; a simulation must resolve the scale $t\,m^{-3/2}$ to see it.

3.3. **From single arcs to the uniform bound.**

Remark 3.78 (From arc bounds to the circle). The proof of Theorem 1.2 is the bookkeeping suggested by the single-arc scale. Partition $\mathbb{T}$ into $K = \lfloor c_1 m \rfloor$ arcs $I_1, \dots, I_K$ of arclength $\Theta(1/m)$. After enlarging the two endpoint arcs by a fixed number of neighbors, every remaining arc lies at distance $\ge \gamma_0/m$ from $\{\pm 1\}$, and Theorem 1.1 gives

$$\mathbb{P}\{\min_{I_k} |G_m| \le t\} \le C\,\frac{t}{\sqrt{m}}.$$

A union bound over $O(m)$ such arcs returns the whole-circle scale $C\sqrt{m}\,t$. The endpoint arcs are no longer a conditional input: Proposition 3.24 proves their contribution is $O(t/\sqrt{m})$, and Proposition 3.25 shows this local order is sharp. Corollary 3.54 records the resulting global endpoint floor. Thus the proved upper bound has the correct local one-dimensional rates at both the non-real arcs and the two parity endpoints; what remains for the sharp two-sided global law is the second-moment step that turns the $O(m)$ non-real local crossing opportunities into a lower-tail probability and an eventual limiting law.

Remark 3.79 (Decorrelation and the matching lower bound). The matching lower bound, and with it the exponential law, should follow from a second-moment argument in the spirit of the good-family estimate of [2]: with $Y_t = \#\{k : \min_{I_k} |G_m| \le t\}$ one needs

(3.10)
$$\mathbb{E}[\mathbf{1}\{\min_{I_k} |G_m| \le t\}\,\mathbf{1}\{\min_{I_{k'}} |G_m| \le t\}] \le (1+o(1))\,\mathbb{P}\{\min_{I_k} |G_m| \le t\}\,\mathbb{P}\{\min_{I_{k'}} |G_m| \le t\}$$

for arcs at angular separation $\gg 1/m$, the analogue of the two-slab estimate [2, Lem. 4.3]; the off-diagonal Dirichlet mass (2.1) between two separated arcs is $o(1)$ after normalization, so the joint field on $I_k \cup I_{k'}$ should factorize up to $o(1)$. Proposition 3.18 records the corresponding count-level implication: once the factorial second moment of the bulk down-crossing count is controlled, the fixed-bulk lower tail follows immediately. We do not need this lower bound for the conditioning theorem, which uses only an upper bound: Hypothesis 1.4 for general laws and Corollary 1.3 for Gaussian polynomial scales. The full two-sided law is the target of the minimum-modulus literature cited above.

4. From the boundary minimum to the finite section

We now turn the boundary control into a bound on $\sigma_{\min}(T_n)$. The mechanism (1.3) is made precise as follows.

Proposition 4.1 (Finite-section reduction). *Let $P(z) = z^{-r}G_m(z)$ be a banded symbol with $G_m(\zeta) \neq 0$ for all $\zeta \in \mathbb{T}$, and assume $\mathrm{wind}(P) = 0$; equivalently, in the balanced case, $\nu = r$. Set*

$$\mu := \min_{z\in\mathbb{T}} |G_m(z)| > 0, \qquad \rho := \min\{|\,|\zeta|-1\,| : G_m(\zeta) = 0\} > 0, \qquad M := \max_j |\xi_j|\,.$$

There is a constant $c_2 = c_2(m)$, polynomial in m, and a threshold $n_0 = n_0(m,\rho)$ such that for all $n \ge n_0$

$$\sigma_{\min}(T_n(P)) \;\ge\; c_2(m)^{-1}\,\mu\,\rho^{r}. \tag{4.1}$$

Consequently $\kappa(T_n) = \sigma_1(T_n)\sigma_{\min}(T_n)^{-1} \le c_2(m)\,(m+1)\,M\,\mu^{-1}\rho^{-r}$, since $\sigma_1(T_n) \le \|T_n\|_\infty \le (m+1)M$.

Proof sketch. This is the quantitative form of the Gohberg–Feldman inversion for banded symbols, e.g. [3, Ch. 3 and Thm. 3.7]. Since G_m has no zeros on $\mathbb{T}$ and $\mathrm{wind}(P) = 0$, P admits a Wiener–Hopf factorization $P = P_-P_+$

with $P_{\pm}^{\pm 1} \in H^{\infty}$ of the two half-planes; the finite-section inverses converge to $T(P^{-1})$ with an error that decays like ρ^n once $n \geq n_0$. The norm of the limiting inverse is controlled by $\left\|P_{+}^{-1}\right\| \left\|P_{-}^{-1}\right\|$, and for a degree-$m$ polynomial symbol these factors are bounded by the reciprocal boundary minimum μ^{-1} times a factor accounting for the location of the zeros, which contributes ρ^{-r} from the r zeros nearest the circle from inside. Collecting the polynomial-in-m prefactors into $c_2(m)$ gives (4.1). The norm bound $\sigma_1(T_n) \leq (m+1)M$ is the row-sum bound. □

Corollary 4.2 (Uniform threshold under a clearance floor)**.** *Fix m and $0 < \eta < 1$. There is a deterministic threshold $N_0(m, \eta) < \infty$ such that every degree-m balanced symbol $P = z^{-r} G_m$ with $G_m \neq 0$ on $\mathbb{T}$, $\operatorname{wind}(P) = 0$, boundary minimum $\mu > 0$, and circle clearance $\rho \geq \eta$ satisfies*

$$\sigma_{\min}(T_n(P)) \geq c_2(m)^{-1}\, \mu\, \rho^r \qquad (n \geq N_0(m, \eta)),$$

with the same polynomial-in-m constant $c_2(m)$ as in Proposition 4.1. Consequently, after fixing a deterministic clearance floor η_m, the finite-section part of the argument is uniform over all zero-winding symbols with $\rho \geq \eta_m$.

Proof. The proof of Proposition 4.1 is deterministic. If $\rho \geq \eta$, then the zeros of G_m avoid the annulus $\{1 - \eta/2 \leq |z| \leq 1 + \eta/2\}$. The Wiener–Hopf factors and their inverses are therefore rational functions whose poles stay a distance $\gtrsim \eta$ from $\mathbb{T}$. For fixed m this gives a uniform geometric tail bound for the finite-section error, of the form

$$\left\|T_n(P)^{-1} - T(P)^{-1}_{[n]}\right\| \leq C(m, \eta)\, \mu^{-1} \rho^{-r} q(\eta)^n, \qquad 0 < q(\eta) < 1,$$

where $T(P)^{-1}_{[n]}$ denotes the $n \times n$ corner of the limiting inverse. Choosing $N_0(m, \eta)$ so that this tail is at most one half of the limiting inverse bound gives the stated estimate. The coefficient size is not part of the threshold; it enters only later in the condition-number upper bound through $\sigma_1(T_n) \leq (m+1)M$. □

Proof of Theorem 1.6. Work in the balanced regime $r = s$, $m = 2r$. We prove the result on the zero-winding event $E0 := \{\nu = r\}$, intersected with three analytic events whose complements have probability $o(1)$.

(E1) **Clearance.** Let $E1 := \mathcal{C}_m$ be the event from Hypothesis 1.5. Its complement has probability $o(1)$, and on $E0 \cap E1$,

$$\rho \geq \frac{1}{C_0\, m^2 (\log m)^3},$$

so G_m has no zeros on $\mathbb{T}$. Put

$$\eta_m := \frac{1}{C_0 m^2 (\log m)^3}.$$

By Corollary 4.2, the finite-section estimate (4.1) is available uniformly on $E0 \cap E1$ for all $n \geq N_0(m, \eta_m)$.

(E2) **Boundary minimum.** Choose $A_0 > 2$ and set $E2 := \{\mu \geq m^{-A_0}\}$. Corollary 3.60 gives $\mathbb{P}(E2) = 1 - o(1)$. This is the only boundary-minimum input needed by the finite-section reduction.

(E3) **Coefficient size.** Set $E3 := \{M = \max_j |\xi_j| \leq m\}$. Since $\mathbb{E}\,|\xi_j|^2 = 1$, Chebyshev and a union bound give $\mathbb{P}(E3) = 1 - o(1)$, and $M \leq m$ is more than sufficient.

Set $\Omega_m := E1 \cap E2 \cap E3$. Then $\mathbb{P}(\Omega_m) = 1 - o(1)$, and by the union bound

$$\mathbb{P}(E0 \cap \Omega_m) = \mathbb{P}(E0) - o(1).$$

On $E0 \cap \Omega_m$, Corollary 4.2 gives, for $n \geq N_0(m, \eta_m)$,

$$\sigma_{\min}(T_n) \geq c_2(m)^{-1}\,\mu\,\rho^r \geq c_2(m)^{-1}\,m^{-A_0}\left(\frac{1}{C_0 m^2 (\log m)^3}\right)^{m/2}.$$

On Ω_m, the lower bound is the deterministic quantity

$$\delta_m := c_2(m)^{-1} m^{-A_0} \big(C_0 m^2 (\log m)^3\big)^{-m/2} > 0,$$

and the condition-number estimate in Proposition 4.1 gives the deterministic upper bound

$$\kappa(T_n) \leq \Gamma_m := c_2(m)(m+1)m\,m^{A_0}\big(C_0 m^2 (\log m)^3\big)^{m/2}.$$

Given the prescribed exponent $b > 0$, take $n_1(m, b)$ so that $n^{-b} \leq \delta_m$ and $n^b \geq \Gamma_m$ for all $n \geq n_1(m, b)$. Then

$$n_{0,b}(m) := \max\{N_0(m, \eta_m), n_1(m, b)\}$$

is deterministic and gives (1.10). □

Proof of Corollary 1.7. Repeat the proof of Theorem 1.6, but keep the losses explicit. Put

$$E1 := \mathcal{C}_m, \qquad E2 := \{\mu \geq m^{-A_0}\}, \qquad E3 := \{M = \max_j |\xi_j| \leq R_m\}.$$

The assumed clearance estimate gives $\mathbb{P}(E1^c) \leq q_m$. By Corollary 3.60,

$$\mathbb{P}(E2^c) \leq C_{A_0,K} m^{-2(A_0-1)/3}.$$

Finally, Chebyshev's inequality and the union bound give

$$\mathbb{P}(E3^c) \leq \sum_{j=0}^{m} \mathbb{P}\{|\xi_j| > R_m\} \leq \frac{m+1}{R_m^2}.$$

Let $\Omega_{m,A_0,R} := E1 \cap E2 \cap E3$. The displayed probability bound for $\Omega^c_{m,A_0,R}$ follows by the union bound, and intersecting with $\{\nu = r\}$ costs no more.

On $\Omega_{m,A_0,R} \cap \{\nu = r\}$, set

$$\eta_m = (C_0 m^2 (\log m)^3)^{-1}.$$

Corollary 4.2 applies for all $n \geq N_0(m, \eta_m)$. Hence

$$\sigma_{\min}(T_n) \geq c_2(m)^{-1} m^{-A_0} \big(C_0 m^2 (\log m)^3\big)^{-m/2} =: \delta_{m,A_0} > 0,$$

and the condition number is at most

$$\Gamma_{m,A_0,R} := c_2(m)(m+1)R_m\, m^{A_0}\big(C_0 m^2(\log m)^3\big)^{m/2}.$$

Choose $n_{0,b,A_0,R}(m) \geq N_0(m,\eta_m)$ so large that

$$n_{0,b,A_0,R}(m)^{-b} \leq \delta_{m,A_0}, \qquad n_{0,b,A_0,R}(m)^{b} \geq \Gamma_{m,A_0,R}.$$

Then the same inequalities hold for every $n \geq n_{0,b,A_0,R}(m)$. This deterministic threshold gives the claimed singular-value and condition-number bounds. □

Proof of Theorem 1.10. For Gaussian coefficients, Hypothesis 1.5 is supplied by the Konyagin–Schlag clearance theorem [4]. The conclusion is therefore Theorem 1.6. The sharper Gaussian boundary estimate Corollary 1.3 remains useful for the full boundary-law discussion, but is no longer needed for finite-section conditioning. □

Remark 4.3 (The honest content of the bound). The bound (1.10) is a statement about the n-asymptotics at fixed bandwidth m on the stable subevent: for any prescribed polynomial exponent $b > 0$, the balanced finite sections do *not* suffer the exponential-in-n blow-up of the unbalanced regime once the winding number is zero and the symbol stays off $\mathbb{T}$, after increasing the deterministic threshold $n_{0,b}(m)$. The theorem deliberately makes no high-probability claim for stability itself, which is constrained by the parity obstruction (1.2). The prefactors are exponentially small in m, reflecting the rare-but-possible near-degeneracies of the boundary minimum and the circle clearance; we make no claim that the m-dependence is sharp.

Remark 4.4 (Single arc versus the whole circle). The single-arc bound (Theorem 1.1) controls $|G_m|$ only on one arc I_0, not on all of $\mathbb{T}$, and so by itself does not give (4.1), which needs the global minimum μ. Its fixed-arc consequence would be a lower bound on the symbol restricted to a favorable arc, equivalently a localized resolvent estimate; promoting it to the full $\sigma_{\min}(T_n)$ is exactly the role of the uniform bound (1.8).

5. The bridge to the parity obstruction

We make precise the moral asserted in Section 1.4: the parity observable of [1] and the boundary small-ball of this paper are two readings of the same boundary evaluations. For Rademacher coefficients in the balanced convention $m = 2r$, the exact zero story is subtler than in adjacent parities because $m+1$ is odd.

5.1. **The parity observable is a boundary evaluation.** Recall from [1, Lem. 2.1] that, since G_m has real coefficients, its non-real zeros come in conjugate pairs of equal modulus, so

(5.1)
$$\nu \equiv \#\{\text{real zeros of } G_m \text{ in } (-1,1)\} \equiv \mathbf{1}\{G_m(1)G_m(-1) < 0\} \pmod 2,$$

the last congruence holding when $G_m(\pm 1) \neq 0$. The two numbers $G_m(1) = \sum_j \xi_j$ and $G_m(-1) = \sum_j (-1)^j \xi_j$ are precisely the evaluations of the boundary

field $G_m|_{\mathbb{T}}$ at the two real points $z = \pm 1$ — the same two points at which, in Lemma 3.1, the field collapses from two real dimensions to one and the small-ball exponent degrades from t^2 to t. Thus the parity obstruction lives at the two real boundary points, and the boundary small-ball estimate is its quantitative refinement: (5.1) reads only the *sign* of the boundary field at ± 1, while $\min_{\mathbb{T}} |G_m|$ reads its *magnitude* along all of $\mathbb{T}$.

5.2. **The Bernoulli lattice obstruction.**

Proposition 5.1 (Endpoint lattice scale for Rademacher coefficients)**.** *Let $\varepsilon_0, \dots, \varepsilon_m$ be independent Rademacher signs and let $G_m(z) = \sum_{j=0}^m \varepsilon_j z^j$. Put $N = m + 1$. There is an absolute constant $C < \infty$ such that, for each $\zeta \in \{1, -1\}$, each $a \in \mathbb{R}$, and each $t \geq 0$,*

$$\mathbb{P}\{|G_m(\zeta) - a| \leq t\} \leq C \, \frac{1+t}{\sqrt{N}}.$$

In particular,

$$\mathbb{P}\{|G_m(\zeta)| \leq t\} \leq C \, \frac{1+t}{\sqrt{N}}.$$

Moreover, if N is even then

$$\mathbb{P}\{G_m(\zeta) = 0\} \asymp N^{-1/2},$$

whereas if N is odd then

$$\mathbb{P}\{|G_m(\zeta)| = 1\} \asymp N^{-1/2}.$$

Thus the endpoint small-ball scale for lattice coefficients is governed by the lattice spacing; a density-type bound $Ct/\sqrt{N}$, uniformly in the center, cannot hold below that spacing.

Proof. For $\zeta = 1$, $G_m(1) = \sum_{j=0}^m \varepsilon_j$. For $\zeta = -1$, the signs $(-1)^j \varepsilon_j$ are again independent Rademacher signs, so $G_m(-1)$ has the same law. It is therefore enough to consider $S_N = \sum_{j=1}^N \varepsilon_j$. The variable S_N is supported on the lattice $\{k \in \mathbb{Z} : k \equiv N \pmod 2\}$, and

$$\mathbb{P}\{S_N = k\} = 2^{-N} \binom{N}{(N+k)/2}$$

whenever $N + k$ is even. By Stirling's formula, uniformly in k,

$$\mathbb{P}\{S_N = k\} \leq C N^{-1/2}.$$

The interval $[a - t, a + t]$ has length $2t$, hence contains at most $\lfloor t \rfloor + 2$ points of this parity sublattice. Therefore

$$\mathbb{P}\{|S_N - a| \leq t\} \leq C(\lfloor t \rfloor + 2) N^{-1/2} \leq C' \frac{1+t}{\sqrt{N}}.$$

When N is even, $\mathbb{P}\{S_N = 0\} = 2^{-N} \binom{N}{N/2} \asymp N^{-1/2}$. When N is odd, the two central values are ± 1, and $\mathbb{P}\{|S_N| = 1\} = 2^{1-N} \binom{N}{(N+1)/2} \asymp N^{-1/2}$. □

Proposition 5.2 (Root-of-unity lattice scale for Rademacher coefficients)**.** *Let $\varepsilon_0, \dots, \varepsilon_m$ be independent Rademacher signs, let $G_m(z) = \sum_{j=0}^{m} \varepsilon_j z^j$, and put $N = m+1$. Fix a root of unity $\zeta \notin \{1, -1\}$, of order $q \geq 3$. There is an absolute constant $C < \infty$ such that, for every $a \in \mathbb{C}$ and every $t \geq 0$,*

$$\mathbb{P}\{|G_m(\zeta) - a| \leq t\} \leq C\, \frac{q(1+t)^2}{N}.$$

Thus, for each fixed order q, a non-real root of unity has the two-dimensional fixed-point lattice scale N^{-1}, in contrast to the one-dimensional endpoint scale $N^{-1/2}$ in Proposition 5.1.

Proof. It is enough to consider $N \geq 4q$, since otherwise the right-hand side is at least a positive absolute constant and the claim follows by increasing C. Choose an integer $k \in \{1, \dots, q-1\}$ such that

$$\frac{\pi}{3} \leq \frac{2\pi k}{q} \leq \frac{2\pi}{3};$$

such a k exists for every $q \geq 3$. Then $\left|\Im(\zeta^k)\right| \geq \sqrt{3}/2$. For $0 \leq \ell < q$, let

$$J_\ell = \{0 \leq j \leq m : j \equiv \ell \pmod q\}, \qquad S_\ell = \sum_{j \in J_\ell} \varepsilon_j.$$

Then the S_ℓ's are independent Rademacher sums, the two classes J_0, J_k satisfy $|J_0|, |J_k| \geq N/(2q)$, and

$$G_m(\zeta) = \sum_{\ell=0}^{q-1} \zeta^\ell S_\ell.$$

Because $\zeta^k \notin \mathbb{R}$ with imaginary part bounded below, the real-linear map $(x, y) \mapsto x + \zeta^k y$ from $\mathbb{R}^2$ to $\mathbb{C}$ has inverse norm bounded by an absolute constant. Condition on all S_ℓ with $\ell \notin \{0, k\}$. For the resulting center $b \in \mathbb{C}$, the set of real pairs (x, y) satisfying $\left|x + \zeta^k y - b\right| \leq t$ has diameter $O(1+t)$ in both coordinate directions, hence contains at most $C(1+t)^2$ points of the parity lattice on which (S_0, S_k) lives.

By Stirling's formula, a Rademacher sum of length M has point probabilities bounded by $CM^{-1/2}$. Therefore each admissible pair $(S_0, S_k) = (x, y)$ has probability at most

$$C\, |J_0|^{-1/2}\, |J_k|^{-1/2} \leq C\, \frac{q}{N}.$$

Multiplying by the $O((1+t)^2)$ admissible pairs and then averaging over the conditioned variables gives the stated bound. □

Corollary 5.3 (Rademacher anti-concentration on a rational grid)**.** *Let $G_m(z) = \sum_{j=0}^{m} \varepsilon_j z^j$ with independent Rademacher signs, put $N = m+1$, and let $\Omega_q = \{\zeta \in \mathbb{C} : \zeta^q = 1\}$. There is an absolute constant $C < \infty$ such*

that, for every $q \geq 1$ and $t \geq 0$,

$$\mathbb{P}\Big\{\min_{\zeta\in\Omega_q\setminus\{1,-1\}} |G_m(\zeta)| \leq t\Big\} \leq C\,\frac{q^2(1+t)^2}{N},$$

and, including the endpoint roots,

$$\mathbb{P}\Big\{\min_{\zeta\in\Omega_q} |G_m(\zeta)| \leq t\Big\} \leq C\,\frac{1+t}{\sqrt{N}} + C\,\frac{q^2(1+t)^2}{N}.$$

Thus rational grids of order $q = o(\sqrt{N})$ have no small Rademacher boundary value with high probability at any fixed lattice scale, apart from the endpoint contribution.

Proof. For the non-real roots, apply Proposition 5.2 to each $\zeta \in \Omega_q \setminus \{1, -1\}$ and sum the resulting $O(q(1+t)^2/N)$ bounds over at most q roots. The endpoint roots are handled by Proposition 5.1; there are at most two of them, and their contribution is $O((1+t)/\sqrt{N})$. □

Corollary 5.4 (Bounded-denominator roots)**.** *Let $G_m(z) = \sum_{j=0}^m \varepsilon_j z^j$ with independent Rademacher signs and put $N = m+1$. For $Q \geq 1$, set*

$$\mathcal{R}_Q := \{\zeta \in \mathbb{T} : \zeta^q = 1 \text{ for some } 1 \leq q \leq Q\}.$$

There is an absolute constant $C < \infty$ such that, for every $Q \geq 1$ and $t \geq 0$,

$$\mathbb{P}\Big\{\min_{\zeta\in\mathcal{R}_Q\setminus\{1,-1\}} |G_m(\zeta)| \leq t\Big\} \leq C\,\frac{Q^3(1+t)^2}{N},$$

and, including the endpoint roots,

$$\mathbb{P}\Big\{\min_{\zeta\in\mathcal{R}_Q} |G_m(\zeta)| \leq t\Big\} \leq C\,\frac{1+t}{\sqrt{N}} + C\,\frac{Q^3(1+t)^2}{N}.$$

Thus all rational angles of denominator $Q = o(N^{1/3})$ are separated at any fixed lattice scale with high probability, apart from the endpoint contribution.

Proof. Apply Corollary 5.3 for each $1 \leq q \leq Q$ and take a union bound. The duplicate representation of the same root at several denominators only makes this union bound larger. Since $\sum_{q\leq Q} q^2 \leq CQ^3$, the non-real bound follows. The two endpoint roots are handled once by Proposition 5.1. □

Proposition 5.5 (Fixed-root exact atom local limit)**.** *Let $G_m(z) = \sum_{j=0}^m \varepsilon_j z^j$ with independent Rademacher signs and put $N = m+1$. Let $q \geq 3$ be fixed, let ζ be a primitive q-th root of unity, and put $d = \varphi(q)$. For $0 \leq \ell < q$, let*

$$M_\ell(N) := \#\{0 \leq j < N : j \equiv \ell \pmod q\}, \qquad a_\ell(N) \equiv M_\ell(N) \pmod 2,$$

and let

$$L : \mathbb{Z}^q \to \mathbb{Z}[\zeta], \qquad L(s_0, \dots, s_{q-1}) = \sum_{\ell=0}^{q-1} s_\ell \zeta^\ell, \qquad K := \ker L.$$

Let $\operatorname{covol}(K)$ *denote the Euclidean covolume of* K *in* $K \otimes_{\mathbb{Z}} \mathbb{R} \subset \mathbb{R}^q$*, and set*

$$\gamma_q := \frac{(2q/\pi)^{\varphi(q)/2}}{\operatorname{covol}(K)}.$$

Then, for all N*,*

$$\mathbb{P}\{G_m(\zeta) = 0\} \le C_q N^{-d/2}.$$

Moreover fix a residue class $R \pmod{2q}$*, and write* $a^R = a(R)$ *for the constant value of the parity vector* $a(N)$ *on* $N \equiv R \pmod{2q}$*. If*

$$K \cap \{s \in \mathbb{Z}^q : s_\ell \equiv a_\ell(R) \pmod 2 \text{ for all } \ell\} \neq \varnothing,$$

then, along that residue class,

$$N^{\varphi(q)/2}\mathbb{P}\{G_m(\zeta) = 0\} \longrightarrow \gamma_q.$$

If the displayed intersection is empty, then $\mathbb{P}\{G_m(\zeta) = 0\} = 0$ *for every* $N \equiv R \pmod{2q}$*. In particular the old divisible case* $q \mid N$ *is always admissible with this same local limit constant, the primitive quadratic roots* $q = 3, 4, 6$ *have possible exact atom scale* N^{-1}*, and a primitive root of prime order* p *has possible exact atom scale* $N^{-(p-1)/2}$*.*

Proof. Group the coefficients by residues modulo q:

$$S_\ell = \sum_{\substack{0 \le j \le m \\ j \equiv \ell \pmod q}} \varepsilon_j, \qquad 0 \le \ell < q.$$

The variables $S_0, \dots, S_{q-1}$ are independent Rademacher sums of lengths $M_\ell(N)$, and $M_\ell(N) = N/q + O_q(1)$. The image of L has rank $d := \varphi(q)$, because $\mathbb{Z}[\zeta]$ is generated by $1, \zeta, \dots, \zeta^{d-1}$ and has rank d. Hence the kernel lattice K has rank $q-d$. The event $\{G_m(\zeta) = 0\}$ is exactly $\{(S_0, \dots, S_{q-1}) \in K\}$.

For the upper bound, choose a set $J \subset \{0, \dots, q-1\}$ of size $q-d$ such that the coordinate projection $\pi_J : K \to \mathbb{Z}^J$ is injective; this is possible by choosing a nonzero maximal minor in a basis matrix for the fixed lattice K. If $p_\ell(s) = \mathbb{P}\{S_\ell = s\}$, Stirling's formula gives $\sup_s p_\ell(s) \le C_q N^{-1/2}$ for all large N. Therefore

$$\sum_{s \in K} \prod_{\ell=0}^{q-1} p_\ell(s_\ell) \le C_q N^{-d/2} \sum_{u \in \pi_J(K)} \prod_{j \in J} p_j(u_j) \le C_q N^{-d/2}.$$

The last sum is at most 1, since it is a sub-sum of the product law of the $q-d$ free coordinates.

Now fix a residue class $R \pmod{2q}$. The parities $a_\ell(N)$ are constant on this class. If no point of K has these coordinate parities, then every point in the support of $(S_0, \dots, S_{q-1})$ lies outside K, so the atom is zero.

Assume instead that $a^R \in K$ has the required coordinate parities. Then the admissible support points in K are exactly the coset $a^R + 2K$: if $x \in K \cap 2\mathbb{Z}^q$, then $x = 2y$ with $y \in \mathbb{Z}^q$, and torsion-freeness of $\mathbb{Z}[\zeta]$ gives $y \in K$.

We now apply the local central limit theorem on this fixed lattice coset. The Hoeffding bound makes the contribution of $\max_\ell |s_\ell| > N^{2/3}$ exponentially small. Uniformly for $\max_\ell |s_\ell| \le N^{2/3}$ on the support parities,

$$p_\ell(s_\ell) = \sqrt{\frac{2}{\pi M_\ell(N)}} \exp\left(-\frac{s_\ell^2}{2M_\ell(N)}\right)(1+o(1)).$$

Since $M_\ell(N) = N/q + O_q(1)$, uniformly in the same central range,

$$\prod_{\ell=0}^{q-1} p_\ell(s_\ell) = \left(\frac{2q}{\pi N}\right)^{q/2} \exp\left(-\frac{q\|s\|_2^2}{2N}\right)(1+o(1)).$$

Let $k = q - d = \operatorname{rank} K$. The Riemann-sum limit for the coset $a^R + 2K$ in $K \otimes \mathbb{R}$ gives

$$\sum_{s \in a^R + 2K} \exp\left(-\frac{q\|s\|_2^2}{2N}\right) = \frac{N^{k/2}}{\operatorname{covol}(2K)} \left(\frac{2\pi}{q}\right)^{k/2} + o(N^{k/2}).$$

As $\operatorname{covol}(2K) = 2^k \operatorname{covol}(K)$, multiplying the last two displays yields

$$\mathbb{P}\{G_m(\zeta) = 0\} = \frac{(2q/\pi)^{d/2}}{\operatorname{covol}(K)} N^{-d/2} + o(N^{-d/2}).$$

Finally, if $q \mid N$, then all $M_\ell(N) = M = N/q$ have the same parity. When M is even the zero vector is an admissible point of K; when M is odd the vector $\mathbf{1} = (1, \dots, 1)$ is admissible because $\sum_{\ell=0}^{q-1} \zeta^\ell = 0$. Thus every divisible residue class is admissible and has the same local-limit constant γ_q. □

Corollary 5.6 (Finite cyclotomic-layer exact-zero asymptotic)**.** *Let $\mathcal{S}$ be a fixed nonempty finite set of integers $q \ge 3$, and for each $q \in \mathcal{S}$ let E_q be the event $G_m(\zeta_q) = 0$, where ζ_q is any primitive q-th root of unity. This event is independent of the chosen primitive root. Put*

$$d_* = \min_{q \in \mathcal{S}} \varphi(q),$$

and let γ_q be the covolume constant of Proposition 5.5. Then

$$\mathbb{P}\Big(\bigcup_{q \in \mathcal{S}} E_q\Big) = N^{-d_*/2} \sum_{\substack{q \in \mathcal{S} \\ \varphi(q) = d_*}} \gamma_q \, \mathbf{1}_{\{N \bmod 2q \text{ is admissible for } q\}} + o_{\mathcal{S}}(N^{-d_*/2}).$$

Thus every fixed finite rational grid has a complete leading exact-zero asymptotic on its first nonempty cyclotomic degree layer.

Proof. Because G_m has integer coefficients, $G_m(\zeta_q) = 0$ for one primitive q-th root if and only if the cyclotomic polynomial Φ_q divides G_m, hence if and only if G_m vanishes at every primitive q-th root. Thus E_q is well defined.

The single-order asymptotic is Proposition 5.5. The total contribution of orders with $\varphi(q) > d_*$ is $o_{\mathcal{S}}(N^{-d_*/2})$. It remains only to show that

intersections between distinct orders in the first layer are lower order. Let $q \neq r$ and put $L = \operatorname{lcm}(q, r)$. Group the signs by residue classes modulo L:

$$S_a = \sum_{\substack{0 \le j < N \\ j \equiv a \pmod{L}}} \varepsilon_j, \qquad 0 \le a < L.$$

The two equations defining $E_q \cap E_r$ are the kernel condition for the $\mathbb{Z}$-linear map

$$T : \mathbb{Z}^L \to \mathbb{Z}[\zeta_q] \oplus \mathbb{Z}[\zeta_r], \qquad T(s) = \left(\sum_{a=0}^{L-1} s_a \zeta_q^a, \sum_{a=0}^{L-1} s_a \zeta_r^a \right).$$

After tensoring with $\mathbb{Q}$, this is the product quotient map from $\mathbb{Q}[x]/(x^L - 1)$ to $\mathbb{Q}[x]/(\Phi_q) \oplus \mathbb{Q}[x]/(\Phi_r)$. Since Φ_q and Φ_r are distinct irreducible factors of $x^L - 1$, its rank is $\varphi(q) + \varphi(r)$. The same projection-and-largest-atom argument used for Proposition 5.5 therefore gives

$$\mathbb{P}(E_q \cap E_r) \le C_{q,r} N^{-(\varphi(q)+\varphi(r))/2}.$$

For q, r in the first layer this is $O_{\mathcal{S}}(N^{-d_*})$, hence $o_{\mathcal{S}}(N^{-d_*/2})$. Inclusion–exclusion over the finite first layer completes the proof. □

Corollary 5.7 (Primitive quadratic exact atom constants)**.** *Let $G_m(z) = \sum_{j=0}^{m} \varepsilon_j z^j$ with independent Rademacher signs, and put $N = m + 1$. If ζ is a primitive third or sixth root of unity, then*

$$\mathbb{P}\{G_m(\zeta) = 0\} = 0 \qquad \textit{unless } 3 \mid N,$$

while, as $N \to \infty$ through multiples of 3,

$$N\mathbb{P}\{G_m(\zeta) = 0\} \longrightarrow \frac{2\sqrt{3}}{\pi}.$$

If ζ is a primitive fourth root of unity, then

$$\mathbb{P}\{G_m(\zeta) = 0\} = 0 \qquad \textit{unless } 4 \mid N,$$

while, as $N \to \infty$ through multiples of 4,

$$N\mathbb{P}\{G_m(\zeta) = 0\} \longrightarrow \frac{4}{\pi}.$$

Thus the cyclotomic degree-two exact-zero floor is completely visible at the level of divisibility classes and constants: cubic and primitive sixth roots require $3 \mid N$ and have constant $2\sqrt{3}/\pi$, while quartic roots require $4 \mid N$ and have constant $4/\pi$.

Proof. For $q = 3$, write $S_\ell = \sum_{j \equiv \ell \pmod 3} \varepsilon_j$. Since $1 + \zeta + \zeta^2 = 0$, the identity

$$S_0 + S_1\zeta + S_2\zeta^2 = 0$$

is equivalent to $S_0 = S_1 = S_2$. Such equality is compatible with the support parities exactly when $M_0(N), M_1(N), M_2(N)$ have the same parity. Writing $N = 3a + r$, the counts are

$$(M_0, M_1, M_2) = (a + \mathbf{1}_{r \ge 1},\ a + \mathbf{1}_{r \ge 2},\ a), \qquad r = 0, 1, 2,$$

so the parities agree exactly when $r = 0$, i.e. $3 \mid N$. When $N = 3L$, the three residue sums are independent copies of a length-L Rademacher sum. If $p_L(s)$ denotes the mass function of such a sum, then

$$\mathbb{P}\{G_m(\zeta) = 0\} = \sum_s p_L(s)^3.$$

Stirling's formula gives, uniformly for $|s| \le L^{2/3}$ on the support parity,

$$p_L(s) = \sqrt{\frac{2}{\pi L}} \exp\left(-\frac{s^2}{2L}\right)(1 + o(1)),$$

and the Hoeffding tail makes the complementary range contribute $o(L^{-1})$ to the cubic sum. Since the support lattice has spacing 2,

$$\begin{aligned}\sum_s p_L(s)^3 &= \frac{1}{2}\int_{\mathbb{R}} \left(\frac{2}{\pi L}\right)^{3/2} \exp\left(-\frac{3x^2}{2L}\right) dx + o(L^{-1}) \\ &= \frac{2}{\pi\sqrt{3}}\,\frac{1}{L} + o(L^{-1}).\end{aligned}$$

Multiplying by $N = 3L$ gives the limit $2\sqrt{3}/\pi$.

For $q = 4$, a primitive root is $\zeta = \pm i$. Grouping the residue sums modulo 4,

$$G_m(\zeta) = S_0 + \zeta S_1 - S_2 - \zeta S_3,$$

so $G_m(\zeta) = 0$ is equivalent to $S_0 = S_2$ and $S_1 = S_3$. The necessary and sufficient parity conditions are therefore

$$M_0 \equiv M_2 \pmod 2, \qquad M_1 \equiv M_3 \pmod 2.$$

For $N = 4a + r$,

$$(M_0, M_1, M_2, M_3) = (a + \mathbf{1}_{r\ge 1},\, a + \mathbf{1}_{r\ge 2},\, a + \mathbf{1}_{r\ge 3},\, a),$$

and the two congruences hold exactly when $r = 0$. When $N = 4L$, the four residue sums are independent copies of a length-L Rademacher sum, so

$$\mathbb{P}\{G_m(\zeta) = 0\} = \left(\sum_s p_L(s)^2\right)^2.$$

Vandermonde's identity and Stirling's formula give

$$\sum_s p_L(s)^2 = 4^{-L}\binom{2L}{L} = \frac{1}{\sqrt{\pi L}}(1 + o(1)).$$

Thus $\mathbb{P}\{G_m(\zeta) = 0\} = (\pi L)^{-1} + o(L^{-1})$, and multiplying by $N = 4L$ gives the limit $4/\pi$.

For $q = 6$, let ζ be primitive, so $\zeta^3 = -1$ and $\zeta^2 - \zeta + 1 = 0$. Put

$$D_0 = S_0 - S_3, \qquad D_1 = S_1 - S_4, \qquad D_2 = S_2 - S_5.$$

Then

$$G_m(\zeta) = D_0 + D_1\zeta + D_2\zeta^2 = (D_0 - D_2) + (D_1 + D_2)\zeta,$$

so $G_m(\zeta) = 0$ is equivalent to $D_0 = D_2 = -D_1$. The parity of D_j is $M_j + M_{j+3} \pmod 2$. For $N = 6a + r$,

$$\begin{aligned} M_0 + M_3 &\equiv \mathbf{1}_{r\geq 1} + \mathbf{1}_{r\geq 4}, \\ M_1 + M_4 &\equiv \mathbf{1}_{r\geq 2} + \mathbf{1}_{r\geq 5}, \\ M_2 + M_5 &\equiv \mathbf{1}_{r\geq 3} \qquad \pmod 2. \end{aligned}$$

These three parities are equal exactly for $r = 0$ and $r = 3$, equivalently $3 \mid N$. When they are unequal, no zero is possible. When $N = 3L$, the random variables D_0, D_1, D_2 are independent and each has the law of a length-L Rademacher sum: if $N = 6a$ then $L = 2a$ and D_j is the difference of two length-a sums, while if $N = 6a + 3$ then $L = 2a + 1$ and D_j is the difference of a length-$a + 1$ sum and a length-a sum. By symmetry,

$$\mathbb{P}\{G_m(\zeta) = 0\} = \sum_s p_L(s)p_L(-s)p_L(s) = \sum_s p_L(s)^3.$$

The $q = 3$ computation therefore gives the same limit $2\sqrt{3}/\pi$. □

Corollary 5.8 (Pairwise primitive-quadratic overlap constants)**.** *Let E_q denote the event that G_m vanishes at a primitive q-th root, for $q \in \{3, 4, 6\}$. Then*

$$\mathbb{P}(E_3 \cap E_6) = 0 \qquad \textit{unless } 6 \mid N,$$

and, as $N \to \infty$ through multiples of 6,

$$N^2\mathbb{P}(E_3 \cap E_6) \longrightarrow \frac{48}{\pi^2}.$$

Moreover

$$\mathbb{P}(E_3 \cap E_4) = \mathbb{P}(E_4 \cap E_6) = 0 \qquad \textit{unless } 12 \mid N,$$

and, as $N \to \infty$ through multiples of 12,

$$N^2\mathbb{P}(E_3 \cap E_4) \longrightarrow \frac{8\sqrt{3}}{\pi^2}, \qquad N^2\mathbb{P}(E_4 \cap E_6) \longrightarrow \frac{8\sqrt{3}}{\pi^2}.$$

Proof. First group the coefficients modulo 6, and write the corresponding sums as $T_0, \dots, T_5$. The event E_3 is

$$T_0 + T_3 = T_1 + T_4 = T_2 + T_5,$$

while the calculation in the preceding proof shows that E_6 is

$$T_0 - T_3 = T_2 - T_5 = -(T_1 - T_4).$$

These two systems are equivalent to

$$T_0 = T_2 = T_4, \qquad T_1 = T_3 = T_5.$$

The parity requirements for the three even residue classes and the three odd residue classes are simultaneously compatible exactly when $6 \mid N$. If $N = 6L$, the six sums are independent length-L Rademacher sums. Hence

$$\mathbb{P}(E_3 \cap E_6) = \left(\sum_t p_L(t)^3\right)^2 = \left(\frac{2}{\pi\sqrt{3}}\, L^{-1} + o(L^{-1})\right)^2,$$

which gives the constant $48/\pi^2$ after multiplying by $N^2 = 36L^2$.

For $E_3 \cap E_4$, the individual admissibility criteria in Corollary 5.7 already force $12 \mid N$. Assume $N = 12L$. Identify the residue classes modulo 12 with the 3×4 grid $(a, b) \in \mathbb{Z}/3\mathbb{Z} \times \mathbb{Z}/4\mathbb{Z}$, and let $X_{a,b}$ be the corresponding independent length-L sums. Let

$$R_a = \sum_{b=0}^{3} X_{a,b}, \qquad C_b = \sum_{a=0}^{2} X_{a,b}.$$

Then $E_3 \cap E_4$ is the fixed-rank lattice event

$$R_0 = R_1 = R_2, \qquad C_0 = C_2, \qquad C_1 = C_3.$$

Equivalently $AX = 0$, where

$$AX = (R_0 - R_1,\ R_0 - R_2,\ C_0 - C_2,\ C_1 - C_3).$$

The four rows of A have Gram matrix

$$AA^T = \begin{pmatrix} 8 & 4 & 0 & 0 \\ 4 & 8 & 0 & 0 \\ 0 & 0 & 6 & 0 \\ 0 & 0 & 0 & 6 \end{pmatrix},$$

whose determinant is 1728. The 4×4 minor using columns

$$(0,0), \qquad (0,1), \qquad (1,0), \qquad (2,2)$$

has determinant 1. Thus the standard covolume formula for the kernel of a full-rank integer map gives

$$\operatorname{covol}(\ker A) = \sqrt{\det(AA^T)} = 24\sqrt{3}.$$

The same fixed-lattice local central limit argument used in Proposition 5.5 gives

$$\mathbb{P}(E_3 \cap E_4) = \frac{(2/\pi)^2}{24\sqrt{3}}\, L^{-2} + o(L^{-2}).$$

Since $N = 12L$, this is $8\sqrt{3}/(\pi^2 N^2) + o(N^{-2})$.

Finally, for $E_4 \cap E_6$, the individual criteria force $12 \mid N$ again. With residues grouped modulo 12, the four defining equations have Gram matrix

$$\begin{pmatrix} 6 & 0 & 0 & 0 \\ 0 & 6 & 0 & 0 \\ 0 & 0 & 8 & -4 \\ 0 & 0 & -4 & 8 \end{pmatrix},$$

and the minor using residues $0, 1, 2, 3$ has determinant 1. The kernel covolume is again $24\sqrt{3}$, and the same local limit gives the same constant $8\sqrt{3}/\pi^2$. □

Corollary 5.9 (Primitive-quadratic atom expansions to second order)**.** *Let E_q denote the event that G_m vanishes at a primitive q-th root, for $q \in \{3, 4, 6\}$. Then*

$$\mathbb{P}(E_3) = \mathbf{1}_{\{3 \mid N\}} \left(\frac{2\sqrt{3}}{\pi N} - \frac{2\sqrt{3}}{\pi N^2} \right) + o(N^{-2}),$$

$$\mathbb{P}(E_6) = \mathbf{1}_{\{3|N\}} \left(\frac{2\sqrt{3}}{\pi N} - \frac{2\sqrt{3}}{\pi N^2} \right) + o(N^{-2}),$$

and

$$\mathbb{P}(E_4) = \mathbf{1}_{\{4|N\}} \left(\frac{4}{\pi N} - \frac{4}{\pi N^2} \right) + o(N^{-2}).$$

Proof. The inadmissible classes are exactly those identified in Corollary 5.7. It remains to sharpen the admissible binomial sums. Let p_L be the mass function of a length-L Rademacher sum. Stirling's formula, uniformly on the support for $|s| \le L^{1/2+\eta}$ with any fixed $0 < \eta < 1/6$, gives

$$p_L(s) = \sqrt{\frac{2}{\pi L}} \exp\left(-\frac{x^2}{2}\right) \left(1 - \frac{x^4 - 6x^2 + 3}{12L} + O_\eta(L^{-2}(1 + |x|^{12}))\right),$$
$$x = \frac{s}{\sqrt{L}}.$$

The complement contributes $O(e^{-cL^{2\eta}})$ to the cubic sum. Using the spacing $2/\sqrt{L}$ of the support lattice and the Gaussian Riemann-sum limit,

$$\begin{aligned}\sum_s p_L(s)^3 &= \frac{1}{2} \left(\frac{2}{\pi L}\right)^{3/2} L^{1/2} \int_{\mathbb{R}} e^{-3x^2/2} \left(1 - \frac{x^4 - 6x^2 + 3}{4L}\right) dx + o(L^{-2}) \\ &= \frac{2}{\pi\sqrt{3}} L^{-1} - \frac{2}{3\pi\sqrt{3}} L^{-2} + o(L^{-2}).\end{aligned}$$

The identities in the proof of Corollary 5.7 show that both E_3 and E_6, on their admissible classes, are given by this cubic sum with $L = N/3$. This yields the first two expansions.

For E_4, on $N = 4L$,

$$\mathbb{P}(E_4) = \left(\sum_s p_L(s)^2\right)^2 = \left(4^{-L}\binom{2L}{L}\right)^2.$$

The central-binomial expansion

$$4^{-L}\binom{2L}{L} = \frac{1}{\sqrt{\pi L}} \left(1 - \frac{1}{8L} + O(L^{-2})\right)$$

therefore gives

$$\mathbb{P}(E_4) = \frac{1}{\pi L} - \frac{1}{4\pi L^2} + o(L^{-2}) = \frac{4}{\pi N} - \frac{4}{\pi N^2} + o(N^{-2}).$$

□

Corollary 5.10 (Primitive degree-four exact atom constants)**.** *Let $G_m(z) = \sum_{j=0}^m \varepsilon_j z^j$ with independent Rademacher signs, and put $N = m + 1$. If ζ is a primitive fifth or tenth root of unity, then*

$$\mathbb{P}\{G_m(\zeta) = 0\} = 0 \qquad \textit{unless } 5 \mid N,$$

while, as $N \to \infty$ *through multiples of* 5,

$$N^2 \mathbb{P}\{G_m(\zeta) = 0\} \longrightarrow \frac{20\sqrt{5}}{\pi^2}.$$

If ζ *is a primitive eighth root of unity, then*

$$\mathbb{P}\{G_m(\zeta) = 0\} = 0 \qquad \textit{unless } 8 \mid N,$$

while, as $N \to \infty$ *through multiples of* 8,

$$N^2 \mathbb{P}\{G_m(\zeta) = 0\} \longrightarrow \frac{64}{\pi^2}.$$

If ζ *is a primitive twelfth root of unity, then*

$$\mathbb{P}\{G_m(\zeta) = 0\} = 0 \qquad \textit{unless } 6 \mid N,$$

while, as $N \to \infty$ *through multiples of* 6,

$$N^2 \mathbb{P}\{G_m(\zeta) = 0\} \longrightarrow \frac{48}{\pi^2}.$$

Proof. Let p_L denote the mass function of a length-L Rademacher sum. We first record the local-limit sum used for the fifth and tenth roots:

$$\sum_t p_L(t)^5 = \frac{1}{2} \int_{\mathbb{R}} \left(\frac{2}{\pi L}\right)^{5/2} \exp\left(-\frac{5x^2}{2L}\right) dx + o(L^{-2}) = \frac{4}{\sqrt{5}\,\pi^2}\, L^{-2} + o(L^{-2}).$$

The Hoeffding tail and Stirling approximation justify the same central-range Riemann-sum replacement as in the proof of Corollary 5.7.

For $q = 5$, the identity $1 + \zeta + \cdots + \zeta^4 = 0$ makes

$$S_0 + S_1\zeta + \cdots + S_4\zeta^4 = 0$$

equivalent to $S_0 = \cdots = S_4$. The support parities of the five residue sums agree exactly when the five residue-class counts modulo 5 have the same parity, which for $N = 5a + r$ happens exactly at $r = 0$. When $N = 5L$, therefore,

$$\mathbb{P}\{G_m(\zeta) = 0\} = \sum_t p_L(t)^5,$$

and multiplying the displayed asymptotic by $N^2 = 25L^2$ gives $20\sqrt{5}/\pi^2$.

For $q = 10$, write

$$D_a = S_a - S_{a+5}, \qquad 0 \le a < 5.$$

Since $\zeta^5 = -1$ and $1 - \zeta + \zeta^2 - \zeta^3 + \zeta^4 = 0$, the equation $G_m(\zeta) = 0$ is equivalent to

$$D_a = t(-1)^a, \qquad 0 \le a < 5,$$

for some integer t. The parity of D_a is $M_a + M_{a+5} \pmod 2$, the count of the residue class a modulo 5. These five parities agree exactly when $5 \mid N$. If $N = 5L$, each D_a has the law of a length-L Rademacher sum, and symmetry gives again

$$\mathbb{P}\{G_m(\zeta) = 0\} = \sum_t p_L(t)^5.$$

Thus the tenth-root constant is the same $20\sqrt{5}/\pi^2$.

For $q = 8$, put $D_a = S_a - S_{a+4}$ for $0 \le a < 4$. A primitive eighth root has degree 4, so $1, \zeta, \zeta^2, \zeta^3$ are linearly independent over $\mathbb{Q}$, and $G_m(\zeta) = 0$ is equivalent to $D_a = 0$ for every $a < 4$. This requires $M_a + M_{a+4}$ to be even for all a, i.e. every residue-class count modulo 4 is even; for $N = 4a + r$ this is equivalent to $r = 0$ and a even, or $8 \mid N$. When $N = 8L$, the four equalities are independent and

$$\mathbb{P}\{G_m(\zeta) = 0\} = \left(\sum_t p_L(t)^2\right)^4.$$

By Vandermonde's identity,

$$\sum_t p_L(t)^2 = 4^{-L}\binom{2L}{L} = \frac{1}{\sqrt{\pi L}}(1 + o(1)),$$

and multiplying by $N^2 = 64L^2$ gives $64/\pi^2$.

For $q = 12$, put $D_a = S_a - S_{a+6}$ for $0 \le a < 6$. The two relations

$$1 - \zeta^2 + \zeta^4 = 0, \qquad \zeta - \zeta^3 + \zeta^5 = 0$$

show that $G_m(\zeta) = 0$ is equivalent to

$$D_0 = D_4 = -D_2, \qquad D_1 = D_5 = -D_3.$$

The needed parity equalities say that the three even residue-class counts modulo 6 have the same parity and that the three odd residue-class counts modulo 6 have the same parity. For $N = 6a + r$, a direct check of $0 \le r < 6$ shows that this happens exactly when $r = 0$, i.e. $6 \mid N$. If $N = 6L$, each D_a has the law of a length-L Rademacher sum, so

$$\mathbb{P}\{G_m(\zeta) = 0\} = \left(\sum_t p_L(t)^3\right)^2.$$

The cubic local-limit sum computed in Corollary 5.7 is

$$\sum_t p_L(t)^3 = \frac{2}{\pi\sqrt{3}}\, L^{-1} + o(L^{-1}),$$

and multiplying its square by $N^2 = 36L^2$ gives $48/\pi^2$. □

Corollary 5.11 (Bounded-denominator exact zeros)**.** *Let $G_m(z) = \sum_{j=0}^m \varepsilon_j z^j$ with independent Rademacher signs, put $N = m + 1$, and let*

$$\mathcal{R}_Q^\circ := \{\zeta \in \mathbb{T} : \zeta^q = 1 \text{ for some } 1 \le q \le Q\} \setminus \{1, -1\}.$$

For every fixed $Q \ge 1$ there is $C_Q < \infty$ such that, for all N,

$$\mathbb{P}\{\exists\, \zeta \in \mathcal{R}_Q^\circ :\ G_m(\zeta) = 0\} \le \frac{C_Q}{N}.$$

Moreover,

$$\mathbb{P}\{\exists\, \zeta \in \mathcal{R}_Q^\circ : \varphi(\operatorname{ord}\zeta) > 2,\ G_m(\zeta) = 0\} \le \frac{C_Q}{N^2}.$$

The first bound is sharp at fixed Q as soon as $Q \geq 3$: if $3 \mid N$, then

$$\mathbb{P}\{\exists\, \zeta \in \mathcal{R}_Q^\circ :\ G_m(\zeta) = 0\} \geq \frac{c}{N}$$

with an absolute $c > 0$.

Proof. Every point of $\mathcal{R}_Q^\circ$ is a primitive root of some order $3 \leq q \leq Q$. Proposition 5.5 gives

$$\mathbb{P}\{G_m(\zeta) = 0\} \leq C_q N^{-\varphi(q)/2}$$

for a primitive root of order q. Summing over the finitely many roots with order at most Q gives the C_Q/N bound, since $\varphi(q) \geq 2$ for every non-real root of unity. If $\varphi(q) > 2$, then $\varphi(q) \geq 4$, giving the C_Q/N^2 bound after the same finite union.

For sharpness, take a primitive cubic root ζ. When $3 \mid N$, the divisible case in Proposition 5.5 is admissible and $\varphi(3) = 2$, so

$$\mathbb{P}\{G_m(\zeta) = 0\} \geq cN^{-1}$$

after decreasing c to absorb the finitely many small N. This event is contained in the bounded-denominator union whenever $Q \geq 3$. □

Corollary 5.12 (Peeled bounded-denominator cyclotomic layers)**.** *With the notation of Corollary 5.11, fix $D_0 \geq 0$ and assume that there is an order $3 \leq q \leq Q$ with $\varphi(q) > D_0$. Put*

$$d_{Q,D_0} := \min\{\varphi(q) : 3 \leq q \leq Q,\ \varphi(q) > D_0\}.$$

Then

$$\begin{aligned}&\mathbb{P}\{\exists\, \zeta \in \mathcal{R}_Q^\circ : \varphi(\operatorname{ord}\zeta) > D_0,\ G_m(\zeta) = 0\}\\ &\quad = N^{-d_{Q,D_0}/2} \sum_{\substack{3\leq q\leq Q\\ \varphi(q)=d_{Q,D_0}}} \gamma_q\, \mathbf{1}_{\{N \bmod 2q \text{ is admissible for } q\}} + o_Q(N^{-d_{Q,D_0}/2}).\end{aligned}$$

Thus every fixed bounded-denominator exact-zero grid has a leading coefficient on each peeled cyclotomic degree layer; only continuum or growing-denominator questions remain outside this finite lattice expansion.

Proof. The event in the first line is the union of the primitive-order events E_q from Corollary 5.6 over the finite set

$$\mathcal{S}_{Q,D_0} := \{q : 3 \leq q \leq Q,\ \varphi(q) > D_0\}.$$

Indeed, integer coefficients make vanishing at one primitive q-th root equivalent to vanishing at every primitive q-th root. Applying Corollary 5.6 to $\mathcal{S}_{Q,D_0}$ gives the displayed formula. □

Corollary 5.13 (Bounded-denominator exact-zero leading coefficient)**.** *With the notation of Corollary 5.11,*

$$\mathbb{P}\{\exists\, \zeta \in \mathcal{R}_Q^\circ :\ G_m(\zeta) = 0\} = \frac{c_Q(N)}{N} + o_Q(N^{-1}),$$

where the periodic coefficient is

$$c_Q(N) = \frac{2\sqrt{3}}{\pi}\mathbf{1}_{\{Q\geq 3\}}\mathbf{1}_{\{3|N\}} + \frac{4}{\pi}\mathbf{1}_{\{Q\geq 4\}}\mathbf{1}_{\{4|N\}} + \frac{2\sqrt{3}}{\pi}\mathbf{1}_{\{Q\geq 6\}}\mathbf{1}_{\{3|N\}}.$$

Equivalently, the only order N^{-1} exact-zero contributions in a fixed bounded-denominator non-real grid are the primitive cubic pair, the primitive quartic pair, and the primitive sixth pair; all higher-degree roots and all overlaps among these three pairs are lower order.

Proof. Let E_q be the event that $G_m(\zeta) = 0$ for a primitive q-th root ζ. Since the coefficients are real, the two conjugate primitive roots of a fixed order give the same event. Corollary 5.7 gives

$$\begin{aligned}
\mathbb{P}(E_3) &= \frac{2\sqrt{3}}{\pi N}\mathbf{1}_{\{3|N\}} + o(N^{-1}),\\
\mathbb{P}(E_4) &= \frac{4}{\pi N}\mathbf{1}_{\{4|N\}} + o(N^{-1}),\\
\mathbb{P}(E_6) &= \frac{2\sqrt{3}}{\pi N}\mathbf{1}_{\{3|N\}} + o(N^{-1}).
\end{aligned}$$

The remaining primitive orders have $\varphi(q) > 2$, and their total contribution is $O_Q(N^{-2})$ by Corollary 5.11.

Corollary 5.8 gives the sharper explicit N^{-2} constants for all pairwise intersections among E_3, E_4, E_6, in particular these intersections are $O(N^{-2})$. Inclusion–exclusion over the subset of $\{3, 4, 6\}$ with order at most Q, together with the $O_Q(N^{-2})$ higher-degree contribution, gives the displayed asymptotic. □

Corollary 5.14 (First nonquadratic bounded-denominator coefficient)**.** *With the notation of Corollary 5.11,*

$$\mathbb{P}\{\exists\, \zeta \in \mathcal{R}_Q^\circ : \varphi(\operatorname{ord}\zeta) > 2,\ G_m(\zeta) = 0\} = \frac{d_Q(N)}{N^2} + o_Q(N^{-2}),$$

where

$$\begin{aligned}
d_Q(N) =& \frac{20\sqrt{5}}{\pi^2}\mathbf{1}_{\{Q\geq 5\}}\mathbf{1}_{\{5|N\}} + \frac{64}{\pi^2}\mathbf{1}_{\{Q\geq 8\}}\mathbf{1}_{\{8|N\}}\\
&+ \frac{20\sqrt{5}}{\pi^2}\mathbf{1}_{\{Q\geq 10\}}\mathbf{1}_{\{5|N\}} + \frac{48}{\pi^2}\mathbf{1}_{\{Q\geq 12\}}\mathbf{1}_{\{6|N\}}.
\end{aligned}$$

Equivalently, after the primitive quadratic roots are removed, the only order-N^{-2} exact-zero contributions in a fixed bounded-denominator non-real grid come from primitive orders $5, 8, 10, 12$; all higher cyclotomic orders and all overlaps are lower order.

Proof. This is Corollary 5.12 with $D_0 = 2$, together with the explicit degree-four constants and admissible classes from Corollary 5.10. The only orders with $\varphi(q) = 4$ are $q = 5, 8, 10, 12$. □

Corollary 5.15 (Bounded-denominator exact-zero expansion through N^{-2})**.** *With the notation of Corollary 5.11,*

$$\mathbb{P}\{\exists\,\zeta \in \mathcal{R}_Q^\circ :\ G_m(\zeta) = 0\} = \frac{c_Q(N)}{N} + \frac{e_Q(N)}{N^2} + o_Q(N^{-2}),$$

where $c_Q(N)$ is the coefficient in Corollary 5.13. Let $d_Q(N)$ be the degree-four coefficient displayed in Corollary 5.14. Then

$$\begin{aligned} e_Q(N) = d_Q(N) &- \frac{2\sqrt{3}}{\pi}\mathbf{1}_{\{Q\ge 3\}}\mathbf{1}_{\{3|N\}} - \frac{4}{\pi}\mathbf{1}_{\{Q\ge 4\}}\mathbf{1}_{\{4|N\}} \\ &- \frac{2\sqrt{3}}{\pi}\mathbf{1}_{\{Q\ge 6\}}\mathbf{1}_{\{3|N\}} - \frac{8\sqrt{3}}{\pi^2}\mathbf{1}_{\{Q\ge 4\}}\mathbf{1}_{\{12|N\}} \\ &- \frac{48}{\pi^2}\mathbf{1}_{\{Q\ge 6\}}\mathbf{1}_{\{6|N\}} - \frac{8\sqrt{3}}{\pi^2}\mathbf{1}_{\{Q\ge 6\}}\mathbf{1}_{\{12|N\}}. \end{aligned}$$

Thus the full unpeeled second-order term is not just the first nonquadratic cyclotomic layer: it also contains the second-order corrections to the degree-two single atoms and the pairwise overlaps among the degree-two order events.

Proof. Let E_q be the primitive-order event as above. The single-event contributions from $q = 3, 4, 6$ are given to order N^{-2} by Corollary 5.9. The single-event contributions from $\varphi(q) = 4$ sum to $d_Q(N)N^{-2} + o_Q(N^{-2})$ by Corollary 5.14. All primitive orders with $\varphi(q) \ge 6$ contribute $O_Q(N^{-3})$ by Proposition 5.5.

For intersections, Corollary 5.8 gives the three degree-two pairwise terms. They enter with a minus sign in inclusion–exclusion and give exactly the last three displayed terms in $e_Q(N)$. Every other pairwise intersection contains either one degree-two and one degree-four-or-higher event, or two degree-four-or-higher events. The product-quotient rank argument used in Corollary 5.6 gives rank at least 6 in the first case and at least 8 in the second, hence these intersections are $O_Q(N^{-3})$. The same argument gives $O_Q(N^{-3})$ for every triple intersection, including $E_3 \cap E_4 \cap E_6$, because the three distinct cyclotomic factors Φ_3, Φ_4, Φ_6 have total rank 6. Higher inclusion–exclusion terms are smaller. Combining these terms gives the displayed expansion. □

Corollary 5.16 (Bounded-denominator exact zeros including endpoints)**.** *Let $G_m(z) = \sum_{j=0}^m \varepsilon_j z^j$ with independent Rademacher signs, put $N = m + 1$, and let*

$$\mathcal{R}_Q := \{\zeta \in \mathbb{T} : \zeta^q = 1 \text{ for some } 1 \le q \le Q\}.$$

Define

$$b_N := \mathbf{1}_{\{2|N\}} 2^{-N}\binom{N}{N/2}, \qquad h_N := \mathbf{1}_{\{4|N\}}\left(2^{-N/2}\binom{N/2}{N/4}\right)^2.$$

For every fixed $Q \ge 1$,

$$\mathbb{P}\{\exists\,\zeta \in \mathcal{R}_Q :\ G_m(\zeta) = 0\} = (1+\mathbf{1}_{\{Q\ge 2\}})b_N - \mathbf{1}_{\{Q\ge 2\}}h_N + \frac{c_Q(N)}{N} + o_Q(N^{-1}),$$

where $c_Q(N)$ is the non-real coefficient in Corollary 5.13. Thus the full bounded-denominator root-of-unity grid splits, through order N^{-1}, into the exact endpoint-union term and the leading non-real cyclotomic contribution.

Proof. Let $E_+ = \{G_m(1) = 0\}$, $E_- = \{G_m(-1) = 0\}$, and

$$F_Q = \{\exists\, \zeta \in \mathcal{R}_Q^\circ :\ G_m(\zeta) = 0\}.$$

The endpoint variables $G_m(1) = \sum_{j=0}^{N-1} \varepsilon_j$ and $G_m(-1) = \sum_{j=0}^{N-1} (-1)^j \varepsilon_j$ both have the law of a length-N Rademacher sum, hence

$$\mathbb{P}(E_+) = \mathbb{P}(E_-) = b_N.$$

For the joint endpoint event, write

$$A = \sum_{\substack{0 \le j < N \\ j \text{ even}}} \varepsilon_j, \qquad B = \sum_{\substack{0 \le j < N \\ j \text{ odd}}} \varepsilon_j.$$

Then $G_m(1) = A + B$ and $G_m(-1) = A - B$, so $E_+ \cap E_-$ is the event $\{A = 0,\ B = 0\}$. This is impossible unless both parity classes have even cardinality. Since the two parity classes have equal size $N/2$ when N is even, this happens exactly when $4 \mid N$, and then independence gives

$$\mathbb{P}(E_+ \cap E_-) = \left(2^{-N/2} \binom{N/2}{N/4}\right)^2.$$

Therefore the endpoint contribution is

$$\mathbb{P}(E_+ \cup E_-) = 2b_N - h_N$$

when $Q \ge 2$, while for $Q = 1$ it is just b_N.

It remains to check that intersections between endpoint zeros and non-real bounded-denominator zeros are $o_Q(N^{-1})$. Fix a primitive non-real q-th root with $3 \le q \le Q$. The event $G_m(1) = 0$ is the divisibility condition $\Phi_1 \mid G_m$, and $G_m(-1) = 0$ is $\Phi_2 \mid G_m$. Grouping coefficients modulo the relevant least common multiple, the simultaneous endpoint and primitive-q conditions are the kernel condition for the product quotient map onto the distinct cyclotomic factors Φ_1, Φ_q, or Φ_2, Φ_q. Its rational rank is $1 + \varphi(q)$. The projection-and-largest-atom argument used in Corollary 5.6 gives

$$\mathbb{P}(E_\pm \cap E_q) \le C_q N^{-(1+\varphi(q))/2} = O_q(N^{-3/2}).$$

With both endpoints and E_q the rank is $2+\varphi(q)$, so the bound is even smaller. Summing over the finite set $3 \le q \le Q$ gives $\mathbb{P}((E_+ \cup E_-) \cap F_Q) = o_Q(N^{-1})$. Finally, Corollary 5.13 gives

$$\mathbb{P}(F_Q) = \frac{c_Q(N)}{N} + o_Q(N^{-1}),$$

and inclusion–exclusion with the endpoint union proves the formula. □

Proposition 5.17 (Fixed-radius lower bound at rational angles)**.** *Let $G_m(z) = \sum_{j=0}^{m} \varepsilon_j z^j$ with independent Rademacher signs and put $N = m+1$. Let $q \geq 3$ be fixed, and let ζ be a primitive q-th root of unity. There are constants $c_q, t_q > 0$ such that, for all $N \geq 1$,*

$$\mathbb{P}\{|G_m(\zeta)| \leq t_q\} \geq \frac{c_q}{N}.$$

Thus the fixed-point upper bound $O_q(N^{-1})$ from Proposition 5.2 is sharp at every fixed non-real rational angle at a constant lattice scale, even though the exact atom may be smaller when $\varphi(q) > 2$.

Proof. It is enough to prove the estimate for $N \geq 4q$; finitely many smaller N are absorbed by increasing t_q and decreasing c_q. Choose $k \in \{1, \dots, q-1\}$ with $\left|\Im(\zeta^k)\right| \geq \sqrt{3}/2$, as in the proof of Proposition 5.2, and group the residue-class sums

$$S_\ell = \sum_{\substack{0 \leq j \leq m \\ j \equiv \ell \pmod q}} \varepsilon_j, \qquad 0 \leq \ell < q.$$

Let $M_\ell = \#\{0 \leq j < N : j \equiv \ell \pmod q\}$. Then

$$\frac{N}{2q} \leq M_\ell \leq \frac{2N}{q} \qquad (0 \leq \ell < q)$$

for $N \geq 4q$. Condition on all S_ℓ with $\ell \notin \{0, k\}$, and write

$$B = \sum_{\ell \notin \{0,k\}} \zeta^\ell S_\ell.$$

The event

$$E = \{|S_\ell| \leq C_q \sqrt{N} \text{ for every } \ell \notin \{0, k\}\}$$

has probability at least $c_q > 0$, by the central binomial lower bound on a window of length $\asymp_q \sqrt{M_\ell}$, applied independently to the $q-2$ conditioned residue classes. On E, $|B| \leq C_q \sqrt{N}$.

The real-linear map $T(x, y) = x + \zeta^k y$ has inverse norm bounded in terms of q. The image of the parity lattice on which (S_0, S_k) lives is one of the four translates of the fixed lattice $2\mathbb{Z} + 2\zeta^k\mathbb{Z}$ in $\mathbb{C}$. Let t_q be a common covering radius for these four translates. For every realized B, there is an admissible parity pair (x_B, y_B) such that

$$\left|x_B + \zeta^k y_B + B\right| \leq t_q.$$

If E holds, the inverse-norm bound gives $|x_B| + |y_B| \leq C_q \sqrt{N}$, after increasing C_q. Since $M_0, M_k \asymp_q N$, the local central binomial estimate at bounded standard-deviation scale gives

$$\mathbb{P}\{S_0 = x_B,\ S_k = y_B\} \geq c_q N^{-1}.$$

Consequently, conditionally on E,

$$\mathbb{P}\{|G_m(\zeta)| \leq t_q \mid (S_\ell)_{\ell \notin \{0,k\}}\} \geq c_q N^{-1}.$$

Multiplying by $\mathbb{P}(E) \geq c_q$ proves the claim. □

Remark 5.18 (Fixed rational angles versus the continuum). Proposition 5.2 settles the simplest discrete analogue beyond the two parity endpoints: a root of unity of order q has small-ball probability $O(q/N)$, and fixed non-real rational angles therefore have the expected two-dimensional N^{-1} fixed-point anti-concentration; Proposition 5.17 shows that this scale is sharp at every fixed order at a constant lattice radius. Corollary 5.3 packages this into an all-points bound on a fixed rational grid, and Corollary 5.4 extends the same elementary method to all denominators $q \leq Q$ at the cost Q^3/N. Proposition 5.5 identifies the exact zero atom at every fixed primitive root: on the admissible parity residue classes it has a local limit $\gamma_q N^{-\varphi(q)/2}$, with γ_q given by a kernel covolume, while on the inadmissible classes it is absent. Corollary 5.6 sums these local limits over any fixed finite set of cyclotomic orders and shows that only the first nonempty totient layer contributes to the leading exact-zero asymptotic. Corollary 5.11 promotes this to a bounded-denominator exact-zero grid statement: for fixed Q, non-real roots of denominator at most Q have total exact-zero probability $O_Q(N^{-1})$, this is sharp when the cubic roots are present, and after removing the cyclotomic degree-two roots the bound improves to $O_Q(N^{-2})$. Corollary 5.12 gives the general peeled-layer formula after removing all primitive orders below any fixed cyclotomic degree threshold. Corollary 5.13 then identifies the exact leading N^{-1} coefficient: it is the sum of the primitive quadratic pair constants that are present on the relevant divisibility classes. Corollaries 5.8 and 5.9 compute the degree-two pairwise overlaps and the second-order corrections to the degree-two single atoms, and Corollary 5.15 combines these with the degree-four layer to give the full unpeeled fixed-Q expansion through N^{-2}. Corollaries 5.10 and 5.14 separately identify the first nonquadratic peeled layer, where primitive orders $5, 8, 10, 12$ give the complete order-N^{-2} coefficient after the degree-two roots are removed. Corollary 5.16 then adds back $z = \pm 1$: through order N^{-1}, the full bounded-denominator root-of-unity grid is the exact endpoint-union central-binomial term plus the same non-real coefficient $c_Q(N)/N$. Thus the two-dimensional N^{-1} scale is a possible genuine atom floor exactly at the primitive quadratic roots $q = 3, 4, 6$, every fixed bounded-denominator cyclotomic layer has a leading coefficient, and the unpeeled non-real bounded-denominator exact-zero union is explicit through order N^{-2}, with the endpoint-inclusive grid explicit through order N^{-1}. Fixed-radius small balls therefore cannot, in general, be read off from the zero atom alone. This is still far from a Rademacher analogue of Theorem 1.2. The continuum problem requires uniform control over moving arcs, large denominators, and irrational angles, where the relevant input is a density-level local central limit theorem for the boundary field, not just the binomial atom bound used above.

Remark 5.19 (Boundary lattice obstruction). For Rademacher coefficients and the balanced convention $m = 2r$, both $G_m(1)$ and $G_m(-1)$ are odd

sums of signs, hence neither can vanish. Thus the two parity points do *not* give an atom at $t = 0$ in the balanced model considered in Theorem 1.6. What remains true, and is the point relevant here, is that the boundary values are lattice-valued rather than density-valued: Proposition 5.1 gives the correct endpoint scale and shows that small-ball estimates for $|G_m(\pm 1)|$ cannot behave like bounded-density estimates below the lattice spacing, while Proposition 5.2 gives the corresponding two-dimensional fixed-point scale at each fixed-order non-real root of unity, with an explicit q-loss when the order is q. In adjacent parity conventions, where the number of signs is even, exact boundary zeros at $z = \pm 1$ occur with probability of order $m^{-1/2}$. This is why any discrete analogue of Theorem 1.2 must be formulated only above the appropriate lattice scale.

5.3. **The shared moral with phase retrieval.** The same continuity warning appears in [2]. There, for ± 1 measurement matrices $A \in \{\pm 1\}^{(2m-1)\times m}$, a pigeonhole argument in the first two coordinates shows that at least m of the $2m - 1$ rows lie in one of the two hyperplanes $\{x_1 = x_2\}$, $\{x_1 = -x_2\}$, so some $m \times m$ submatrix is singular and the least-singular-value functional $\omega(A)$ vanishes *deterministically*, for every sign matrix [2, Rem. 6.2]. The conclusion drawn there is that continuity of the entry law is necessary for any lower bound on ω. The present paper exhibits the Toeplitz mirror of this warning at the boundary-evaluation level: lattice-valued boundary fields cannot be treated by bounded-density small-ball estimates all the way down to zero. We regard this as the unifying moral of the two companions: the parity coin of [1], the Bernoulli pigeonhole of [2], and the boundary lattice obstruction of Remark 5.19 are three faces of a single boundary-evaluation obstruction.

6. Open problems

We close with the problems left open, in decreasing order of how much of the present program they would complete.

Problem 6.1 (Sharp constants and the lower tail)**.** *Determine the matching lower tail and limiting law behind Theorem 1.2. The proof gives the correct upper scale* $\mathbb{P}\{\min_{\mathbb{T}} |G_m| \le t\} \le C\sqrt{m}\, t$ *and, at the continuum threshold* $t = \tau/\sqrt{N}$*, the sharp whole-circle probability upper constant* $\sqrt{\pi/3}\,\tau$*, but it does not prove the matching lower bound or the exponential minimum-modulus law expected from* [5, 6]*. Proposition 3.25 proves the endpoint contribution at order* $t/\sqrt{m}$*, and Corollaries 3.26, 3.27, and 3.28 give the exact bivariate endpoint formula, sharp second-order endpoint expansion, and continuum-scale* $O_\tau(N^{-1})$ *endpoint-neighborhood contribution before Corollary 3.54 makes the resulting global endpoint floor explicit. Proposition 3.14 proves the correct away-from-endpoint first moment for level crossings, and Proposition 3.15 computes the fixed-bulk down-crossing density as* $t\sqrt{N}/\sqrt{12\pi}$ *per unit arclength, and Corollary 3.16 converts it at* $t = \tau/\sqrt{N}$ *into the explicit fixed-bulk*

limit $(2\pi - 4\delta)\tau/\sqrt{12\pi}$ *for down-crossings, while Corollary 3.17 gives the corresponding fixed-bulk first-moment probability upper bound with the same down-crossing constant. Proposition 3.18 proves that a bound on the factorial second moment of the fixed-bulk down-crossing count gives a Paley–Zygmund lower tail with leading small-*τ *constant* $(2\pi - 4\delta)/\sqrt{12\pi}$*, and Corollary 3.19 combines this with the sharp fixed-bulk upper bound to give the complete first-order fixed-bulk law as* $\tau \downarrow 0$*. Corollary 3.29 then combines the fixed-bulk upper bound with endpoint-neighborhood separation to give the sharp whole-circle continuum upper constant* $\sqrt{\pi/3}$*, and Corollary 3.20 shows that uniformity as* $\delta \downarrow 0$ *gives the whole-circle lower-tail constant* $\sqrt{\pi/3}$*; together, Corollary 3.30 gives the whole-circle first-order law under that uniform factorial second-moment input. Thus the remaining global lower-tail problem is not the one-dimensional endpoint small ball, the two-endpoint expansion, the continuum-scale endpoint neighborhood separation, the endpoint floor, the bulk first-moment constant, the sharp global upper constant, the resulting subcritical and logarithmic inverse-moment control, or the fixed-bulk/global first-order transfer, but the anti-clustering and decorrelation analysis that proves the required second-factorial-moment bound for the* $O(m)$ *local down-crossing opportunities uniformly through the endpoint-cutoff removal and ultimately upgrades it to an exponential minimum-modulus law, including the transition beyond the exact two-endpoint expansion at the parity points* $z = \pm 1$*.*

Problem 6.2 (Non-Gaussian universality for boundary separation)**.** *Prove Hypothesis 1.4, or the sharper analogue of (1.8), for bounded-density laws at full generality. Proposition 3.2 proves the fixed-point and deterministic-grid anti-concentration, and Corollary 3.57 shows that a deterministic* $Lm^{3/2}$ *global speed cutoff gives the correct uniform continuum rate with* L *as the speed parameter. Corollary 3.67 reduces the stated polynomial-scale hypothesis to proving a polynomially small speed tail at the natural* $\sqrt{\log m}$ *level; Corollary 3.70 verifies that tail under uniformly bounded moments of all orders, and Corollary 3.71 verifies it under a subexponential bounded-density assumption. Corollary 3.60 and Theorem 1.6 show that this natural-scale tail is not needed for the finite-section conditioning conclusion under circle clearance: the tail-free polynomial separation floor* $\min_{\mathbb{T}} |G_m| > m^{-A}$*, for every* $A > 1$*, already supplies the one boundary-minimum threshold needed there. Thus the missing non-Gaussian step is now the full cutoff-free continuum speed/crossing law for bounded-density coefficients, or a local-limit replacement for it, not point anti-concentration and not the clearance-conditional conditioning theorem. For discrete laws such as Rademacher, the rational-angle lattice cases are now separated: Proposition 5.1 gives the one-dimensional endpoint scale, Proposition 5.2 gives an* $O(q/N)$ *bound at a non-real root of unity of order* q*, Proposition 5.17 gives the matching* N^{-1} *fixed-radius lower bound at every fixed non-real rational angle, Corollary 5.3 gives the resulting* q^2/N *fixed-grid union bound, and Corollary 5.4 controls all denominators* $q \le Q$ *with loss*

Q^3/N. *Proposition 5.5 identifies the exact atom local limit at every fixed primitive q-th root as $\gamma_q N^{-\varphi(q)/2}$ on the admissible parity residue classes and zero on the inadmissible classes, and Corollary 5.11 shows that a fixed bounded-denominator non-real grid has total exact-zero probability $O_Q(N^{-1})$, sharp once cubic roots are included, improving to $O_Q(N^{-2})$ away from cyclotomic degree two. Corollary 5.16 adds the endpoint zeros back through order N^{-1}, as an exact central-binomial endpoint-union term plus the same non-real coefficient. Corollary 5.7 makes those degree-two admissible classes explicit: $q = 3, 6$ require $3 \mid N$, while $q = 4$ requires $4 \mid N$. The remaining discrete analogue is therefore a continuum local-CLT theorem above the relevant lattice scale, uniformly over moving arcs, large denominators, and irrational angles, not a fixed-rational-angle or exact-zero calculation. The minimum-modulus universality of Cook–Nguyen [6] establishes an analogous law for random trigonometric polynomials; transporting it to the banded boundary field $G_m|_{\mathbb{T}}$ requires the density-level delocalization input flagged in* [2, Rem. 6.3]*, since a Berry–Esseen estimate is insufficient at the microscopic scale and a local CLT for the boundary field is needed. The note of caution recorded in Remark 3.77 applies a fortiori to any numerical study of the discrete case: the Gaussian law is $\sqrt{m}\,t$, with excursions of angular length $\asymp t\,m^{-3/2}$, and the apparent $m\,t^2$ behaviour seen on coarse grids is a sampling artifact, not the small-ball exponent. Corollary 5.7 also records the corresponding leading constants on those classes: $2\sqrt{3}/\pi$ for $q = 3, 6$ and $4/\pi$ for $q = 4$. Corollaries 5.8 and 5.9 compute the pairwise N^{-2} overlap constants and the second-order single-event corrections among these three degree-two events. More generally, Corollary 5.6 gives the leading exact-zero asymptotic on the first nonempty cyclotomic degree layer for any fixed finite set of orders. Corollary 5.13 specializes this to the full leading coefficient for every fixed bounded-denominator exact-zero grid, and Corollary 5.15 upgrades it to the full unpeeled expansion through N^{-2}. Corollary 5.16 adds the two real endpoints and records the full bounded-denominator root-of-unity expansion through order N^{-1}. Corollary 5.14 separately gives the first nonquadratic layer after removing primitive orders $3, 4, 6$. More generally, Corollary 5.12 gives the next coefficient after peeling off all orders below any fixed cyclotomic degree threshold. These finite-layer calculations remove the fixed-rational exact-zero problem from the remaining discrete continuum problem.*

Problem 6.3 (Sharp m-dependence of the conditioning)**.** *Determine the correct m-dependence of $\sigma_{\min}(T_n)$ in the balanced regime on the stable event $\nu = r$. Proposition 4.1 loses a factor ρ^r with $\rho \asymp m^{-2}$, hence $\sigma_{\min} \gtrsim m^{-O(m)}$ at fixed large n; is the true typical size only polynomially small in m after conditioning on stability? This connects to the conjectural disk-count fluctuation and local-limit problem isolated in* [1, Conj. 1.3]*.*

Problem 6.4 (Joint law of boundary minimum and parity)**.** *Quantify the dependence between the boundary minimum $\mu = \min_{\mathbb{T}} |G_m|$ and the parity $\mathbf{1}\{G_m(1)G_m(-1) < 0\}$. Both are driven by the boundary field; (5.1) and*

Lemma 3.1 suggest that small μ is correlated with the field being small near a real point, where the parity is decided. A joint local limit theorem for $(G_m(1), G_m(-1), \mu)$ would unify the anti-concentration of [1] with the conditioning bound here.

Acknowledgments. The author thanks the Randomstrasse 101 project for the circle of questions around stable phase retrieval, and acknowledges that this paper is a companion to [1] and [2], from which it borrows, respectively, the operator-theoretic framework and the negative-second-moment small-ball machinery. The research was supported by the European Research Council (ERC) under the European Union's Horizon Europe research and innovation programme (grant agreement No. 101041711), by the Israel Science Foundation (grant number 2258/19) and by the Israel Science Foundation (ISF Grant 4101/25).

Declaration on generative AI. During the preparation of this work the author used generative AI tools to accelerate drafting and revision. All mathematical claims, proofs, and bibliographic information were subsequently reviewed and edited by the author, who takes full responsibility for the content of the manuscript.

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